PATENT PENDING

COORDINATED EMBER AND TERRAIN DEFENSE SYSTEM FOR THE WILDLAND-URBAN INTERFACE

> *A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.*

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THE ARCHIVAL RECORD

This document is the complete specification and drawings as filed, reproduced without alteration. It is a fixed record of the disclosure on its filing date and is not revised. Where any later edition of the Open Source Firefighter family differs from this document, this document states what was filed; the current edition states what the work has since become.

Specification — wrapper and Exhibits A through K	83 sheets
Drawings — FIG. 1 through FIG. 7	7 sheets
Total as filed	**90 sheets**

The two documents are recorded in the USPTO electronic acknowledgment receipt under the SHA-512 message digests below. Any copy of this material may be verified against them.

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Provisional Application — Coordinated Ember and Terrain Defense System for the Wildland-Urban Interface

PROVISIONAL APPLICATION FOR PATENT

SPECIFICATION

Title of Invention: Coordinated Ember and Terrain Defense System for the Wildland-Urban Interface

Inventor: Andrew Morton, Palmer Lake, Colorado, United States

Dedication: *This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.*

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

Filing basis: 35 U.S.C. § 111(b) provisional application

FIELD OF THE INVENTION

The invention relates to distributed fire-defense infrastructure for the wildland-urban interface, and more particularly to a coordinated system by which multiple independent water-application installations — residential, commercial, governmental, and expeditionary — negotiate shared water supply, coordinate protective wetting during the ember-cast window of a wildfire, integrate with incident command, and produce a verifiable operational record, over a transport-independent coordination protocol constrained by a publication-policy trust layer that enforces an operator-prevailing control discipline.

BACKGROUND

Ember cast, rather than radiant heat from the flame front, is the dominant cause of structure loss in wildland-urban interface fires. Wetting receiving surfaces during the ember-cast window prevents ignition, and property-scale sprinkler and cannon systems demonstrably reduce loss. What has not previously been addressed is the coordination problem at neighborhood and mixed-jurisdiction scale: multiple installations drawing on a common water supply cannot all operate at maximum draw without exhausting it; federal facilities and remote installations require appropriate handling of their operational data; and professional incident command arriving on scene requires a unified operational picture across installations built to different radio and network standards. Existing approaches are single-property, single-vendor, or dependent on cloud connectivity and thus fail precisely when infrastructure is compromised by the fire. The invention addresses coordination, degradation, trust, and integration as the primary problems.

SUMMARY OF THE INVENTION

The invention is a system and method for coordinated fire defense in which participating installations, each locally autonomous, form a shared addressable coordination domain and negotiate a rotating activation schedule that keeps aggregate water draw within sustainable supply while maintaining protective coverage across all participants during the ember-cast window. The system is organized in independent layers — physical/link, network, coordination, trust, and application — such that any technology at one layer may be substituted without modification to the layers above or below, provided the interface contract is met.

The following aspects are disclosed, each of which is described in detail in the documents attached hereto and identified in the Detailed Description below. Each is intended to be understood both in combination and independently:

1. A **transport-independent coordination protocol** carrying a closed, enumerated set of message categories, wherein a message lacking a category defined by the protocol has no representation and cannot be transported by a compliant installation, regardless of the underlying network.
2. A **publication-policy trust layer** situated above transport, which enforces at the boundary between local and published data — before any radio or wire is reached — that data the policy does not release cannot be transported, and which cryptographically signs and attests to what is published.
3. An **operator-prevailing control discipline** wherein automation manages resources but does not override an operator physically present at the equipment; conflicts at the point of use resolve in favor of the operator, and

no automated lockout of an operator's physically present resources or controls is permitted.

4. A **rotating coordination schedule** computed by a distributed algorithm from each node's reported water-source class, supply state, coverage pattern, and activation state, maintaining neighborhood coverage while allowing individual nodes to conserve water and pump runtime.
5. A **graceful-degradation architecture** in which every failure mode returns to the unprotected baseline the world already accepts: an isolated node operates autonomously on conservative defaults; a node with failed water supply serves as a network repeater; a missed coordination message is answered by supply-pressure physics as the fallback delivery mechanism.
6. A **connectionless command-broadcast mechanism** by which incident command issues best-effort domain-wide directives carrying scope filters (by supply class, zone, or geography), time-to-live expiry requiring refresh, and cryptographic signing, wherein a cease-draw directive to a node holding stored water constrains external draw rather than defense.
7. A **single command surface** presenting all enclosure classes — residential through tactical — as one coordination domain distinguished by capability metadata rather than class-specific software, integrable into existing incident-command systems.
8. A **cryptographically verifiable operational ledger** producing after-event reports from signed, chained snapshots rather than operator narrative, such that the record carries evidentiary weight independent of operator credibility.
9. An **expeditionary configuration** providing pump-site telemetry, duty-cycled protection extending a fixed fuel or water load across the ember-cast window, a published pump-control interface, and prepositioned units requiring no external activation, registration, or connectivity to enter service.
10. A set of **coordination-capable water appliances** — remotely actuated valves, gates, wyes, manifolds, and a buffered supply point providing managed standing storage from intermittent tender loads — each preserving manual operation on actuator failure and attaching to utility infrastructure without permanent modification.
11. A **node-waypoint aerial patrol** in which a domain census auto-generates an inspection route flown as observation-only, substituting a scheduled flyby for a fixed observation tower and reporting detection events, never imagery, over the coordination protocol.
12. A **thermal-signature verification method** in which an aerial thermal pass over the coordination domain reads protection state directly from physical surfaces — a wetted surface evaporatively cools and a dry or failed surface does not — producing a human-readable map of which structures are actually being protected and which have failed or gone unprotected, independent of any node's self-reported state, and stacking with the signed operational ledger to provide command-plus-confirmation (command record and independent physical confirmation) as well as an overflight-scale maintenance inspection.
13. A **utility right-of-way corridor-defense** deployment geometry in which the coordination domain is arrayed along an electric utility corridor, using the utility's own ignition and vegetation history as the pre-surveyed siting map, with the signed ledger serving as the utility's verifiable mitigation record.
14. A **local operator interface discipline** by which a physically present operator can initiate protection, lock out automation, command each zone independently, and signal distress to the coordination domain, through an interface that functions without a personal device, account, network connection, display interpretation, or prior instruction, that is operable in heavy gloves and in darkness, and that no software configuration may disable — disclosed in a reference embodiment of discrete panel-mounted controls with indicator lamps, and equally in any physical interface meeting the same capability, including rotary selectors, keyed switch sets, membrane panels, or other arrangements.
15. A **governance and adoption-by-reference discipline** by which external standards are incorporated to their current issued edition rather than restated, and by which the coordination protocol, closed vocabulary, and trust layer constitute a protected core that revision may not bind to any single transport, vendor, or mandatory external dependency.

The system is buildable from commodity components available through standard industrial supply channels, requires no novel electronic engineering, and operates when fully isolated from any external network. Specific values recited anywhere in this disclosure or its exhibits — radio band and power, direct-current bus voltage, enclosure rating and dimensions, zone count, control count, and component makes and models — are the applicant's reference implementation and are disclosed as examples rather than as limitations; the invention resides in the coordination, trust, degradation, and operator-control architecture, which is disclosed as operable across the range of physical implementations that satisfy the described capabilities, including in jurisdictions whose radio allocations differ from those of the reference. The threat model is a non-cognitive adversary — fire — such that the trust layer is scoped to human data-boundary and authority concerns rather than to electronic warfare, and commodity hardware is the correct and sufficient engineering answer rather than a compromise.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings, submitted herewith, are instructional schematics of a residential-class node and illustrate the system by function; component names appearing therein are class examples and not limitations.

- **FIG. 1 — Coordination domain:** a plan view of a neighborhood sharing one supply, showing equipped and unequipped structures, node states (drawing, yielded, degraded-relay), the coordination mesh, the gateway publishing outward to incident command, and the rotating schedule by which aggregate draw is held within sustainable capacity.
- **FIG. 2 — Layer model and trust boundary:** the layered architecture with module assignment, the substitution line below which transport may be replaced without effect above it, and the publication-policy chokepoint through which every outbound message passes, with the inbound verification sequence.
- **FIG. 3 — Node state machine:** the operating states including isolated, degraded, and repeater, with the transitions between them and the return path from every degraded state.
- **FIG. 4 — Node control flow:** the same behavior expressed as sequence, with the decision points, their failure destinations, and the asynchronous operator interrupts.
- **FIG. 5 — Node master schematic:** the complete node organized as a physical enclosure — antenna penetrations, dry control section, door-mounted operator controls and indicators, sloped wet/dry drip-shield boundary, and wet plumbing section — showing signal, power, and fluid paths together.
- **FIG. 6 — Terminal schedule:** the field-to-panel terminal interface, wherein panel wiring and field wiring meet at a labeled strip and lamp logic reports aggregate system state.
- **FIG. 7 — Reference installation, concept view:** an illustrative rendering of a residential-class node as installed, showing the wall-mounted enclosure, its external supply connection, the distribution line routed to the roof, and ridge-mounted heads wetting the roof plane and the structure’s immediate surroundings during an ember cast; provided as concept art rather than as a construction drawing.

DETAILED DESCRIPTION OF THE INVENTION

The complete detailed description of the invention is set forth in the following documents, attached hereto and forming part of this specification as filed, in their entirety. Together they constitute the enabling disclosure, providing capability requirements, example implementations, coordination protocol behavior, trust-layer and governance discipline, employment doctrine, and a worked operational scenario sufficient for a person of ordinary skill in the relevant arts — industrial controls, mesh networking, and fire-protection engineering — to make and use the invention across the described deployment classes.

The applicant intends the broadest disclosure consistent with the written description. Each numbered aspect in the Summary, and each capability, variation, appliance, deployment geometry, and method described in the incorporated documents, is disclosed both in combination and standing alone, and this specification is intended to establish priority as to all subject matter described in the attached documents.

Documents attached hereto and forming part of this specification as filed:

- **Exhibit A — Main Specification:** system architecture, layered model, control and enclosure requirements, radio-network classes, coordination protocol, event interface and enumerated message categories, command broadcast, backhaul and gateway nodes, expeditionary deployment, cryptographic event log, incident-command interface, governance charter, and — at Appendix C — the wire format, closed verb vocabulary, and outbound cross-reference to external exchange standards.
- **Exhibit B — Hardware Annex A:** reference hardware specification, enclosure and wet/dry cohabitation, wire-color and control discipline, local operator interface, accessory water appliances, tactical radio reference and power budget, thermal-signature verification, node-waypoint aerial patrol, the utility right-of-way corridor-defense disclosure, and — at Appendix A — the common parts catalog.
- **Exhibit C — Tactics Paper, including the Supply and Coverage Planning Table:** employment doctrine, the three deployment tiers, combined operation, and the supply, coverage, and head-spacing planning method.
- **Exhibit D — Deadman Fire Scenario:** an end-to-end worked operational narrative anchored to real fire behavior, demonstrating enablement and degradation behavior under fire.
- **Exhibit E — Design Philosophy:** the design rules governing the system, including structural safety, operator supremacy, commodity sourcing, and comprehensibility.
- **Exhibit F — Public Overview:** plain-language description of the system and its purpose.
- **Exhibit G — Community Adoption Briefing:** deployment and adoption method, including proof-of-function by thermal overflight.
- **Exhibit H — Public Roadmap (development path):** extension path for the specification family and the

shared-configuration model by which deployments are reproduced.

- **Exhibit I — Inspector's Review Draft:** the system expressed as numbered installation requirements in code-adoption language, including operator-interface and accessory-appliance requirements.
- **Exhibit J — The Training Range:** deployment and employment on instrumented range land, and the coordination-as-infrastructure argument.
- **Exhibit K — Software Requirements and Development Roadmap:** the layered software model, fixed invariants, node state machine, coordination algorithm properties, trust-layer enforcement, degradation ladder, and coverage relations.
- **Exhibit L — Drawings, FIG. 1 through FIG. 7:** the schematics described above.

Where any statement in this wrapper and a statement in an incorporated document differ, the incorporated document governs as to technical detail.

CONCLUSION

This provisional application is filed to establish priority as of its filing date for all subject matter disclosed herein and in the documents attached hereto and forming part of this specification as filed. No claims are submitted, none being required for a provisional application. The applicant reserves the right to pursue claims to any and all disclosed subject matter, in combination and severally, in one or more later-filed nonprovisional applications within the pendency period.

Respectfully submitted,

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The Open Source Firefighter

A specification for coordinated neighbourhood ember defense

v1.0 · August 2026 · The spine of the specification family

*Document ID: **OSF-1** · one document of the Open Source Firefighter specification family.*

> **A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.**

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

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Preface

This document specifies the Open Source Firefighter, a coordinated defensive infrastructure system designed to protect residential structures, federal facilities, and remote installations during the ember-cast window of wildland-urban interface fires. The specification defines capability requirements, example implementations, coordination protocols, integration interfaces, and operational logging sufficient for competent operators or contractors to build compliant installations across the range of deployment environments the actual problem requires.

The specification is published under permissive open licence. Implementations may be built by homeowners, by contractors, by federal land management agencies, or by manufacturers. Substitution of components meeting the capability requirements is expressly permitted and expected. This is a family of systems defined by function rather than by specific product.

This specification is the spine of a document family. Hardware detail lives in Annex A; wire format and vocabulary in Appendix C of this document; a known-good reference configuration in the parts catalogue at Annex A, Appendix A; employment doctrine in the Tactics Paper.

This specification incorporates in full: the Layered Architecture, the Event Interface and Message Category Enumeration, Backhaul and Gateway Nodes, Expeditionary Deployment, and the Governance Charter, each at the location its original front matter directed.

The Problem Addressed

Ember cast, not radiant heat, destroys most structures in wildland-urban interface fires. Embers travel ahead of the flame front and land on roofs, in gutters, in dry vegetation adjacent to structures, and in other ignitable receiving sites. Small ignitions grow into structural fires while the flame front is still distant. Keeping receiving surfaces wet during the ember-cast window prevents these ignitions.

The defensive principle at the individual property level is well understood. Sprinkler and cannon systems that wet roof, gutter, and perimeter surfaces during fire events reduce structure loss substantially. The engineering has been demonstrated. What has not been demonstrated at scale is coordination of individual property systems across whole neighbourhoods sharing a common water supply, or integration of residential systems with federal facility and remote installation defenses in mixed-jurisdiction fire environments.

This specification addresses the coordination problem across the range of participants that actual wildland-urban interface events involve. Multiple houses on a common water supply cannot simultaneously operate at maximum draw without exhausting the supply. Federal facilities surrounded by residential development benefit from neighbourhood-scale coordination but require appropriate handling of their internal operational data. Remote installations on federal land require radio infrastructure appropriate to their geographic scale and reliability requirements. Professional incident command arriving on scene requires unified operational picture across all participating systems regardless of the underlying radio infrastructure.

System Architecture

Overview

Each participating installation has an autonomous local system consisting of a water supply (municipal connection, cistern, well, or combination), distribution plumbing, coverage sprinklers or cannons, an industrial controller, a mesh network node, and an integration router. Where the water source requires a pump or pump starter, the controller provides a pump-control interface (a dry-contact start/stop signal, or a controller adapter card for larger starters) exactly as it interfaces any other controlled resource; a node on adequate pressurized supply requires no pump and populates no pump interface. Local systems operate independently when network coordination is unavailable and coordinate with neighbouring systems when the mesh network is operational. The intelligence in the system is resource management and coordination, not autonomous fire detection or targeting.

Layered Architecture

The system is organized as independent layers. Each layer is defined by the service it provides to the layer above and the interface it requires from the layer below. No layer depends on the internal implementation of any other. Technology at any layer may be substituted without modification to the layers above or below it, provided the interface contract is met. This is the architectural mechanism by which the specification absorbs new technology without revision to its core, and by which installations built decades apart remain interoperable.

The layering is organized along the lines of the OSI reference model, and the layer names below are given with their OSI positions. This is a **conceptual frame rather than a conformance claim**: the model is used because it is the one most readers already have, so that a network engineer, a controls integrator, and a fire officer can point at the same part of the system and mean the same thing. Nothing in this specification asserts conformance to OSI, and the architectural properties that matter — substitutability below the trust boundary, and dependencies that point in one direction only — do not depend on the layer numbering being canonical.

Physical and link layer (OSI 1–2). The radio, wire, or backhaul that moves bits between nodes. LoRa mesh via Meshtastic protocol and tactical-MANET-class StreamCaster MANET are the current reference classes. Ethernet, commercial LEO satellite service (Starlink class), government satellite transport (Starshield class or successor programs), cellular transport, and future radio classes are equally valid implementations of this layer. The specification imposes no requirement here beyond the ability to deliver coordination-domain traffic between participating nodes. Nothing above this layer knows or cares which technology carries the traffic.

Network layer (OSI 3). Participating nodes must share a common addressable coordination domain. A neighbourhood LoRa mesh satisfies this requirement. So does a set of remote installations joined by VPN over satellite backhaul. So does a hybrid in which residential mesh nodes and a federal facility on agency radio infrastructure are bridged through a gateway router. The requirement is the domain, not the medium. Routing, gateway selection, and addressing within the domain use standard mechanisms; this specification defines no novel protocols at this layer.

Envelope layer (OSI 4). The compact binary envelope and its delivery semantics. Delivery is best-effort by design; the system maintains no sessions and no connections. A missed message is a normal event rather than an error condition, and supply-pressure physics is the fallback coordination mechanism when messages do not arrive.

Vocabulary and trust layer (OSI 5–6). The closed, enumerated message vocabulary; quantized field encoding; cryptographic signing; and publication policy enforcement. This layer decides what leaves the node and attests to what does, and it is where the enumerated-message discipline becomes structural: a message whose verb is not in the enumeration has no representation and cannot be constructed. Data that the publication policy does not release cannot be transported, regardless of the capability of the underlying network.

Application layer (OSI 7). The coordination protocol — schedule negotiation, node state reporting, heartbeats, override handling, degradation behaviour — together with the event interface, incident command integration, the neighbourhood ledger, and after-event reporting. The coordination protocol is the invention. It is transport-independent by construction and shall remain so: no revision to this specification may bind it to a specific physical, link, or network technology. Consumers at this layer see coordination-domain participants and enumerated message categories. They do not see radios, transports, or local installation internals.

Two properties follow from this arrangement, and both are load-bearing.

Substitution. Everything at OSI 1–3 may be replaced — a different radio, a different mesh, a wire, a technology that does not yet exist — without modification above it. The trust layer and the coordination protocol cannot name a transport, and therefore cannot depend on one.

The publication chokepoint. Every outbound message, whatever application-layer function produced it, traverses the vocabulary and trust layer on its way out. Signing, scope resolution, and any rendering into an external exchange language occur there and nowhere else. The boundary between local data and published data is therefore enforced before any radio or wire is reached, and is independent of how the installation is networked. There is no path from the application layer to a transport that bypasses it; an implementation that provides one has not implemented this specification.

Control Architecture

The controller is industrial-grade PLC-based hardware mounted in a NEMA-rated enclosure with DIN rail mounting and standard terminal block wiring. This equipment class is chosen for reliability, long service life, availability through standard industrial supply channels, and compatibility with the installer pool of industrial electricians and controls technicians. Consumer-grade proprietary controllers are expressly outside the specification.

Controller requirements:

- IEC 61131-3 compatible programming environment or equivalent open programming interface
- Minimum sixteen digital I/O for pump control, valve control, flow sensing, and status indication
- Analog inputs sufficient for the installation’s sensing set, which may include supply pressure, expansion-tank or reservoir level, temperature, and optional acoustic or auxiliary sensing; the number of channels follows from the sensors actually fitted rather than a fixed minimum
- Serial or Ethernet interface for mesh node and router integration
- DIN rail mounting compatible with 35mm standard rail
- Operating temperature range appropriate for local installation environment

Example products meeting these requirements include the M-Duino series from Industrial Shields, the Kincony line, the Norvi controllers, and OpenPLC-compatible ESP32-based industrial controllers. Substitution of any controller meeting the capability requirements is compliant.

Enclosure

The controller, router, and associated interface hardware install in a NEMA 4X rated enclosure appropriate for outdoor and wet mounting. Standard hinged industrial enclosures meeting NEMA 4X (polycarbonate or stainless) satisfy the requirement. Enclosure size is not fixed: 16 by 20 by 8 inches is the reference size for a residential node, and any size that houses the internal panel layout and preserves the wet/dry separation is compliant. Detailed enclosure, mounting, grounding, labeling, and internal layout practice is specified in Annex A.

Radio Network Layer

The specification supports multiple radio infrastructure classes appropriate to different deployment environments. The specification does not prescribe a single radio technology because different participants have different operational requirements, different existing infrastructure investments, and different environmental conditions. Compliant installations select the radio class appropriate to their deployment context and integrate through the router layer.

LoRa mesh via Meshtastic protocol. Low-power mesh radio operating on the LoRa physical layer in the unlicensed ISM band allocated for the purpose in the jurisdiction of installation, using the Meshtastic protocol. The reference implementation operates in the 902–928 MHz band available in North America; installations elsewhere use the corresponding regional allocation (for example the 863–870 MHz band in Europe) at the power limit their regulator permits. Band and power are properties of the jurisdiction, not of the specification. **Operation is confined to unlicensed allocations.** A compliant node shall not operate on amateur radio spectrum. Amateur service rules require a licensed control operator and periodic station identification, restrict unattended automatic operation, bar communications that could reasonably be furnished through another radio service on a regular basis, and prohibit use where the operator holds a pecuniary interest — none of which an unattended, continuously operating, community-deployed node satisfies. Transmit power is bounded by the certification of the radio module as installed, not by the theoretical maximum of the allocation. Appropriate for residential and small commercial installations at neighbourhood scale. Low cost per node. Adequate coverage for typical suburban geometry. Suitable for homeowner-buildable installations following the RepRap tradition of specification-driven construction from documented parts.

Requirements:

- LoRa transceiver operating in the unlicensed ISM allocation of the installation jurisdiction at the maximum power that allocation permits; 902–928 MHz at 100 mW is the North American reference
- Meshtastic protocol firmware or protocol-compatible equivalent
- External antenna capability with weatherproof antenna mounting
- Battery backup sufficient for minimum 72 hours autonomous operation
- Serial or I2C interface to the local controller and router

Example products include the LILYGO T-Beam, the Heltec Wireless Stick Lite series, the RAK4631, and various other ESP32-based LoRa development boards packaged for industrial deployment.

Tactical MANET radio class. Agency-furnished mobile ad-hoc mesh data radios of the hardened interagency class (Silvus StreamCaster, Persistent Systems Wave Relay, TrellisWare TSM, and peers), deployed across federal land management, wildland firefighting operations, emergency medical services, law enforcement, and public safety broadly. Appropriate for federal facilities, remote installations on federal land, forestry cabins, ranger stations, and installations requiring range and reliability beyond LoRa mesh capability. Substantially higher throughput than LoRa enables coordination of larger installations with more comprehensive sensor and status telemetry. Compatible with existing agency radio infrastructure. The class is defined by capability — self-forming mesh, IP presentation, MIL-STD-810-class environmental qualification — never by vendor; any radio the furnishing agency fields and trusts is compliant at this layer.

Requirements:

- Tactical MANET-class radio (Silvus StreamCaster, Wave Relay, TSM, or interface-compatible peer)
- Self-forming mesh waveform with standard agency-compatible antenna configuration
- Integration through the router layer with layer abstraction from the coordination protocol

Interagency governance, adopted by reference: radio equipment operating on interagency wildland fire networks is governed by the National Interagency Incident Communications Division (NIICD) approved-equipment listings, and interoperability channel use by the National Interoperability Field Operations Guide (NIFOG). Nothing in this specification transmits on, contends for, or requires those networks; where agency-furnished radios are themselves NIICD-listed or incident-assigned, they operate under the furnishing agency’s authority and the incident communications plan. This specification’s coordination traffic rides its own unlicensed or agency-furnished data layers and never the fire voice nets.

Router Integration Layer

The router abstracts radio-layer differences from the coordination protocol. Participating installations exchange coordination messages through the router regardless of the underlying radio infrastructure. Homeowner installations with LoRa mesh nodes and federal installations with tactical MANET-class radios exchange the same coordination messages through the router layer. The router handles protocol translation, priority queuing, and network abstraction.

The router also serves as the policy enforcement point for publication decisions. Each installation configures its router to identify which log categories publish to the neighbourhood ledger and which stay local. This enables federal facilities, commercial installations, and privacy-conscious residential participants to participate in coordinated defense while maintaining appropriate boundaries on their internal operational data. Publication policy is specified in the Cryptographic Event Log section.

Router requirements:

- Radio layer abstraction supporting LoRa mesh and tactical MANET class at minimum
- Extensibility for additional radio classes as they are added to the specification
- Configurable publication policy engine with per-category filtering
- Cryptographic signing capability for outbound published messages
- Standard Ethernet or serial interface to the local controller
- Standard interfaces to supported radio hardware

The router is specified as functional capability. Reference implementations may run on industrial computing hardware, dedicated network appliances, or integrated within the controller platform. Substitution of any router implementation meeting the capability requirements is compliant.

Backhaul and Gateway Nodes

3.1 Satellite and External Backhaul

Installations and coordination domains may connect to external networks via commercial LEO satellite service (Starlink class, including compact low-power terminals suitable for expeditionary power budgets), government satellite transport (Starshield class or successor programs), or any transport presenting a standard IP interface to the network layer. The specification is transport-generic; named services are examples of the class.

Backhaul is an enhancement, never a dependency. A node or domain with no external connectivity coordinates locally, sequences, and performs its fire defense function in full. The backhaul extends the coordination domain and publishes the ledger; it is not required for fire defense. No revision may alter this.

3.2 Gateway Nodes

A gateway node is an ordinary node whose router additionally bridges the coordination domain to a backhaul transport. Gateway is a role, not a model; any node may serve as one. One gateway serves an entire coordination domain: because the coordination protocol's message categories are compact by design, an entire valley of participating nodes reaches the outside world through mesh hops to a single terminal. Multiple gateways are permitted for redundancy. Zero gateways is fully operational.

Gateway nodes advertise external reachability into the coordination domain using standard routing protocol mechanisms; nodes select among advertised gateways per normal route metric behavior, and failed gateways age out and traffic reconverges per those same mechanisms. **This specification defines no novel gateway election protocol.** At the IP tier this is standard router configuration; radio classes with native gateway mechanisms meet the requirement through the router integration layer.

The gateway is also a policy point: it sits at the trust layer boundary where domain traffic meets the outside world, and publication policy applies to it as to any router.

Water Supply and Reservoir

The water supply may be a dedicated cistern, connection to municipal supply, connection to a well system with adequate delivery capacity, or a combination. Minimum reservoir capacity for cistern-based installations is 2500 gallons, sized to sustain individual system operation during coordination scheduling for the duration of the anticipated ember-cast window. Neighbourhoods with reliable municipal supply may use municipal connection as primary source with a smaller reservoir for buffering. Remote installations without municipal supply require cistern capacity appropriate to their coordination participation and response scenario.

Water supply requirements:

- Delivery capacity is sized to the installation's coverage set and supply class. A node on pressurized supply (municipal or gravity) operates at the available service pressure and flow, from a single hose-bib sprinkler upward. A node on a pumped source (cistern or draft) is sized by its own pump and line selection; the pump and plumbing dictate valve and outlet sizing, not the reverse.
- Standard NPT threaded connections appropriate for the source and distribution outlets
- Backflow prevention consistent with local plumbing code
- Isolation valves for maintenance and emergency shutoff

Pump

The pump provides the flow and pressure required to operate the coverage sprinklers or cannons during activation. Gas or diesel drive is preferred to eliminate dependence on grid electric power during events. Electric pumps with adequate battery backup or generator support are permitted where fuel-driven pumps are impractical.

Pump requirements:

- Self-priming, capable of lift from reservoir mounting configuration
- Rated flow minimum 40 gallons per minute at 80 psi discharge pressure
- Fuel drive: gasoline or diesel, minimum four hour continuous runtime on onboard fuel
- Electric drive: 240V single phase or 208V three phase, appropriate motor protection
- Remote start capability compatible with PLC control output

Example products include the Honda WX10 and WX15 series, the Predator 6.5 HP gas water pumps, various trash pump configurations rated for continuous operation, and industrial centrifugal pumps from Grundfos, Goulds, or equivalent manufacturers for electric installations.

Coverage Distribution

Coverage distribution places water on structure surfaces and adjacent receiving sites where embers land. Standard configurations include roof-line sprinklers, gutter-line spray heads, perimeter rotating sprinklers, and directable cannons for larger structures or specific coverage requirements. The specific configuration depends on structure geometry, roof material, adjacent vegetation, and property topology.

Coverage distribution requirements:

- Roof-line coverage sufficient to maintain wet surface across full roof area
- Gutter coverage sufficient to prevent ember accumulation in gutter debris
- Perimeter coverage extending minimum ten feet from structure walls
- Zone valving allowing selective activation of coverage areas
- Materials rated for radiant heat exposure during fire events

Coordination Protocol

Local Autonomous Operation

Each local system operates autonomously when the mesh network is unavailable or when coordination messages are not received within specified timeout intervals. Autonomous operation defaults to conservative water use appropriate for extended individual protection without regard to neighbourhood coordination. The system does not fail when isolated; it operates at individual capability without the coordination advantages the mesh network provides.

Coordinated Operation

When the mesh network is operational and coordination messages are being exchanged, participating systems synchronize their activation schedules. The coordination algorithm allocates operating time across participating systems such that aggregate water draw stays within the sustainable supply capacity while ensuring that no participating structure has an extended gap in coverage during the ember-cast window.

The reference coordination algorithm operates on the following principles. Each participating node reports its water source category, current supply state, coverage pattern, and current activation state to the mesh network through its router. The distributed algorithm computes a rotating schedule that keeps sufficient nodes active to maintain neighbourhood coverage while allowing individual nodes to conserve water and pump runtime during their off-cycles. Schedule recalculation happens continuously as node states change.

Graceful Degradation

The specification anticipates realistic degradation patterns. Individual nodes may fail during operation. Water supplies may be interrupted. Mesh network connectivity may vary. Fire department override may or may not be available. The system continues providing value at each degradation level. A node with failed water supply but functional mesh capability continues serving as a network repeater. A neighbourhood with reduced participation continues providing coordination for the participating subset. A neighbourhood operating without professional response integration continues operating on its autonomous coordination schedule.

Event Interface and Message Category Enumeration

2.1 The Event Interface

The system exposes one outbound integration port: the event interface. It is a read-only tap of the system’s internal event messaging — the same enumerated messages the system already generates for its own operation. Through this interface, external systems may be informed that an activation has started, that a zone is cycling, that supply pressure is degraded, that an event has ended, that the system requires maintenance.

The event interface is deliberately narrow:

- **Outbound only.** External systems listen. The event interface accepts no inbound commands and transports no inbound payloads. Control of the installation remains with the installation’s own controller, its operator, and — where integrated — the incident command interface defined in the main specification.
- **Enumerated messages only.** The interface carries the message categories enumerated in this specification and nothing else. It is not a general-purpose bus.
- **Lightweight by construction.** Specified as a simple application-layer event protocol suitable for serial, Ethernet, or dry-contact presentation. A home automation system, generator controller, building management system, or hand-built display panel may consume it with minimal integration effort.

A smart home may listen to the Firefighter. The Firefighter does not join the smart home.

2.2 Enumerated Message Categories

Published message categories are enumerated by this specification. The coordination mesh transports enumerated categories only. A message without a category defined in this section has no representation in the coordination protocol and cannot be transported by a compliant installation, regardless of the publication policy configured at the router.

Residential and light commercial tier:

- System status: heartbeat, readiness, fault, maintenance-required
- Distress: node identity, position, timestamp — published on operator activation of the hardware emergency control; deliberately minimal
- Node position: GNSS-derived location; coarse-position-only publication is a supported policy option
- Activation events: event start, event end, trigger source class
- Zone state: zone open/close transitions, cycle schedule position
- Supply state: pressure, flow aggregate, reservoir level where equipped
- Coordination traffic: schedule negotiation, node state reports, override acknowledgments
- Command broadcast: directive class (cease water operations, resume, priority protect, broadcast alert), scope filter, issuing authority, expiry — see Section 2.4
- Energy events: power source transitions, battery state, low-voltage disconnect
- Environmental telemetry: temperature, wind speed and direction, humidity, ember detection *events* where equipped

Expeditionary tier — additional categories:

- Pump state: running/stopped, RPM class, load state, fuel level or acoustic burn-rate forecast, start/stop command acknowledgment where pump control is supported
- Distribution state: lay pressure, flow, burst/loss detection events

Government and high-specification commercial tier — additional categories:

- Thermal detection events: detection occurrence, bearing class, confidence class — detection *events*, not imagery
- Triage state: internal triage strategy selection, contingency template activation
- Extended sensor telemetry as defined in the applicable tier annex

Expressly non-transportable at all tiers:

- Video and imagery of any kind
- Audio of any kind, including raw or compressed audio from acoustic machinery sensors; acoustic sensing publishes derived machinery state events only
- Physical security data: camera-derived data, motion detection intended for security purposes, access logs
- Personnel data of any kind

These categories are not merely “typically not published” — they have no message category and no representation in the coordination protocol. There is no compliant configuration in which they transit the mesh.

2.3 Local Sovereignty

Local input/output belongs to the operator. Nothing in this specification prevents an operator from connecting cameras, motion detectors, microphones, or any other equipment to spare controller inputs to trigger local behavior — starting a local recording, sounding a local alert, opening a local valve. Such integrations are compliant. They terminate at the router. The router’s publication policy engine has no category in which to place them, and the coordination protocol has no message type to carry them.

The rationale is stated plainly so future revisions preserve it. First, bandwidth: the residential radio class cannot carry rich media, and the specification treats that constraint as a design asset rather than a limitation to be engineered away — the same message discipline that enforces the mission boundary is what allows an entire coordination domain to share a single thin backhaul. Second, attack surface: data that is never transported cannot be intercepted, and a sensor that is not networked cannot be reached; the residential installation’s security posture rests on carrying almost nothing. Third, mission: this is a fire defense specification. Capacity for general-purpose home monitoring is architecturally adjacent and deliberately unbuilt. Operators with sophisticated custom systems should find in this specification a model to learn from, not a platform attempting to absorb them.

2.4 Command Broadcast

Incident command may issue domain-wide directives as connectionless broadcasts — fire-and-forget datagrams to the coordination domain, with no acknowledgment round and no per-node addressing. The motivating case: an engine company drafting from a municipal main needs the main clear immediately; the automatic pressure-sag response yields the same result but across sensing and convergence lag. The broadcast is the human jumping ahead of the control loop.

Broadcast semantics:

- **Delivery is best-effort by design.** Nodes that receive the directive comply; nodes that miss it continue local operation and yield through the normal pressure-sag response — the physics is the fallback delivery mechanism. Graceful degradation requires no additional machinery.
- **Scope filters ride in the message:** by supply class (the primary case — municipal-supplied nodes cease while cistern, pond, and reservoir nodes continue protecting), by zone, or by geography. Nodes evaluate the filter locally against their own configuration; no central registry exists. Nodes outside the filter, or without the resources the directive concerns, ignore it.
- **A cease-operations directive to a node with stored water is a source switch, not a shutdown:** the node drops external draw and continues coverage on its own reserves. The directive constrains draw, not defense.
- **Directives carry expiry, not release.** Every command broadcast has a time-to-live and must be refreshed by the issuing authority; on expiry, nodes resume automatically. A stale silencing order cannot outlive the attention of the commander who issued it.
- **Directives are signed and audited:** issued under the incident command interface's existing override authority, verified at the trust layer, and logged to the ledger by every complying node. The after-event record shows what was ordered, by whom, when, and what each node did.
- Operator hardware controls are unaffected, per the main specification's override provisions and Charter Clause 7.

2.5 Single Command Surface

All enclosure classes present through a single command surface. A residential node on a hose bib and a tactical node at a pond appear in the same operational picture, distinguished by capability metadata carried in their status messages, never by separate applications. No enclosure class shall require class-specific command software. The incident command interface, including its presentation into existing agency systems (TAK-class and ICS software per the main specification), sees a coordination domain — one protocol, one message enumeration, one pane of glass. Class differences end at the enclosure wall.

2.6 Common Exchange Languages (Adopted at the Boundary)

The tactical, incident-command, and public-alerting worlds already maintain common exchange languages, and this specification adopts them at its publication boundary rather than inventing competitors:

- **Cursor-on-Target (CoT) / TAK protocol** — the common operating picture grammar of the TAK ecosystem and the tactical MANET world. Node presence, status, position, detections, and events publish outbound as CoT events with appropriate type codes; this is the formal identity of the "TAK-class publication" named elsewhere in this specification.
- **Common Alerting Protocol (CAP)** — the OASIS/IPAWS standard for public alerting. Where a coordination domain's detection layer contributes to community warning, the domain publishes outbound as CAP.
- **SIA Contact ID / DC-09** — the fire-alarm industry's device-status and trouble grammar, for consumers that are alarm-monitoring systems rather than tactical or public-alert pictures.
- **TIA-102 (P25) data services** — the exchange grammar of public safety trunked data, noted per the P25 section; integration, where an agency elects it, presents through the same translation boundary.

Translation is outbound-only and occurs at the router. The governing principle is adoption by reference: wherever a destination system already has an established standard for a concept the domain must express, OSF publishes in that standard rather than inventing a synonym — CoT toward a tactical picture, CAP toward public alerting, Contact ID toward alarm monitoring, and additional destination standards as they arise. The publication policy layer renders the closed OSF vocabulary into whichever exchange language the consumer speaks, at the gateway boundary. Inbound command remains exclusively the closed, signed, enumerated OSF command set — external systems read the domain in their native grammar but command it only through ours. The security

architecture is preserved by construction: the mesh never carries a rich external format, and no external language can express a command the vocabulary does not contain. The mapping tables (OSF↔CoT, OSF↔CAP, OSF↔Contact ID, and further destinations as adopted) are an specification appendix deliverable, maintained under charter governance alongside the vocabulary itself.

Incident Command Interface

The specification requires incident command interface capability sufficient for professional response integration during multi-jurisdiction wildland-urban interface events. The interface presents unified operational picture across all participating systems regardless of underlying radio infrastructure. Incident commanders arriving on scene operate one interface to see and coordinate with the entire distributed defense operation.

Interface Presentation

The interface presents information consistent with Incident Command System conventions. Standard incident command information appears in standard positions. Water system status integrates into overall tactical picture. Override authorities are available with appropriate confirmation for consequential actions. Radio-class differences abstract away so the commander sees participating nodes as a unified list with capability metadata rather than as separate networks requiring separate management.

Override Authorities

The interface supports commander override of automated coordination. Priority activation of specific structures. Exclusion of zones where fire department personnel are operating. Modification of aggregate draw when municipal supply is stressed. Broadcast alerts to all participating systems. Suspension of automated operation during commanded operations. Command authority mapping respects actual authority structures at events. Homeowners retain override of their own systems even during commanded operations, but coordination happens through the incident command structure when it is operating.

Existing System Integration

The interface supports integration with existing incident command systems where they are deployed. Federal wildland incidents use specific ICS software and coordination systems. Municipal departments use various commercial and open systems. The interface can present its coordination picture into these existing systems so the water defense operation appears as one more resource type in the commander’s overall tactical picture rather than as a separate system requiring separate attention.

Optional Deployment

Neighbourhoods may deploy the specification without incident command integration. Fire departments may choose not to integrate with participating neighbourhoods. The system operates at its designed capability with or without integration; integration is an enhancement rather than a requirement. Where deployed, the integration operates through the router layer without requiring modification of existing agency infrastructure.

Cryptographic Event Log

The specification requires cryptographically verifiable operational logging sufficient for post-event trust establishment. The log architecture uses change log format with periodic snapshots, balancing bandwidth constraints during active events against public-facing verifiability after events.

Local Operational Log

Each participating system maintains its own operational log at whatever resolution its local storage supports. The log captures everything the system knows about its own operation: trigger events, sensor readings, PLC state changes, coordination messages sent and received, override commands accepted, water flow measurements, pump runtime, PLC errors, and maintenance events. High-resolution local record lives on the local system.

Snapshot Publication

Periodic snapshots roll up the local log into committed states published to the neighbourhood ledger. Snapshot frequency depends on operational context. During normal operation, daily or weekly snapshots capture routine

status. During active events, snapshot frequency increases to whatever the mesh bandwidth supports, capturing operational state changes as they occur. Snapshots are compact by design: minimal traffic when nothing is changing, summary state when changes are occurring.

Log Data Categories

The specification defines log data categories sufficient for unambiguous filtering by publication policy. Each installation configures which categories publish to the neighbourhood ledger and which stay local.

- Fire defense operational data: system activation, coordination participation, water use, coverage status. Minimum publication requirement for meaningful participation.
- Environmental sensor data: temperature, wind, humidity, ember detection when equipped.
- Coordination messages: mesh network exchanges, override commands, scheduling decisions.
- Physical security data: cameras, motion sensors, access logs. Typically not published.
- Personnel data: any information about individuals present at the installation. Typically not published.
- Installation-specific operational data: proprietary process data, agency internal operations. Typically not published.

The message category enumeration in the Event Interface section governs what the coordination mesh can transport; the categories above govern what the ledger records and what publication policy filters. Categories marked expressly non-transportable in the enumeration have no representation on the mesh regardless of ledger configuration.

Publication Policy

Each installation’s router enforces publication policy. Federal facility routers filter camera feeds, personnel data, and installation-specific operational data. They publish the fire defense subset. Commercial facility routers filter based on what they need to protect. Residential routers publish according to homeowner preferences. Publication policy is configurable at installation and modifiable by the installation operator without requiring changes to the coordination protocol.

The specification defines minimum publication requirements for meaningful participation: activation events, water use aggregate, coordination participation acknowledgment, and system status heartbeat. Everything else is at the participant’s discretion. This ensures the ledger has enough data to be useful for post-event reconstruction while respecting the flexibility that different participants need.

Cryptographic Verification

Snapshots are cryptographically signed at creation and appended to a chain that makes tampering detectable. Each entry references the previous entry’s hash. Signing keys are held by participating systems and by trusted witnesses appropriate to the neighbourhood’s trust structure. Fire departments, insurance carriers, and community coordinators may participate as witnesses. The ledger is distributed across nodes so no single point of control can rewrite history.

The specification identifies capability requirements for the ledger technology rather than prescribing specific implementation. Cryptographic signature verification, append-only chain structure, distributed storage, and threshold witnessing are the required capabilities. Implementations may use established distributed-ledger technologies or purpose-built ledger implementations meeting the capability requirements.

Public Access

The neighbourhood ledger provides queryable access to published snapshots. Time range selection, node selection, event category filtering, and appropriate detail level are supported through the query interface. Consumers of the interface build their own report views appropriate to their needs. Insurance systems, incident command systems, community coordination interfaces, and specification refinement research all access the same underlying ledger data through views appropriate to their purposes.

After-event reports generate from ledger queries rather than from operator narratives. The reports show what happened when, in what sequence, with what system states, and what outcomes resulted. Because the ledger is verifiable, the after-event reports carry the trust weight of the cryptographic verification rather than depending on the operators’ credibility.

4.1 Deployment Class

The expeditionary configuration serves forestry, National Guard, and emergency management deployments to areas without installed coverage. Current structure-protection practice deploys pump-and-hose kits — a portable pump drafting from a river or pond, hose lays, tripod sprinklers, a fuel supply — running until dry or until a crew returns to check. The expeditionary node ties into this practice without modifying it: dropped alongside the kit, it provides real-time pump-site intelligence, deliberate water and fuel scheduling, and coordination-domain participation, with zero contact with agency pump equipment required.

4.2 Capabilities

- **Pump-site telemetry:** discharge pressure and flow via an inline sensor spool clamped into the hose lay; engine state via non-contact acoustic sensing (derived events only, per Section 2.2); fuel forecasting from burn-rate inference. Hose bursts, pump stoppages, and fuel exhaustion become immediate events instead of discoveries on the next crew visit.
- **Duty-cycled protection:** zone valves in the distribution lays allow the sequencer to time-share pump runtime across coverage points, extending a fixed fuel load across the ember-cast window. Continuous run-to-dry operation is replaced by deliberate consumption.
- **Pump control interface:** published in advance of industry support — 12V nominal control convention compatible with existing portable-pump control panel practice, start/stop/state message categories, relay-drive electrical interface. Current-generation portable pumps are sense-only; the interface definition provides procurement language through which agencies can specify coordination-interface pump control in future tenders. Battery-electric portable pumps, when available, integrate under the same interface; duty-cycled sprinkler protection is the operating mode that makes battery-electric portable pumping viable at current energy density.

4.3 Automated Fire Lookout Configuration

The composite kit — expeditionary node, pump and hose kit, weather station, and (at government tier) thermal detection payload — constitutes an automated fire lookout that defends its own site. The historical staffed lookout could see fire but not survive it; this configuration wets its own perimeter. Local weather measurement feeds wet-cycle scheduling directly and provides incident command with spot weather at the point of greatest forecast uncertainty. Detection events ride the coordination protocol per Section 2.2; imagery, where an agency requires it, rides agency radio infrastructure as agency traffic outside this specification’s enumeration. Agency security or surveillance assets deployed adjacent to a node coexist on the same network without being absorbed by this specification.

4.4 The Coordinated Defense Module

The expeditionary configuration composes into a deployable resource type assembled entirely from existing fire service elements: one Type 6 engine, one trailer of cased units, a crew of three to four, and a UAS operator. Nothing in the module is new — the crews, engines, typing conventions, and dispatch mechanisms already exist; the module is a new line item in the resource catalog, orderable and demobilized like any typed asset.

Employment concept — seed and move. The module arrives ahead of the fire at a threatened structure cluster and, through the open-lid commissioning ritual, stands up persistent nodes where structure protection would otherwise stand people. Because the nodes hold ground autonomously, the module does not stay: it leapfrogs to the next cluster and seeds it, laying persistent protection at multiple communities per operational period. The staffing arithmetic inverts from crews-per-site to sites-per-crew. At demobilization, cases are inventoried, signed, and recovered per the joint-inventory doctrine — the resource that lays resources, and takes them home.

Network self-assembly. Every unit carries the common mesh radio regardless of radio tier. Where participating residential systems exist, module nodes join the standing coordination domain on power-up — no integration step, by construction. The exchange runs both directions: the community’s mesh delivers established topology, supply intelligence, and a coverage map to the arriving module; the module delivers command presence, detection, backhaul, and tactical nodes placed at exactly the gaps the domain census shows. **Prepared communities are cheaper to defend, and the network is how the module knows it on arrival** — residential adoption converts measurably into tactical capacity.

Range convergence. Coordination density follows interface density by design: the system’s hydraulic reach (hose-

lay scale), realistic terrain mesh range, and the structure-spacing envelope at which coordinated defense matters all converge at neighborhood scale. Field heuristic: *if a hose could reach it, the mesh reaches it.* Sparse and remote deployments operate as autonomous clusters with reporting reach, per the graceful-degradation architecture; the sweet spot is the interface, which is where the problem lives.

4.5 Prepositioning

Expeditionary units are designed for palletized warehouse storage and drop deployment. **Prepositioned units shall require no external activation, registration, provisioning service, or connectivity to enter service.** A unit sealed in its case enters service years later by being opened, connected, and powered. Warehouse maintenance is limited to battery inspection and commodity-part replacement on standard storage intervals. A gateway kit — satellite terminal, power supply, solar panel — is a single additional pallet line item per coordination domain.

4.6 Terrain-Anchored Deployment and Non-Destructive Suppression

The preceding sections defend structures. This section defines the system's employment against fire at terrain scale: holding ground rather than protecting buildings. It is the highest tier of expeditionary employment and carries the strictest command and bounding conditions.

Terrain-anchored placement. A node's target may be the land, not a structure. Fuel corridors, drainages, chokepoints, saddles, fuel breaks, and field edges — the ground a fire must cross — are legitimate node placements, wetted as coordinated defensive elements ahead of a front. A structure at the mouth of a drainage that cannot be defended at its own walls (for example, one taking direct radiant load from a crowning slope at close range) may nonetheless be defended by denying the fire the corridor upstream.

Pre-surveyed siting. Deployment locations are sited in advance, not improvised under fire. A century of fire history, fuel and terrain data, prevailing wind, and satellite imagery identify where a given landscape burns before any specific fire exists. Chokepoints, wet-line corridors, and green-island/landing-zone sites are surveyed, walked, and mapped in the off-season, embedded in the Community Wildfire Protection Plan, and cached against the named corridors they will hold. Model-and-imagery-based siting is decision support; a qualified person commands placement. The fire executes a plan drawn when the ground was quiet.

Green islands. A pre-wetted patch of ground ahead of a front serves two functions at once: a break point where the fire arrives with reduced energy and can be stopped or steered, and — once committed and safe — a helicopter landing zone. The system generates tactical terrain and aviation infrastructure from the same water it uses to defend.

Wet line as reversible fireline. A continuous wetted strip is a constructed fireline that builds itself faster than hand or dozer line and is held by the nodes rather than by crews. Its tactical function is identical to fuel-removal line — deny the fire the next foot of receiving fuel — but achieved by temporarily raising fuel moisture rather than permanently removing fuel. The line exists when needed and evaporates when the threat passes; the vegetation lives, the ground is unscarred, and the community is not destroyed to be saved. This reversibility is the doctrine's central advantage: the political and ecological objections that limit destructive line construction (bulldozed property, felled trees, speculative damage) do not apply. Wetting a fine (1-hour) fuel below ignition threshold is forestry-manual-level calculation; the supply required to hold a narrow line against a surface fire is small, and modest static sources (a drum uphill, a small cache, a tender-fed porta-tank layer) suffice.

Passive until triggered. Unlike an active hose lay, which commits a crew for its entire duration, a terrain-anchored network is emplaced and dormant — present at the dry corridor at negligible standing cost, doing nothing on the great majority of days, and deploying automatically alongside the static residential network only when conditions and a front arrive. This separation of *presence* from *staffing* is what makes landscape-scale coverage economically possible where continuous hose-kit staffing is not.

Bounding conditions (mandatory). - This is a surface- and moderate-intensity-fire tactic, and a flanking/backing-phase tactic. A wet line denies fire spreading by contact with surface fuel. It does **not** hold a wind-driven crown fire spreading by radiant heat and long-range spotting. Doctrine shall not claim otherwise, and no crew's egress shall be staked on a wet line as though it were a guaranteed barrier. - The wet line is best-effort infrastructure and may be lost like any line; it buys time and probability, not certainty. - Employment is crew-directed and human-commanded. The system is a tool a qualified incident commander or division supervisor wields; it does not conduct autonomous suppression. Where automated action and command judgment meet, command prevails (Charter Clause 7). Placing "containment" without a qualified human owning the tactic is expressly non-compliant.

The force it serves. This doctrine brings a technical advantage to the existing wildland firefighting community — it does not replace it. The line is held by nodes and patrolled, tended, and commanded by people: seasonal hand-crew labor matched to light distributed work (patrol, valve-tending, spot-chasing, node-minding), with qualified leaders on

Type 6 engines running the tactical picture. It is easier and lower-risk than cutting hand line and requires fewer specialized bodies, because the boxes hold the line the crews would otherwise dig and man. When the fire is gone, the crew leaves and the ground is intact. This is a force multiplier for the Pulaski community, built by one of its own — not autonomy displacing the firefighter, but reach and leverage handed to the firefighter who already does the work.

Installation and Maintenance

Installer Qualifications

Installation is within the capability of any competent industrial electrician, licensed plumber, or capable homeowner with appropriate mechanical and electrical skills. The specification does not require licensed installers beyond what local building codes require for the plumbing and electrical work involved. Retrofit of existing residential irrigation systems is within the capability of any licensed irrigation contractor. Federal facility installations follow the agency’s normal contracting and installation practices with the specification providing the technical requirements.

Building Code Compliance

Installations comply with local building codes for plumbing, electrical, and structural work. Backflow prevention meets local water district requirements. Electrical installation meets the applicable National Electrical Code requirements or local equivalent. Structural mounting of roof-line coverage meets local roofing and structural requirements. Standard permitting processes apply.

Operator Responsibility

The specification treats the installation as operator-owned equipment operating under operator responsibility. Maintenance, testing, component replacement, and continued operational readiness are operator obligations. This is the smoke alarm model of property protection: the equipment is installed, the operator maintains it, the system operates when needed if the maintenance has been kept up. No subscription service, no monitoring contract, and no ongoing external obligation is required for the equipment to function.

Cost Estimation

Total installed cost varies substantially with local labor rates, existing infrastructure, specific component selections, and radio class selected. Representative cost ranges for new construction from bare property, residential installation with LoRa mesh:

- Reservoir and plumbing: $1,500 to $3,500 depending on capacity and configuration
- Pump: $500 to $1,500 depending on drive type and flow capacity
- Controller, enclosure, router, and LoRa mesh node: $700 to $1,500
- Coverage distribution: $800 to $2,000 depending on structure size and complexity
- Installation labor: $1,000 to $3,000 depending on local rates
- Total residential range: $4,500 to $11,500 for complete new installation

Federal facility and remote installation costs vary substantially based on tactical radio infrastructure requirements, existing radio investment leverage, and installation environment. Facilities with existing tactical MANET infrastructure add the coordination capability at substantially reduced radio cost. Facilities without existing infrastructure incur higher radio costs offset by capability appropriate to the deployment scale.

Retrofit of existing residential irrigation systems reduces cost substantially. Existing controllers may accept mesh coordination modules without replacement. Existing water supply and distribution infrastructure may require only the addition of coverage-appropriate sprinkler heads and zone reconfiguration. Retrofit installations may be completed for $1,500 to $4,000 depending on existing infrastructure.

Licensing

This specification is released under permissive open licence. Any operator, contractor, agency, or manufacturer may implement compliant installations without licence fees or ongoing royalty obligations. Improvements to the specification through community development are welcomed and integrated into subsequent versions.

Governance Charter

5.0 Adoption by Reference (standing rule)

External standards, codes, and regulatory regimes named anywhere in this family — NWCG publications, NFPA codes, UL standards, FAA regulations, NIICD listings, NFDRS, Section 508, AWWA practice, and all others — are **adopted by reference to their current edition as amended by their issuing authority**, never copied out or recreated. Where any document in this family paraphrases or names a value from an external regime, the external regime's current text governs and this family's paraphrase is descriptive only. This is the same mechanism law uses to remain dynamic: if the issuing authority changes a threshold, an altitude, a listing, or a table, this specification follows automatically and requires no amendment. This family invents only in the space no authority has claimed; everywhere an authority exists, we defer to it, by reference, as it evolves.

5.1 Stewardship Model

The specification is maintained in the manner of open network protocol standards. Anyone may propose a revision. Proposals are evaluated in public against this charter. Adoption requires demonstrated need within the specification's mission; novelty alone is not need. The versioned, cryptographically chained change log defined in the main specification is the governance record: proposals, dispositions, and the reasoning for each are committed to it, so that the standard's history is verifiable by the same mechanism as its installations' operations.

Proposals are evaluated first by locating them in the layered architecture. A proposal located at the physical, link, or network layer is typically already accommodated by the existing interfaces or already excluded by layer discipline, and requires no revision to adopt or reject. Only proposals touching the vocabulary and trust layer, the coordination behaviour, or the message enumeration modify the invention, and these receive the fullest scrutiny.

5.2 Charter Clauses

All revisions are evaluated against the following clauses. A proposal in conflict with any clause is not adopted, however capable the technology it introduces.

Clause 1 — Buildability. The core system shall remain buildable from commodity components available through standard industrial supply channels by a competent individual, electrician, or controls technician. No revision may make the core system dependent on components available only from a single manufacturer or only through restricted channels.

Clause 2 — No mandatory dependency. No revision may introduce a mandatory dependency on external network connectivity, cloud services, subscription services, license servers, or any ongoing external obligation for the core system to perform its fire defense function. The system operates when isolated. This clause is absolute.

Clause 3 — Extension by module. New capability enters the specification as an optional module with its own conformance section. It does not enter as a modification of core requirements. The core remains small; the periphery may grow.

Clause 4 — Non-adoption is not prohibition. Rejection of a proposal bars no one from building it. Extensions not adopted into the standard are simply not part of the standard: they receive no message category, no conformance language, and no compatibility guarantee. Their builders remain free to build.

Clause 5 — Mission boundary. The specification defines a coordinated fire defense system. Revisions extending it toward general-purpose home automation, security monitoring, or surveillance are outside its mission and are not adopted, without regard to technical merit. The message category enumeration of Section 2 may be extended only for fire defense purposes.

Clause 6 — Layer discipline. Revisions respect the layered architecture. New physical, link, and network technologies are welcomed at their layer and shall not impose requirements on the layers above them. A new sensor generation that assumes internet connectivity does not thereby make the specification assume it; the specification's answer to new devices is its existing interfaces.

Clause 7 — Operators prevail. Automation in this specification manages resources; it does not overrule operators. Where automated policy and operator action conflict at the point of use, the operator prevails, and the system's role is to make consequences visible — through status indication, published energy and state events, and the operational log — not to refuse. This extends the main specification's existing principle that homeowners retain override of their own systems even during commanded operations. No revision may introduce an automated lockout of an operator's physically present resources or controls.

5.3 Worked Example

A manufacturer introduces a DIN rail device with integrated Bluetooth provisioning. Under this charter: the device is not barred from use — an installer may deploy it where it meets the capability requirements of its component class (Clause 4). Bluetooth does not become part of the specification, because no demonstrated need within the mission requires it (Clause 5) and the core gains no new dependency (Clause 2). If a future revision finds genuine need — say, commissioning tooling — it enters as an optional module with its own conformance section (Clause 3), and the core system remains fully buildable and operable without it (Clauses 1, 2).

5.4 Amendment of this Charter

The charter itself may be amended only by the same public proposal process, recorded in the change log, and no amendment may weaken Clause 2 or Clause 7.

Acknowledgment

The specification represents synthesis of existing engineering practice across residential sprinkler installation, industrial control systems, mesh network protocols, tactical mobile ad-hoc networking, cryptographic ledger technology, and volunteer and professional fire service operations. It does not depend on proprietary technology and does not require novel engineering development. It is arithmetic and integration applied to a defensive coordination problem that has not previously been addressed at the scale and across the participant range that actual wildland-urban interface fire events involve.

Appendix C — Wire Format and Vocabulary

Consolidated into this specification; previously issued separately as Annex C. This appendix is the authority for the closed message vocabulary, envelope layout, and the outbound cross-reference to external exchange standards.

Envelope

The wire format is a compact binary envelope sized to the most constrained transport in the family. LoRa mesh payloads live in the ~200-byte range; the envelope must fit it with room for a body.

Fixed header, every message: **version · message class · node ID · sequence number · timestamp · signature.**

The signature is the large field (Ed25519 class, 64 bytes) and is non-negotiable — the trust layer rides in the envelope. Realistic body budget after header and signature: **60–100 bytes.** That number is the discipline; every verb must fit it.

Shape Decisions (Fixed Now)

1. **Enums everywhere, strings nowhere.** Human-readable names live in the specification document, never on the wire. This serves bandwidth and, more importantly, the mission boundary: a protocol with no string type cannot smuggle rich payloads. The absence of a general-purpose payload field is a structural property, not a policy.
2. **Quantized classes over precise values.** Coordination needs “pressure sagging,” not 47.3 psi. Readings on the wire are single-byte quantized classes; full-resolution values stay in the local log per the main specification.
3. **Capability bits in the heartbeat.** Each node’s periodic heartbeat carries capability flags (supply class, stored-water presence, pump control, detection payload, gateway role). This is how the single command surface knows what each node is without asking, and how command-broadcast scope filters resolve locally without a registry.
4. **The vocabulary is closed and short.** Additions go through charter governance. Smallness is a protected property of the protocol, not a v1 limitation.

V1 Verb List

Grouped per the message category enumeration (Section 2.2):

Group	Verbs	Notes
Presence	HEARTBEAT	status flags, battery class, supply class, capability bits, coarse position
State	ARM, DISARM, ACTIVATE, DEACTIVATE, ZONE_OPEN, ZONE_CLOSE, FAULT, MAINT_REQ	FAULT carries a fault-code enum

Supply	PRESSURE, FLOW, LEVEL	quantized single-byte classes
Coordination	SCHED_STATE, SCHED_PROPOSE, YIELD, RESUME	schedule negotiation per main specification
Command	CEASE_OPS, RESUME_OPS, PRIORITY_PROTECT, BROADCAST_ALERT	scope filter, authority, expiry per Section 2.4; short, audited vocabulary
Safety	DISTRESS	node, position, timestamp — deliberately minimal
Events	DETECT, ENGINE_STATE, ENERGY	detection events (source/bearing/confidence class); acoustic-derived machinery state; power transitions
Ledger	SNAPSHOT_COMMIT	hash chain link; the payload is the commitment

Roughly two dozen verbs. The full annex will define field layouts per verb and four worked byte-level examples: HEARTBEAT, ZONE_OPEN, CEASE_OPS, DISTRESS.

Why This Is Most of the Software Design

The vocabulary list is the data dictionary. An implementer inherits a closed verb set, field definitions with quantization decided, an envelope with the byte budget pre-fought, and a charter that bounds what may be added. The enum table compiles nearly directly to a header file; the verb list is the dispatch switch; the state transitions described in the main specification are the state machine. Two independent implementers working from this table produce interoperable nodes without coordination — convergence by construction, which is the property that makes distributed community implementation work.

Queued for full drafting during the community development window, alongside reference implementations. Constraints above are fixed; encoding details are not.

Vocabulary Cross-Reference (Adopted Terms)

The OSF vocabulary is small and closed. Where a concept it must express already has an established term in the two domains this system publishes toward — the tactical world (Cursor-on-Target / TAK), the fire-alarm world (SIA Contact ID / DC-05, transported by DC-09), and the public-alerting world (Common Alerting Protocol / CAP, for community warning) — OSF adopts the established term rather than inventing a synonym. The principle is general: where a destination system has an established standard for a concept, OSF publishes in it; CoT, Contact ID, and CAP are the destinations developed first, and others are added the same way as they arise. **Rule of precedence: where the same concept exists in all three (OSF, CoT, Contact ID), the tactical (CoT) form is the OSF standard**, because CoT is the grammar OSF publishes into at the incident-command boundary and the one a coordinating system will most often translate against. Contact ID governs where the concept is native to alarm practice and absent or weaker in CoT (device trouble states). **No OSF output is ever rendered as an alarm-class Contact ID event.** Contact ID rendering is limited to supervisory, trouble, and status codes — the device reporting its own condition to an alarm-monitoring consumer. The system does not signal a fire condition to a monitoring station, and does not initiate a response; that function belongs to a listed fire alarm system and is deliberately outside this specification. This adoption applies to **outbound status and telemetry only**; the inbound command enumeration remains closed, signed, and OSF-native — no external vocabulary is adopted on the command path, by design (§ security model).

A structural finding falls out of the mapping: **almost every cross-domain match lands on the outbound/telemetry vocabulary, and almost none on the command verbs.** Status concepts — heartbeat, position, fault, low battery, tamper, detection, time/expiry — are universal and already standardized elsewhere; the command verbs (WET, terrain-line actions) are domain-unique and have no external equivalent to adopt. This is the security model expressing itself in the vocabulary: the words the system *speaks outward* are shared; the words it *accepts inward* are its own.

Outbound status / telemetry — adopted forms:

OSF concept	CoT (tactical)	Contact ID (alarm)	OSF standard adopts

Node presence / heartbeat	`<event>` with `stale` soft-state; periodic self-report	Periodic test (E602)	**CoT** — stale/expiry soft-state model (triplicate → tactical)
Node identity	`uid` (persistent)	Account + zone number	**CoT** — persistent `uid` (triplicate → tactical)
Position / location	`<point lat lon hae ce le>`	(none; panel is fixed)	**CoT** — native geospatial
Time / event validity	`time` / `start` / `stale`	Message timestamp	**CoT** — start/stale window (triplicate → tactical)
Source quality	`how` (e.g. `m-g`)	(none)	**CoT**
Battery / power status	`<status battery=…>`; AC state	Low battery (302), AC loss (301)	**CoT** `<status>` for telemetry value; Contact ID codes carried in detail for alarm-panel consumers (see note)
Device fault / trouble	(weak)	300-series (301 AC, 302 batt, 373 fire trouble, 380 sensor trouble)	**Contact ID** — alarm-native, richer (duplicate → alarm)
Tamper	`<status>` / detection	137 / 383 (tamper)	**Contact ID** term, CoT transport (alarm-native concept)
Detection event (heat/flame/acoustic)	`d-` sensor detection branch (`d-a` acoustic, `d-m` motion)	*(not rendered — alarm-class)*	**CoT** — `d-` sensor detection taxonomy (triplicate → tactical)
Suppression zone active	`t-`/status event; area condition	*(not rendered — alarm-class)*	**CoT** event + OSF detail sub-schema
Restoral / return to normal	new `<event>` supersedes stale	Qualifier (restoral vs. new)	**CoT** — supersede-by-freshness (triplicate → tactical)
Area condition / reservation	`r-` reservation/restriction branch	(none)	**CoT** — `r-` for wetted-zone / green-island area notices

Inbound command verbs — no external adoption (closed enum):

OSF command	Nearest external analog	Disposition
ARM / DISARM	CoT tasking `t-` (loose)	OSF-native; not adopted
WET / STOP (zone)	CoT `t-` tasking (loose)	OSF-native; domain-unique
SET CLASS (wetting class)	(none)	OSF-native
HOLD / YIELD	(none)	OSF-native
TEST / SILENCE	Contact ID test (602)	OSF-native command; publishes result as adopted status

Detail sub-schema: OSF status publishes as a CoT `<event>` whose `<detail>` carries an OSF sub-schema (the pattern CoT is designed for — an application-specific `<detail>` child under a well-known base). Where a downstream consumer is an alarm-monitoring system rather than a TAK picture, the same status renders to a Contact ID / DC-09 message using the alarm-native codes above. Both are outbound translations at the router; neither is ever accepted inbound as a command. The full CoT type-code assignments (the exact `d-`, `t-`, and `r-` leaf strings OSF emits) and the OSF↔CoT and OSF↔Contact ID mapping tables are drafted with the reference implementation and maintained under charter governance, versioned with the vocabulary itself.

The Open Source Firefighter

Annex A — Reference Hardware Specification

v1.0 · August 2026 · Reference hardware · Companion to the Main Specification *Document ID:* ***OSF-1.2*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

All external standards cited in this annex are adopted by reference to their current edition as amended — the issuing authority's text governs; see the specification's Adoption by Reference standing rule.

- ◦

Scope

This annex defines reference hardware for compliant installations. It specifies capability requirements first and names example products second. Substitution of any component meeting the capability requirements is expressly permitted and expected. Every component class is selected for availability through standard industrial supply channels. No component in a compliant installation may depend on a single manufacturer for replacement.

Three enclosure classes, one system. The hardware family consists of three enclosure classes — **Residential**, **Government/Commercial**, and **Tactical** — running identical firmware, identical coordination protocol, and identical message categories. The classes differ in packaging, connectors, power sourcing, and environmental rating only. Class boundaries are environmental exposure and procurement requirements, never capability. A government installation is not a smarter unit; it is the same unit in harder packaging.

Expansions are orthogonal to class. Pump and reservoir expansion, external zone modules, the gateway kit, and detection payloads attach to any class where noted. Capability grows by module; the core unit does not change.

Equipment category and certification pathway. This equipment is a supplemental exterior wetting system for ember defense: an industrial control panel operating irrigation-class hydraulics. It is not a fire alarm system (NFPA 72 / UL 864 class), not a fire suppression system as defined by the sprinkler codes (NFPA 13, 13D, and 13R), and not a life-safety notification device, and shall not be represented, marketed, or certified as any of these. Where a jurisdiction regulates construction and mitigation at the wildland-urban interface — commonly through the International Wildland-Urban Interface Code, NFPA 1140, or NFPA 1144, and in some states through a dedicated chapter such as California Building Code Chapter 7A — this equipment is supplemental to those requirements and replaces none of them. Compliant units are certifiable as industrial control panels under UL 508A-class standards through the existing multi-vendor panel-shop ecosystem; component-level recognition is carried by the commodity bill of materials. This specification's equipment class does not invoke fire-alarm control unit listing regimes (UL 864 class), and no revision shall move it into one.

Design Principles

Sequencing over splitting. A single supply point — a hose bib, a pump, a fuel can — is time-shared across coverage points rather than divided among them. Default operation opens one zone at a time on a rotating schedule. This is the correct hydraulic design for the supply that exists, and the correct fire physics: a wetted surface remains protective for minutes between passes, and duty-cycled coverage defeats ember ignition at a fraction of the water and fuel. Sequencing at the unit level is the same algorithm the mesh runs at neighbourhood level and the domain runs at valley level.

Commodity wear items. Every component with a service life — batteries, valves, fuses, sensors — shall be a commodity item replaceable by any competent person with a supply house account and no documentation beyond the component label. Battery charging and DC bus management shall be provided by standard DIN rail power modules. Integration of charging circuitry onto proprietary control hardware is expressly non-compliant.

One control object at every size. Zone valves are quarter-turn valve bodies with ISO 5211 actuator mounting pads

driven by 24VDC electric actuators. A 3/4” brass ball valve and an 8” lug butterfly valve present identical control interfaces: open, close, limit-switch position feedback. Scaling changes valve body and actuator torque class only.

Minimum viable installation. One garden hose, one sprinkler, one zone. The Residential unit shall provide scheduled surface wetting and full mesh participation with a single consumer-grade sprinkler on garden hose at 40 psi municipal supply. Two to three sprinklers across the three provided zones shall achieve meaningful roof and perimeter coverage for a single-wide manufactured home, tiny home, cabin, or small residence.

No novel protocols where standard ones exist. Routing, gateway selection, addressing, and power delivery negotiation use standard mechanisms with production histories. This specification invents only at the coordination layer.

Operators prevail. Automation in this system manages resources; it does not overrule operators. Where automated policy and operator action conflict at the point of use, the operator prevails and the system’s role is to make consequences visible. (Main specification.)

Designed for the public. The system is designed to be operated, moved, and understood by untrained members of the public; familiarity, not certification, is the intended long-term relationship between communities and this equipment. Voice guidance, hardware controls, one-person weight limits, and the absence of any account, application, or literacy barrier all serve this principle. The long-term benchmark is the fire hydrant: universally recognized, universally trusted, understood without training.

The filter and the floor converge. The design filter is the hardest user — the expert who will overload, improvise with, stand on, and defeat any equipment given to them. The design floor is the newest user — the untrained volunteer encountering the equipment for the first time. These requirements converge: equipment that survives expert contempt and equipment that welcomes inexperience are the same equipment — simple, verbal, tactile, forgiving, heavy-handled, with nothing delicate, nothing clever, and nothing that requires reading. Where a design choice appears to serve only one of these users, it has probably failed both.

Novelty is spent only where the invention lives. The coordination layer is new; everything the eye and hands touch shall be the most familiar possible object. The unit deliberately resembles a fire alarm panel at the component level — NEMA enclosure, 24 VDC bus, DIN internals, shelf-mounted commodity batteries, status LEDs, audible annunciation, a maintenance chirp — because resemblance is inherited trust: inspectors approve what they recognize, the alarm and controls trades already know the work with their hands, and the maintenance habits society runs for panels and smoke detectors transfer whole. The enclosure earns trust by familiarity so the invention inside never has to spend any.

Common Architecture (All Classes)

The following internals are common to every enclosure class.

Controller

- ESP32-class industrial PLC, DIN rail, 24VDC (Industrial Shields ESP32 PLC 21/38 series, Norvi ENET series, or equivalent)
- Minimum I/O: 8 digital outputs (valve drive, pump-start relay, status), 8 digital inputs (valve limit switches, flow pulse, GNSS PPS), 2 analog 4–20mA inputs (pressure, expansion), 1 analog input reserved for acoustic sensing where equipped
- IEC 61131-3 environment or open firmware toolchain (ESP-IDF / Arduino class)
- Serial or Ethernet interface to radio and router function
- Router function (publication policy, message signing, protocol handling) runs as firmware on the controller or radio MCU at Residential class; dedicated router hardware at Government/Commercial and Tactical class where MANET-class integration or publication policy complexity requires it

Zone Valves and Manifold

- Three standalone motorized 3/4” full-port two-piece brass ball valves, 24VDC, ISO 5211 actuator pads, position feedback (CR-series class), fed by a basic 3/4” intake manifold header; inlet enters the wet section side wall, outlets exit the enclosure bottom, one per zone. Valves mount individually — replacement is one valve, not an assembly. (Solenoid valves are a permitted budget variant where position feedback is deliberately waived; the reference design requires feedback.)
- Common inlet gallery with spring check valve and manual isolation valve

Backflow — hazard class and what the check valve is not. The inlet gallery check valve is a system component;

it is **not the backflow prevention device** and shall not be represented as one. Backflow protection is a separate, listed assembly selected and installed to the local plumbing code by the authority having jurisdiction. Note the hazard class: an installation that can draft from a pond, cistern, or other non-potable source through the same manifold that also connects to a potable supply is a **high-hazard cross-connection**, which typically requires a reduced-pressure-zone assembly or a physical air gap rather than a simple check or dual-check device. Where stored and potable sources share plumbing, the air gap is the conservative answer. - Drain-down port at manifold low point for winterization (firmware-assisted routine: all valves open, supply isolated, drained, depressurized) - One 0–200 psi pressure transducer, 4–20mA, on the inlet gallery - All wetted connections NPT; GHT adapters supplied for inlet and each outlet - **Self-calibration:** the controller opens each zone individually at commissioning and periodically thereafter, observes inlet pressure sag, and sets cycle timing automatically. No hydraulic configuration by the user. - **Sequencing default:** one zone open at a time, rotating. Simultaneous operation is not a requirement and shall not be assumed. - Expansion beyond three zones: external zone modules, same valve family, small NEMA enclosure, daisy-chained 4-wire bus (24VDC pair, signal pair)

Radio and Network

- LoRa mesh node, Meshtastic protocol or compatible, operating in the jurisdiction’s unlicensed ISM allocation (902–928 MHz at 100 mW is the North American reference; use the regional allocation and power limit elsewhere), (RAK4631, T-Beam, Heltec class), external antenna via bulkhead connector — reference radio at Residential class
- **Unlicensed allocations only.** A compliant node operates under the unlicensed rules of its jurisdiction (Part 15 in the United States, or the local equivalent) and shall not be configured for amateur-band operation. Amateur allocations are outside this specification for several independent reasons: amateur operation requires a licensed control operator and station identification at intervals, which an unattended automatic node cannot satisfy; amateur rules restrict unattended automatic operation and bar communications that could reasonably be furnished through other radio services on a regular basis; and amateur allocations may not be used for communications in which the operator has a pecuniary interest. A signed, unattended, continuously operating node is not an amateur station. Firmware shall not offer an amateur-band mode, and a node with message signing disabled is non-compliant regardless of band.
- **Tactical MANET radio class** — Government/Commercial and Tactical option: agency-furnished mobile ad-hoc mesh data radios of the hardened interagency class (Silvus StreamCaster, Persistent Systems Wave Relay, TrellisWare TSM, and peers). The class is defined by capability — self-forming mesh, IP presentation, MIL-STD-810-class environmental qualification — never by vendor; any radio the furnishing agency fields and trusts is compliant at this layer.
- **Interagency governance, adopted by reference:** radio equipment operating on interagency wildland fire networks is governed by the National Interagency Incident Communications Division (NIICD) approved-equipment listings, and interoperability channel use by the National Interoperability Field Operations Guide (NIFOG). Nothing in this specification transmits on, contends for, or requires those networks; where agency-furnished radios are themselves NIICD-listed or incident-assigned, they operate under the furnishing agency’s authority and the incident communications plan, per the Spectrum Posture section. This specification’s coordination traffic rides its own unlicensed or agency-furnished data layers and never the fire voice nets.
- **Satellite backhaul (optional, all classes):** commercial LEO (Starlink class, including compact low-power terminals) or government satellite transport (Starshield class or successors), presenting standard IP to the network layer. Backhaul is an enhancement, never a dependency.
- **Gateway node role:** any node may bridge the coordination domain to a backhaul. Gateway is a role, not a model. Gateways advertise external reachability via standard routing protocol mechanisms; nodes select per normal route metrics; failed gateways age out. This specification defines no novel gateway election protocol. Zero gateways is fully operational.
- **Export-control boundary:** this specification defines open interfaces at the network layer; export-controlled radio and transport equipment (tactical MANET class, government satellite transport) integrates as operator-furnished equipment through those interfaces. No controlled technical data is incorporated into this specification or its reference implementations, and the open project ships no controlled articles. Government configurations are assembled by the operator holding the equipment and the authorities.

Spectrum posture. Primary coordination communications shall use unlicensed spectrum under FCC Part 15 (or the local equivalent) or wired/commercial IP transport (Ethernet, copper, fiber, commercial satellite service); no compliant installation shall require a spectrum license to perform its fire defense function, and no compliant installation shall be made to require frequency coordination. The coordination protocol is designed to function correctly on interference-tolerant unlicensed spectrum: degraded delivery is a handled operating condition, not a failure mode. Licensed radio classes integrate solely as operator- or agency-furnished equipment under the license holder’s authority; agency-furnished radio classes operate under the furnishing agency’s spectrum management

(NTIA/FCC coordination, incident communications plans), and compliant equipment shall accept the operating parameters that authority assigns.

Public safety trunked data (P25 class) — noted, not touched. Existing public safety trunked radio systems include narrow-band data services technically sufficient for this specification's coordination traffic, and this specification's message sizes are compatible with them. Integration is architecturally supported through the router layer as an agency-furnished radio class and is left entirely to the owning agency's initiative. This specification does not contend for public safety spectrum or dispatch capacity.

- **Range-convergence doctrine.** The system's three operative ranges converge by physics, not coincidence: realistic terrain LoRa reach, practical hose-lay length, and interface distance (the structure spacing at which coordinated defense matters) all occupy the same neighborhood-scale envelope. The system coordinates best at exactly the distance where coordination matters most, over exactly the reach the hydraulics serve. Employment gradient: **dense interface** — full coordination, mesh multipath, shared supply intelligence (the design center); **scattered rural** — per-site autonomous clusters (a homestead is its own cell) linking opportunistically or by gateway; **true remote** — single-cluster autonomous with reporting reach via higher radio classes or backhaul, per the graceful-degradation architecture. Field rule of thumb: *if a hose could reach it, the mesh reaches it.*

GNSS: Position and Time

Every node carries a GNSS receiver providing node position (system-status-class data; coarse-position-only publication is a supported policy option) and precision time (PPS-disciplined controller clock preferred) for the cryptographic log and coordination scheduling — the only precision time source consistent with Charter Clause 2. Degradation: no fix, fall back to RTC and last-known position; fire defense unaffected.

Capability: *GNSS receiver, serial or USB interface, NMEA output, PPS preferred*. Compliant antenna/receiver patterns:

1. **External active antenna, internal receiver (preferred):** commodity active puck through SMA bulkhead, top face; sky view required; co-location with LoRa antenna acceptable with a few inches of separation.
2. **External sealed puck on data cable:** weatherproof USB/serial puck (BU-353 class, marine NMEA) through a cord grip. Compliant; second choice.
3. **Integrated through-panel unit:** sealed GNSS whose mounting stub forms the penetration, per the assembled-penetration rule. Marine-grade hardware class; the standard configuration for wet, marine, coastal, mobile, and government-procured installations.

Power System (Common Requirements)

- 24VDC bus throughout
- DIN rail supply with DC-UPS/charger function (Mean Well DRC-40/60 class, DR-UPS40 + supply, PULS UB equivalent); low-voltage disconnect required
- Battery: commodity form factor, field-replaceable — class-specific sourcing below
- Standby budget: controller, radio, GNSS draw single-digit watts; valves draw only during travel; 72-hour autonomous requirement met with margin at every class

User Interface (Common Minimum)

- Panel face: status LEDs (power, radio, readiness, fault, active, distress)
- **Local operator interface (controller inputs, not software-only):** manual activation; disable/lockout; independent per-zone command; and distress notification by deliberate long-press. Required capability, not a fixed control count — the interface shall be operable in structural firefighting gloves, in darkness, without instruction, without a personal device, and without network or account. *Reference implementation: discrete panel pushbuttons with indicator lamps; a rotary selector, keyed switch set, or any other physical interface meeting the same capability is equally compliant.*
- Configuration via local-only service interface: the controller serves a plain web page over direct Ethernet or USB connection. No cloud, no app store, no account.
- Rich UI is the incident command interface (main specification); DIN-mount HMI panel is a permitted optional module

Notification Button (All Classes)

The notification button is a controller input, not an emergency-stop and not a safety-rated device; it claims no safety

category. Actuation requires a deliberate press of roughly three to five seconds, which suppresses false triggers from brush contact or a bump. On a registered press the controller, at every class: publishes a distress event to the coordination domain (enumerated category: node identity, position, timestamp — deliberately minimal); latches the local system into full-protection mode (all-zone maximum cycle); and sets the node's status to distress, placing it on the incident command picture where one is operating. Per Charter Clause 7, no software configuration may disable the operator's controls; the actual hardware kill is the disconnect, and loss of power drives the auto-return valves closed independent of any control.

At Residential class, the button signals the coordination domain and anyone watching it; **it is not a substitute for emergency services, and this specification shall never claim otherwise.** At Tactical class, the same press additionally triggers the satellite emergency module where fitted (see Class 3). There is no reason to omit it from the Residential unit; the interface is identical, the tactical unit simply acts on it further.

Audio Annunciation (All Classes)

Every unit carries a small speaker (Mylar cone or piezo element, weather-appropriate mounting or internal with sound port) driven by an I2S codec/amplifier module — commodity board-level parts native to the controller platform. The unit announces state in plain speech: system armed, activation started and ended, specific fault conditions ("zone two fault — valve did not confirm open"), low battery, supply pressure low, communications lost, maintenance required, and distress-transmitted confirmation on notification-button press. LEDs indicate that something is wrong; the voice says what.

- **Voice content is pre-rendered:** clips generated at firmware build time from a text manifest stored in the specification repository. No runtime synthesis dependency exists (Charter Clause 2); the same fault says the same words every time; and **the voice manifest is a specification artifact** — translations and localizations are community contributions like any other. On-device synthesis is permitted for composed values (readings, percentages) where clip permutation is impractical.
- **Annunciation is output only.** This section adds no microphone requirement at any class, and the Tactical acoustic engine sensor remains a machinery-state sensor per the message enumeration. Voice out; never audio in, on the mesh.
- **Maintenance discipline:** periodic audible maintenance reminder per the fire-panel tradition; interval and volume configurable, mute available in local configuration. Fault annunciation defaults on at Tactical class.

Voice interface doctrine. Voice follows the automated external defibrillator pattern: a closed, finite prompt set in plain imperative language, understandable by an untrained person under stress, requiring no training to operate. This is the design discipline codified across AED usability practice (IEC 62366 usability engineering, medical human-factors review) — proven at life-safety stakes on the general public. The prompt set is finite because the failure modes are finite; a few megabytes of pre-rendered audio covers the whole family. All classes share the same interface; the advanced tiers differ only by an extended clip set, not a different interface. Pre-rendered clips per the voice manifest (a specification artifact; translations are community contributions).

Accessibility conformance. Visual and audible indication are each other's required alternative mode, satisfying the Section 508 / Access Board ICT provision that any visual mode have a non-visual equivalent and any audible mode a visual equivalent. Every state announced by voice is also shown by indicator, and every indicator state is available audibly. A means to increase volume is provided. This is adopted by reference to the Section 508 ICT standards and the ADA audible-interface provisions (the talking-ATM and audible-elevator lineage); the Government/Commercial class is thereby accessibility-conformance-report (VPAT/ACR) ready for federal procurement, and the Residential class inherits conformance at no added cost.

Audio is public; no private listening path required. Audible status is public by design — the device announces only enumerated public status ("system armed," "zone two fault") and carries no private information through its audio channel. A standard audio output or coupler point is permitted where an installation wants one (high-noise environments, commercial preference) but is expressly not required: the Section 508 coupler provision protects private information, and a device with no private audio has nothing to couple. Public-access AEDs omit the jack for the same reason. Adopt the pattern; drop the vestige.

Guided Setup (Tactical Class; Optional Elsewhere)

Tactical units commission by voice. Setup is a talking state machine — a fixed if-then checklist, not an intelligent system — in which each instruction gates on a sensor condition the controller verifies itself: "connect water supply to inlet" (advances on inlet pressure detected); "connect zone one" (advances on valve confirmation); "install antenna" (advances on mesh join or GNSS fix); "system ready — coverage active." The operator knows they are done because the unit says so.

Combined with the prepositioning requirements, this means a unit dropped to an operator who has never seen one commissions itself by voice, without a manual, in the dark, in gloves. Guided setup is a firmware feature, not additional hardware, and may be included at any class.

Fasteners, Mounting, and Grounding

- **One interior fastener, one driver — best effort.** Interior mechanical fastening (rail to backplane, valve bracket, battery bracket, duct) standardizes on a single screw size and drive: M5 pan head, Torx T25 reference (Robertson square permitted). Where smaller hardware is unavoidable, the rule degrades to one driver, two bits, kept in the door pocket. Two finish sets: zinc-plated standard; 316 stainless for marine, coastal, and high-exposure builds (anti-seize on stainless-to-stainless — see Workmanship guide).
- **Mounting:** loaded wall-mount units run 45–60 lb wet. Four-point mounting into structure (studs or masonry anchors); drywall anchors shall not be used. External mounting feet or flanges shall be used so field-drilled enclosure penetrations are not required for mounting.
- **Height:** recommended installation places panel controls and indicators in the 48–60 in. AFF service band, which places the battery shelf at knee height (correct lifting) and the manifold drain at the bottom where gravity serves it. Site conditions govern.
- **Grounding and bonding:** a ground bar or bonding terminal shall be provided on the backplane; equipment grounding per the NEC with the supply circuit; antenna bulkheads (N, SMA) shall be bonded per established coax-entry grounding practice, adopted by reference. Electrical safety grounding (required, code-governed) is distinct from lightning protection (site-specific, outside this specification’s scope, addressed by existing standards).

Wiring Practice (Reference Palettes)

Field-built and kit units are recommended to follow the established industrial palettes; single-color wiring with wire numbers and an internal legend is fully acceptable practice per the listing standard’s own terms. Color is the fastest documentation there is; the legend is the requirement.

- **Power (NFPA 79 convention, UL 508A-compatible):** blue — ungrounded 24VDC control (most of this panel); white/blue stripe — DC common; **orange — circuits that remain energized with the disconnect off**, which in this equipment is the battery/DC-UPS bus and its loops: this row is a safety marking, not an aesthetic; black — supply-voltage ungrounded (the 120VAC side); white — AC neutral; green or green/yellow — grounding. Factory sensor pigtails follow the IEC sensor convention (brown +, blue −, black signal) and are not re-colored.
- **Data (TIA-606 administration convention):** blue — horizontal runs to devices; purple — common equipment (controller, switch); green — network demarc/backhaul hand-off; yellow — auxiliary/alarm circuits (the event interface tap); fiber jacket codes adopted verbatim (yellow single-mode, aqua multimode). Labeling per TIA-606 administration is the underlying requirement at any color scheme.
- **Government tier — black-side doctrine:** all equipment under this specification, at every class, resides entirely on the unclassified (BLACK) side of agency red/black infrastructure separation. No RED-side cabling shall terminate on equipment defined by this specification; agency crypto and classified transport sit across that boundary under agency management, per the export-control section. Cable jacket color per agency convention makes this boundary visually inspectable at the open enclosure.

A companion **Workmanship & Best Practices guide** (recommendation tier: ferrules on stranded conductors, lever-terminal field wiring, dressing, labeling, service loops — “not required for compliance; expected in quality work”) is a queued specification-family artifact. The build layer that fire codes delegate to factories, this specification publishes, because its factory is everyone.

Enclosure Water Management

Interior hydraulic/electrical cohabitation shall follow established sprinkler valve-enclosure and irrigation-controller practice, adopted by reference: control electronics mounted above the wet works and never beneath a mechanical joint; drip loops on all conductors descending toward the hydraulic zone; a drainage path from the enclosure low point to the exterior; and no battery placement below the hydraulic zone without a drained tray. Sectioned two-compartment enclosures or dual-enclosure (electrical + valve box) formats are equally compliant and may be preferred for first-time builds.

Enclosure Penetrations — Assembled-Penetration Rule

Where a sealed component forms its own enclosure penetration (through-panel GNSS, combined antenna/radio pods, panel-mount connectors), the penetration is evaluated against the enclosure rating *as assembled*, not against the

reference fitting list. Reference layouts below are reference, not closed sets.

Class 1 — Residential

Wall-mounted unit for homes, manufactured homes, cabins, and small commercial structures on municipal or well supply.

- **Enclosure:** NEMA 4X (enclosure type ratings per NEMA 250 and UL 50E), hinged, lockable, nominal 16 × 20 × 8 in; polycarbonate or powder-coated steel (Hoffman, Rittal, Saginaw, Attabox class); battery shelf at bottom
- **Penetrations:** 3/4” NPT inlet (side); three 3/4” NPT outlets (bottom — the reference position, so the manifold gravity-drains; side outlets are permitted where the site requires, provided a drain path is maintained); N-type LoRa bulkhead (top); SMA GNSS bulkhead (top); cord grip (bottom/side)
- **Connectors:** commodity throughout — cord grips, DIN terminal blocks, NPT/GHT. Field-wireable with a screwdriver and strippers; nothing keyed or proprietary
- **Power:** 120VAC cord-and-plug; two 12V 7–9Ah SLA bricks in series (standard fire/security panel form factor) or drop-in 24V LiFePO4
- **Radio:** LoRa reference
- **Coverage assumption:** consumer-grade distribution — impact/oscillating sprinklers or irrigation heads on garden hose or 3/4” pipe; each zone drives one to four heads at 3–6 GPM; placement per main specification (roof line, gutters, ten-foot perimeter)

Class 2 — Government / Commercial

Wall-mounted unit for federal facilities, commercial structures, critical infrastructure, and any installation procured against environmental-exposure or agency requirements. **Listing consideration:** a UL 508A panel shop generally cannot enclose an unlisted lithium pack; a lithium option should use cells and a battery system carrying UL 1642/UL 2054 (cell and pack) or UL 1973 (stationary battery system) listings, or the 508A route for the finished panel may be foreclosed. Sealed lead-acid is the routine case and carries no such constraint. Identical internals to Class 1; differences are packaging and options.

- **Enclosure:** NEMA 4X minimum; 316 stainless wetted fittings and polymer enclosures preferred for marine/coastal exposure
- **Connectors:** sealed circular connectors of an open multi-vendor standard for all external connections. **M12 reference class** (A-coded sensor/power, D/X-coded Ethernet); Deutsch DT for vehicle/rugged; MIL-DTL-38999 where procurement requires. Field units unplug and swap rather than rewire.
- **GNSS:** pattern 3 (integrated through-panel, marine grade) standard
- **Radio:** LoRa or tactical MANET tier; dedicated router hardware where publication policy or agency integration requires
- **Power:** hardwired or cord-and-plug per site; same DIN/commodity battery architecture
- **Internal networking:** DIN rail unmanaged industrial Ethernet switch, minimum 5 ports, 24VDC input from the existing bus, standard equipment (Phoenix Contact, Moxa, Antaira, Planet class). Port budget: controller, radio/router, backhaul terminal, external IP sensor or detection payload, service. External M12 X-coded Ethernet bulkheads terminate to the switch. **Unmanaged is the requirement; managed is permitted** where agency network policy requires it — the reference design shall never require network configuration to commission.
- **Blue force tracking:** capable of publishing node position and status as Cursor-on-Target messages to TAK infrastructure (ATAK/TAK Server) where deployed. Every node presents as a fixed friendly asset on crew situational awareness: a known point with water, power, communications, and a distress button.
- **Additional message categories:** thermal detection events, triage state, extended telemetry per the main specification’s tier enumeration
- **Options:** DIN-mount HMI, incident command integration, detection payload interface

Maritime and Coastal Use Cases

The Class 2 marine-hardened configuration (316 stainless wetted fittings, sealed connectors, polymer enclosure, marine GNSS) already provisions the unit for salt-water environments; the maritime cases are that hardware pointed at new targets, with an effectively unlimited raw-water supply. The coordination primitive is unchanged. Representative cases, naming the category so coastal and maritime resources are covered by name:

- **Marina and mooring-field defense.** Packed hulls and fuel make a marina a continuous fuel bed with the worst possible structure spacing — vessels in contact spread fire hull-to-hull exactly as structures do below 30-foot separation, and raw water is immediately available. Dockside nodes drafting raw water to wet vessels and finger piers in rotation apply the sequencing thesis with the supply constraint removed.
- **Laid-up, reserve, and idle-fleet vessels.** Reduced-manning layup (reserve fleets, anchored idle tonnage, mothballed hulls) combines high value, low crew, and elevated ignition risk because no one is aboard to catch a start. A prepositioned node with detection and raw-water draft is the fire-lookout configuration afloat: passive until triggered, reporting by satellite, holding until a crew arrives — the Tactical prepositioning doctrine with a hull in place of a conex.
- **Ports, boatyards, and working waterfront.** Fuel docks, haul-out yards, and shoreside storage are the Class 2 unit performing its ordinary role near salt water.

Regulatory boundary. The maritime environment is governed by its own authorities — U.S. Coast Guard, SOLAS, classification societies, ABYC. This specification's maritime claim is confined to coordination, monitoring, and non-regulated wetting; any function touching vessel safety certification or fixed shipboard fire suppression is deferred to those authorities by reference. Shipboard integration, where undertaken, presents the coordination and monitoring layer to a vessel's existing certified systems and does not substitute for them. As on land, the equipment stays deliberately in the open category and out of the regulated-suppression listing regime.

Class 3 — Tactical

Transportable, prepositionable node for forestry, National Guard, and emergency management deployment alongside existing pump-and-hose structure protection practice. Zero contact with agency pump equipment required.

- **Enclosure:** ruggedized transit case (Pelican class); identical internals and firmware
- **Power, primary:** battery/solar — portable power station of an open commodity class (1–2 kWh, solar input, user-replaceable as a unit) or internal LiFePO4 with charge controller; solar panel in kit
- **Power, auxiliary — tool-battery dock:** the unit provides a removable-battery dock accepting a major open tool-platform pack (DeWalt 20V/FlexVolt, Milwaukee M18, Makita LXT class), **selected at order time to match the operating department's existing battery fleet** — the same platform already powering the department's battery extrication tools and charging banks. 18–20V pack through DC-DC boost to the 24V bus; one 5Ah pack sustains controls and radio for a full shift; hot-swap sustains indefinitely. Roles: field power without solar, emergency runtime extension, commissioning power. *Equipment requiring proprietary single-source battery systems is expressly non-compliant with this interface.*
- **User charging (interior):** the case interface is the open lid; charging provisions live inside it, unsealed, where the operator already has access to everything. One or more interior USB Power Delivery outlets (commodity 12/24V-input PD modules, minimum 45W each) on the panel plate, powered from the internal supply. **Outlets deliver rated power whenever energy is physically present, at operator discretion; automated policy may inform — status indication, energy events on the mesh — but shall not refuse.** How to spend the node's energy is an operator's judgment. A charging cable of the prevailing consumer device interface ships in the case and is **treated as a consumable** — inventoried and replaced at warehouse maintenance intervals, so a long-prepositioned case opens with a cable matching the devices of the year it opens. An inductive charging pad of the prevailing wireless standard is a **standalone module** — a discrete unit mounted in the lid or panel recess, powered from an interior outlet, added, removed, or upgraded independently of all other equipment, because wireless charging standards evolve faster than the case does. Justification for the procurement file: field personnel operate on battery-powered personal and issued devices; a protected charging point on prepositioned equipment is an operational capability, not an amenity. The local service connection is direct Ethernet or USB at the controller, inside the case.
- **Connectors:** high-exposure tier (M12/Deutsch class); GNSS pattern 3
- **Internal networking:** DIN rail unmanaged industrial Ethernet switch per Class 2, standard equipment; agency-furnished radios and backhaul terminals plug into switch ports per the export-control boundary
- **Radio:** LoRa for dense sites; tactical MANET tier (Silvus/Wave Relay/TSM class) for range, agency integration, and detection payloads
- **Tactical MANET radio — hardware reference (current, will change):** the radio class is defined by capability, never by vendor (see §Radio-Network Classes); the units below are the best current-generation reference for the Tactical case and are expected to change as the field advances. As of this revision the reference radios are the **Persistent Systems MPU5 (Wave Relay)** and the **Silvus StreamCaster 4200/4400 series**. Their published electrical envelopes drive three integration requirements:
  - *Bus compatibility (validated):* the MPU5 accepts 8–28 VDC and the StreamCaster 9–36 VDC (9–50 VDC on the

4400E). The node's 24 VDC bus lands mid-band for both; a reference radio feeds directly from the bus with no additional converter. The 24 V bus voltage is confirmed correct for this radio class — a lower bus (12 V) would crowd the MPU5 ceiling under vehicle-power transients and double the feed current; a higher bus (48 V) would exceed the MPU5 input.

  - *Power budget (sizing driver):* these radios are a different load class than the residential node. Peak draw is roughly **40 W (MPU5, transmit)** to **65 W (StreamCaster, high-power mode)** for the radio alone. The Tactical supply and battery shall be sized for the reference radio's transmit peak **in addition to** controller, valve, and sensor loads — budget on the order of 100 W-plus continuous headroom, not the residential node's modest control-only budget. A Tactical case built on residential power sizing will brown out the radio on transmit; this is the single most common sizing error for the class.
  - *Feed and interface:* radio power runs on conductor gauge and connectors rated for transmit current (≈2.7 A at 24 V for a 65 W radio), not signal-wiring gauge. Both radios present data over standard Ethernet (StreamCaster adds RS-232 and uses a break-out/ODU cable carrying power, Ethernet, and serial on one connector — allow a gland for it). Both can *source* low-voltage power outward (the MPU5 supplies 5 V on its side connectors); the node shall not contend with the radio over powering shared accessories such as headsets. Integration remains at the Ethernet layer per the export-control boundary; the radio is operator- or agency-furnished.

- **Blue force tracking:** Cursor-on-Target publication to TAK infrastructure per Class 2. A crew working a division sees the node as a fixed rally point that can call for help.
- **Satellite emergency module (optional module; expected Tactical configuration):** dual-constellation satellite SOS — Iridium-network (inReach/RockBLOCK modem class) and Globalstar-network (SPOT modem class) — both fitted, for independent orbits and failure modes. Emergency button press transmits distress with GNSS position by four paths simultaneously: mesh distress event, both satellite constellations, and gateway backhaul where present. Satellite services activate under the operator's existing service arrangements; for government operators these are typically enterprise or interagency agreements rather than per-device subscriptions. Per Charter Clause 2, the fire defense function and the mesh distress event never depend on satellite service; the module is a life-safety enhancement riding above it.
- **Hydraulic/engine sensing (non-invasive):**
  - *Inline sensor spool:* short hose section, 1.5" NH forestry couplings, carrying pressure transducer and flow meter; clamps into the discharge lay in under a minute; pressure-loss signatures identify burst or cut lays immediately
  - *Acoustic engine sensing:* commodity electret/MEMS microphone element on an analog input; portable fire pumps run ≥100 dBA; running/stopped by amplitude threshold, RPM class, load state, and fuel-starvation onset by onboard exhaust-note analysis. **Derived machinery state events only; audio is non-transportable per the message enumeration.** Fuel forecasting by acoustic burn-rate inference against known fuel quantity; fuel float optional.
  - No baseline sensor touches, mounts to, or modifies the pump
- **Duty-cycled operation:** zone valves in the distribution lays let the sequencer time-share pump runtime across coverage points; fuel consumption drops by the duty cycle; a single fuel load extends across the ember-cast window instead of running to dry
- **Pump control interface (published ahead of industry support):** 12V nominal control convention compatible with existing portable-pump control panel and battery-kit practice; start/stop/state via the pump relay and enumerated pump-state categories. Current-generation portable pumps (MARK-3 Watson Edition class) are pull-start; baseline relationship is sense-only. The interface definition is procurement language: agencies specifying coordination-interface pump control in tenders provide the demand signal for next-generation platforms.
- **Future electric pump configuration:** battery-electric portable pumps integrate under the same interface; no starter circuit; battery maintained by the node. Full attack-flow electric pumping exceeds practical battery capacity at current energy density; duty-cycled sprinkler protection — this specification's exact use case — is what makes battery-electric portable pumping viable at all.
- **Weight and carry:** the Tactical node case, as shipped ready to commission, **shall not exceed 40 lb (18 kg)**. Kit components — power station, solar panel, gateway terminal — are separately carried items, each individually under the same limit. The limit exists so that one person can move any element of the kit over broken ground; it is a deployment requirement, not a packaging preference. It also forces the correct power architecture: the case carries only enough internal battery to commission and ride through swaps; the power station travels separately. Kit items are designed to **pair for carry** (balanced loads carry better than single loads of half the weight), and every case carries handles sized and placed for a two-person lift as well as one. The design floor is not the professional firefighter; it is the untrained volunteer who showed up to help.
- **Joint inventory doctrine:** Tactical kits are managed as signed inventories per established deployable-cache practice. Issue and return follow the open-lid ritual: open the case, verify contents against the card, self-test, sign,

move.
    - *Inventory card:* laminated, mounted in the lid, itemized to ninety-second verification depth — node, power station, panel, cables (counted), sensor spool, antenna, adapters, charging cable (consumable), batteries with date. The card, the case contents, and the pattern-library kit manifest are one document with three lives.
    - *Inventory self-test:* the guided-setup system's first mode. On internal battery, without water or antenna mast, the unit walks its own card audibly: radio check, GNSS fix, valve cycle, battery voltage spoken, distress-path self-test without transmission. The event — who signed, when, what was present, what the self-test reported — is logged to the cryptographic chain, extending tamper-evident custody from operation to accountability. Annual warehouse shelf checks use the same mode, by any technician, logged to the same chain.
- **Prepositioning:** palletized warehouse storage, drop deployment. **Prepositioned units shall require no external activation, registration, provisioning service, or connectivity to enter service.** Combined with voice-guided setup, a unit requires no manual to enter service. Warehouse maintenance: battery inspection/swap and charging-cable replacement on standard storage intervals, commodity parts, any technician, verified by inventory self-test.

Coverage Engineering and Wetting Schedule (All Classes)

Home Ignition Zone Framing

Coverage design follows the home ignition zone (HIZ) framework established in federal fire science: whether a structure survives is determined by the structure and its immediate surroundings — the receiving sites where embers land — not by the flame front. Static mitigation (defensible space, home hardening) prepares the ignition zone; this system provides the layer static mitigation cannot: **active ignition zone maintenance during the ember-cast window.** Coverage placement shall prioritize receiving sites — gutters, roof valleys, wall lines, fence lines, and adjacent combustible accumulations — over area wetting.

Receiving Surface Wetting Classes

Each zone shall be assigned a wetting class at commissioning. Wetting classes anchor to the National Fire Danger Rating System (NFDRS) dead-fuel timelag convention, the classification the fire service already operates by.

- **Vegetation zones** (perimeter grass, mulch, shrub lines) inherit NFDRS classes directly. Fine fuels are 1-hour class and re-dry toward hazard rapidly under fire weather; these zones receive the shortest re-wet cycles and full weather sensitivity.
- **Structure zones** are classed by their effective fuel. Untreated wood shake behaves as coarse dead fuel. Class A rated surfaces (asphalt shingle, metal, tile — per ASTM E108/UL 790) are themselves ignition-resistant; **their effective fuel is the accumulated debris** — gutter litter, needle accumulation in valleys — which is 1-hour class material. A rated roof zone therefore still cycles; its schedule follows its debris, not its shingles.

Adopt-by-reference clause: wetting schedules shall follow the current published methodology of the national fire danger rating system; revisions to that methodology are adopted by reference and do not constitute revisions to this specification.

The Cycle Equation

The sequencer's re-wet interval is the solved output of the drying model, not a fixed timer. For each zone: the application pass raises surface moisture above the protective threshold; the NFDRS drying equation (first-order, driven by temperature, relative humidity, and wind) gives the time for that moisture to decay back toward hazard; that interval is the zone's maximum cycle time. Duty cycle is therefore computed continuously as drying time against wetting time. Hot, windy, low-humidity conditions tighten the rotation automatically; recovery conditions relax it and extend supply endurance. Duty-cycled coverage is not a compromise on continuous wetting; it is the solved form of it.

Weather Sources

The drying model runs on every node; its inputs are a layered service with interchangeable sources, best available first:

1. **Local sensor** — Davis/Kestrel-class station on the node or a domain neighbor; truest, works isolated
2. **Domain-shared weather** — one instrumented node serves the coordination domain via the existing environmental telemetry category

3. **Public feed via gateway** — RAWS, NWS, or MesoWest-class aggregator data injected as domain weather by any node with backhaul; example source classes, not dependencies
4. **Conservative default** — with no weather from any source, the node assumes fire-weather drying rates and cycles accordingly. Isolated nodes err wet, which is the correct failure direction.

Weather sources are an enhancement, never a dependency, per Charter Clause 2.

Application Rates and Head Selection

Coverage distribution adopts by reference the application rates, overlap, and placement practice of published wildland structure-protection sprinkler guidance, to its current editions — the national cache sprinkler-kit standards and the structure-protection field literature, alongside Canadian SPU and FireSmart-class practice. The forestry-standard brass impact sprinkler (25PJ class) is the reference head for roof and broad coverage; low-flow plastic impacts and micro-heads are the reference classes for thin supplies and spot targets. The **Supply & Coverage Planning Table** (OSF-2, Appendix A) provides conservative planning figures for supply runs, head classes, spacing, and per-zone head counts; the unit's commissioning self-calibration measures actual supply and governs over all planning figures.

Head placement — damage prevention is a design requirement. Published structure-protection guidance is emphatic on a point that is easy to get wrong: the objective is to wet the receptive fuels adjacent to the structure and raise humidity around it, not to wash the building. A dwelling's roof and wall assemblies are built to shed water falling from the sky; an impact head drives water horizontally with force, past siding and into the wall cavity. The resulting damage may be immediately visible on an exposed assembly, or may remain hidden and surface months later as mould behind interior finishes. Installations following this specification shall observe the following placement practice:

- **Prefer a settable-arc head over a full-circle head** wherever a head is placed near a wall. The arc is set to throw away from the structure, permitting close-in placement without water striking the building. Full-circle heads are permitted; they require more standoff or an angled placement.
- **With a full-circle head near a wall, place close-in and angled** — staked leaning toward the value at risk, approximately three to four feet off a corner — so the pattern dampens adjacent ground cover without saturating siding.
- **Roof wetting is achieved by lofting, not by driving.** Where the roof plane is the target, water is thrown up onto the deck rather than aimed at the surface, avoiding lifted shingles or shakes and keeping the stream off the walls below.
- **Overlap adjacent patterns.** Spacing follows the Planning Table: spacing equal to the effective radius as the baseline, derated in wind. Spacing at the full diameter of throw leaves a dry gap between heads and does not meet the coverage intent.
- **Design pressure for structure-protection heads is 50 psi.** Below approximately 40 psi the effective radius collapses and any spacing derived at design pressure no longer overlaps. Commissioning self-calibration measures delivered pressure and governs.
- **Feed heads from a trunk with laterals** rather than one long straight run of small-diameter hose, so the last head is not starved relative to the first.

Spacing and Participation

At suburban structure spacing (separations under approximately 30 feet), structure-to-structure ignition dominates the loss mechanism, and adjacent-structure protection becomes the controlling variable: a participant's survival is partly a function of parcels they do not control. Published multi-fire analyses confirm that structure spacing is a critical loss factor and that community-scale coordinated mitigation is what maximizes effectiveness. At this density, participation density matters more than individual system quality. This is the coordination protocol's founding case restated as fire science.

Undefended-Parcel Deployment

Coverage may extend beyond ownership. A vacant lot, unoccupied house, or unmitigated parcel between participants may be covered from participants' own supplies: a long supply run (3/4 in. hose or better strongly preferred over distance — see the Planning Table friction figures) feeding one to two low-flow heads placed at the parcel's receiving sites, on its own zone. Self-calibration accommodates the reduced supply automatically; the sequencer schedules the thin zone on its wetting class like any other. The protection delivered is modest by design and sufficient by intent: the gap in the neighborhood's ignition zone is the neighborhood's problem, and this is the tool shaped like the gap.

Coverage Target: Structure or Terrain

A zone's coverage target may be a structure's receiving surfaces or a piece of ground the fire must cross. The hydraulic and wetting-class treatment is identical — application rate against fuel type under live weather — whether the receiving surface is a roof, a gutter, a fence line, or the surface fuel in a drainage bottom. The terrain-target case is developed as doctrine in the specification's expeditionary section (Terrain-Anchored Deployment and Non-Destructive Suppression); at the annex level the relevant fact is that the coverage-engineering math does not distinguish between wetting a structure and wetting a corridor, and the same node, valves, and sequencing serve both. A structure that cannot be defended at its own walls may be defended by wetting the terrain upstream of it.

Worked Example — Post-War Ranch Archetype

Representative case: single-story ranch, approximately 1,100 sq ft, Class A asphalt roof, municipal supply at 50 psi, 18-foot separation to neighboring structure with wood fence in the gap, mature deciduous trees over the roof line.

- **Zone 1 (roof/gutter):** one 25PJ-class brass impact at the ridge or two small plastic impacts at the eave line; wetting class: 1-hour (debris-governed)
- **Zone 2 (perimeter/vegetation):** low-flow heads on the grass ring and foundation line; wetting class: 1-hour
- **Zone 3 (exposure):** heads directed at the shared fence line and the gap face of the structure; wetting class: coarse fuel (fence) — this zone exists because of the 18-foot spacing, per the Spacing and Participation section
- Total draw per active zone: 3–6 GPM — within a single hose bib. The unit self-calibrates, sequences the three zones on live drying intervals, and participates in domain coordination.

Expansions (Orthogonal to Class)

Pump and reservoir expansion — any class. Cistern per main specification (minimum 2,500 gal cistern-primary; buffer tanks on reliable municipal supply). Pump: self-priming, fuel-drive preferred, minimum 40 GPM at 80 psi, remote start via controller relay (Honda WX class, trash pump configurations, industrial centrifugal electric). Valves move external: ISO 5211 pad bodies with separable 24VDC actuators sized to bore. Sensing additions: submersible 4–20mA reservoir level; pulse-output flow meter on discharge (GPI, Seametrics class). Sensor interfaces at expansion tier: 4–20mA loop **or** Modbus TCP, installer's selection, both native to the controller.

Zone expansion modules — any class. Same valve family, small NEMA enclosure, 4-wire daisy-chain bus.

Gateway kit — one per coordination domain, single pallet line item: satellite terminal, power station, solar panel. One gateway gives an entire deployed domain — a valley of nodes — external visibility and ledger publication through standard mesh routing.

Detection payload — Class 2/3, government tier: thermal camera on tripod. Detection *events* (occurrence, bearing class, confidence class) ride the coordination protocol; imagery, where an agency requires it, rides agency radio infrastructure as agency traffic outside this specification's enumeration. Aerial detection platforms (UAS overwatch) integrate as detection payloads under the same pattern: detection events on the coordination protocol, imagery on agency transport. Adjacent agency security/surveillance assets coexist on the same network without being absorbed by this specification.

Automated fire lookout configuration — the named Class 3 composite: Tactical node + pump and hose kit with sensor spool and zone valves + weather station (Davis/Kestrel class; wired/serial preferred; vendor short-range wireless is a local sensor link terminating at the node per Clause 6; local wind feeds wet-cycle scheduling and gives incident command spot weather at the point of greatest forecast uncertainty) + detection payload at government tier. An automated fire lookout that defends its own site — resolving the historical weakness of the staffed lookout, which could see fire but not survive it.

Valve Scaling Table

The control object is constant. Only body and actuator scale.

Bore	Body type	Actuator class (typ. torque)	Application
3/4" – 1"	Full-port brass ball, motorized or ISO 5211	Integrated / ≤ 20 Nm	Class 1 zones, Class 3 lays
1.25" – 2"	Full-port brass/SS ball, ISO 5211	20–50 Nm, 24VDC quarter-turn	Expansion zones, light commercial

2.5” – 4”	Lug or wafer butterfly, ISO 5211	50–150 Nm	Commercial, small municipal
6” – 8”	Lug butterfly, ISO 5211	150–400 Nm	Federal facility, municipal main class

All actuators: 24VDC quarter-turn electric, open/close, dual limit-switch feedback. Modulating (4–20mA positioner) actuators are the drop-in upgrade at any bore. **Spring-return (fail-closed) baseline**: loss of power drives the valve closed without controller action, which is what makes the disconnect a true hardware kill. Fail-in-place actuators are a permitted per-zone variant where a design or code requirement calls for held position, but a node built entirely on fail-in-place actuators does not satisfy the power-loss-closes property this specification relies on.

Class 1 Reference Bill of Materials

Item	Qty	Example products	Est. cost
NEMA 4X enclosure 16×20×8	1	Hoffman, Attabox, Saginaw	$150–300
Motorized 3/4” ball valve, 24VDC, feedback	3	CR-series class	$120–210
Inlet check + isolation valves, gallery fittings	1 set	Commodity brass, NPT	$60–120
Pressure transducer 0–200 psi 4–20mA	1	Commodity industrial	$40–80
ESP32 industrial PLC	1	Industrial Shields ESP32 PLC 21, Norvi ENET	$150–250
LoRa mesh node + antenna + N bulkhead	1	RAK4631, T-Beam	$60–120
GNSS receiver + active antenna + SMA bulkhead	1	u-blox class + commodity puck	$30–60
DIN 24VDC supply w/ DC-UPS	1	Mean Well DRC-60	$70–120
Batteries, 12V 7–9Ah SLA	2	Commodity panel batteries	$35–50
DIN rail, terminals, wiring, cord, strain relief	1 set	Commodity	$60–100
GHT adapters, drain valve, bulkhead fittings	1 set	Commodity brass	$40–70
Class 1 unit total (parts)			**$815–1,480**

Class 2 adds connector and enclosure upgrades and radio options against the same internals. Class 3 pricing is dominated by case, power station, and radio tier selection. Costs representative at July 2026 through standard industrial and alarm-supply channels.

Compliance Statement

An installation is compliant with this annex when every capability requirement above is met, regardless of component substitution. Named products are examples only. Improvements to this annex through community development are welcomed and integrated into subsequent versions under the same permissive licence as the main specification.

Accessories — The Smart Water-Appliance Family

The fire service’s manual water-handling layer — hydrant operation, gate valves, wyes and water thieves, manifolds, inline shutoffs — is a small, closed set of appliances in a small, closed set of sizes, all riding the national standard

hose thread (NFPA 1963)s already universal on hydrants and apparatus. This section discloses their remotely-commanded siblings. The thread standard did the interoperability work a century ago; the smart family inherits it — no new coupling, no new size chart, no modification to any hose, hydrant, or apparatus.

Doctrine: sparse by design. The smart family *follows* the manual system; it does not replace it. The manual layer remains the backbone. Smart appliances deploy only at selected key junctions where remote command pays — the hydrant feeding an operation, the wye at a critical branch, the manifold at a buffer site — chosen by the incident commander or planner, not by default. Every smart appliance retains full manual operation when unpowered; the smart version is a powered hand, never a replacement for one. Placement advice may be drawn from hydraulic calculation and from the system's own accumulated ledger data (an AI or analyst trained on the domain's hydraulics and event history can recommend which junctions earn actuation), but placement is always a human decision.

Remote hydrant valve actuator. A clamp-on, non-invasive rotary tool engaging the hydrant operating nut — no modification to the hydrant — electrically driven, remotely commanded through the standard node interface with a two-verb vocabulary (OPEN / CLOSE). The device drives the valve to thread-bottom in the commanded direction and self-terminates; detection of travel completion is left to the implementation (torque-aware self-terminating drive is established art in commercial impact tools). Operates only under the authority of the agency owning the hydrant. Tactical value includes closing hydrants behind a withdrawing crew without a person standing at the wrench.

Smart appliances (gates, wyes, thieves, manifolds, shutoffs). Remotely commanded versions of the standard manual appliances in the standard hose-thread sizes. Each is electrically actuated, node-standard in communications and command security, commanded through the closed vocabulary (OPEN / CLOSE per port; throttled and position-reporting variants contemplated within this disclosure), retains manual operation when unpowered, and uses the existing thread standard as its entire mechanical interface.

The buffered supply point ("a water tower from tender loads"). A portable-tank drop site — multiple standard portable tanks, a multi-port smart manifold in a transit case, and an observation node (camera and/or level sensing) watching tank levels — becomes managed buffered storage: discrete tender deliveries accumulate into standing capacity, delivered as continuous managed supply, with refill and tank-swap triggers driven by observed actuals rather than guesses. The shuttle loop gains a gauge; supply state becomes visible on the command surface and failures of the D-7 kind (a supply point silently dying) become visible hours before they bite. This is the strongest single argument for cached water: storage plus observation plus a commandable manifold equals a water tower assembled from shuttle loads, anywhere a tender can reach.

Power and compliance. Compliant implementations of every appliance in this family accept the system's common battery interface per the family's commodity-battery doctrine. Proprietary-power and alternative drive-mechanism variants are contemplated within this disclosure and are disclosed herein so that they cannot subsequently be patented by others; as system devices they are non-compliant. Vendors are invited to compete on quality — actuation, sealing, ergonomics, runtime — on shared threads, shared battery, and the shared closed vocabulary.

The node-waypoint patrol (aerial observation without towers). Every node publishes its position; the domain census is therefore also a UAS patrol route — node coordinates plus an altitude offset generate a waypoint mission automatically, and a drone flying the circuit captures a snapshot of each node, tank site, and appliance per pass. Visual level-check by flyby substitutes for a fixed observation node wherever a camera tower cannot be sourced, spared, or justified; automated waypoint inspection from asset positions is established commercial art (docked autonomous patrol systems), and this disclosure claims the integration: domain census as auto-generated patrol mission, flyby observation as a peer of fixed observation on the command surface. Airspace conduct is adopted by reference: on incidents, per NWCG PMS 515 — no launch without positive contact with aerial supervision, operation within the assigned altitude block inside the Fire Traffic Area, manned aviation above; outside incidents, per FAA Part 107 and successor beyond-visual-line-of-sight rules as they mature. The patrol serves observation only; it commands nothing, per the standard security model.

Thermal-signature verification (protection state as a map). A wetted surface evaporatively cools; a dry or failed surface does not. The contrast is stark in the thermal band, so a single thermal pass over a coordination domain renders whole-domain protection state as a human-readable image: cool structures are being wetted, hot structures are not. This is external, physics-based ground truth — it measures actual water on the actual surface rather than asking any node to report its own state, and therefore catches the failure modes self-reporting cannot: a clogged head, a kinked line, a valve that confirmed open and delivered nothing. The thermal read is disclosed as a payload variant of the node-waypoint patrol above: the same auto-generated circuit, flown with a thermal camera, produces a periodic protection-state map at range without any crew setting foot on a property.

It stacks with the signed operational ledger to close the last gap in the record. The ledger attests what was *commanded*; the thermal pass attests what physically *happened*. Together they are command-plus-confirmation, and the pairing is what allows an after-event report to survive a reader who does not trust the operator. Timestamps and zone identities in the ledger are recorded so that an external thermal observation can be correlated against them directly.

As a maintenance instrument the same pass is an inspection. Flown before a season, or during a commissioning test cycle, the hot roofs are the work order — a whole neighbourhood is verified functional in one overflight rather than one property visit at a time. A configuration in the shared library is proven when the roof reads wet. Observation only; it commands nothing, per the standard security model.

Disclosure — Utility Right-of-Way Corridor Defense

A coordination domain may be deployed along electric utility rights-of-way as corridor defense: detection nodes at spans and crossings identified by the utility's own vegetation and ignition history, terrain-defense wet line at high-risk crossings and structure-adjacent segments, buffered supply points at substations and access points, and node-waypoint patrol routes generated from the corridor census for vegetation-encroachment and post-event holdover inspection. The domain's signed operational ledger serves as the utility's verifiable mitigation record: after-event reports demonstrating that corridor defense existed, operated, and responded are generated from the ledger rather than from operator narrative. All equipment, message categories, trust-layer behavior, and charter constraints are unchanged; the corridor is a deployment geometry, not a new system class. Utility-furnished transport, where provided, integrates at the network layer per the layered architecture.

Appendix A — Common Parts Catalog

Reference configuration. Consolidated into this annex; previously issued separately. The parts named here are a reference implementation, not a requirement — any component meeting the capability requirements of this annex is compliant.

What this is (and is not)

This is **a** known-good reference configuration, not **the** design. It names one working part per function — biased toward the better version (a MeanWell supply, not a generic; the oversized enclosure, not the tight one) so a first build is likely to work rather than likely to teach the wrong lesson. Every entry is tagged with the *function and requirement* it fills, so a builder reads it as a starting point and substitutes freely against the requirement (per Annex A capability-substitution). The catalog fills in over time: community builds report alternatives, regional availability, and price tiers under each function. Quality-biased for prototype success; substitution-permitted forever after.

This is a starting place for others to build from. It is not a bill of materials and it is not prescriptive. The reference part is the one we could verify openly and build with confidence — nothing more.

Sourcing rule — vendor accessibility tiering

Reference parts are selected in part by **how openly the vendor publishes specs**, because that property serves the builder exactly as it serves the catalog: an openly-specced part is one the next builder can verify and substitute. Vendors are tiered:

- **Tier A (preferred):** public downloadable datasheet PDF or full web spec, no login; **bonus:** downloadable CAD / STEP / DXF stencils for panel layout. *(e.g. MeanWell — predictable per-model spec PDFs; AutomationDirect — per-part Docs & Drawings tab with datasheet + 3D CAD in 60+ formats.)*
- **Tier B (acceptable):** full specs on the website, no login, no CAD.
- **Tier C (bumped down):** specs behind a login, registration wall, or rep-request. Usable only if no Tier A/B part fills the function.

A Tier-A part with a downloadable stencil is worth more to this catalog than a marginally better part behind a login, because the catalog's job is to be *built from*, and you cannot build easily from a spec you cannot pull.

The Reference Configuration — Datasheet (skeleton)

Columns: subsystem function · requirement it must meet · reference part (to fill) · vendor tier · datasheet/stencil status · notes. Empty cells are open slots to fill in over time.

1. Power

Function	Requirement	Reference part	Tier	Spec / stencil	Notes

Primary AC/DC supply	24 VDC, DIN-rail, UL 508-listed, wide-temp, derated headroom	MeanWell HDR- / NDR- series *(model TBD by load)*	A	PDF ✓	Oversize for margin; step-shape slim body
DC-DC / logic rail	5 V or 3.3 V from 24 V bus, DIN or board	MeanWell DDR series *(TBD)*	A	PDF ✓	4:1 wide input option
DC UPS / battery control	Battery-backed 24 V hold, monitored	MeanWell DR-UPS40 class *(TBD)*	A	PDF ✓	Fills 72-hr autonomy w/ external battery
Battery	LiFePO4 or SLA, sized to autonomy target	*(open)*			Chemistry TBD per class
Solar input / charge (optional)	MPPT, node-scale	*(open)*			Tactical/remote tier
Surge / inrush protection	Per Part 15 / UL 508A practice	*(open)*			

2. Control

Function	Requirement	Reference part	Tier	Spec / stencil	Notes
Compute / controller	Runs closed command enum, low power, field-serviceable	*(open — ESP32-class or industrial equiv.)*			Board vs. industrial PLC TBD
DIN rail	35 mm TS35 steel/aluminum	AutomationDirect DIN rail	A	PDF + CAD ✓	Stencil available
Terminal blocks	Screw/screwless, 24–3/0 AWG range, UL	AutomationDirect DINnector series	A	PDF + CAD ✓	Ground blocks green-yellow
Relays / valve drivers	Drive motorized valve actuators, DIN, isolated	*(open)*			Dry-contact or PLC-card drive per actuator
Fusing / protection	Per subsystem	*(open)*			

3. Comms

Function	Requirement	Reference part	Tier	Spec / stencil	Notes
Mesh radio (baseline)	LoRa class, license-free, neighborhood range	*(open — Meshtastic-compatible module)*			Field-proven range basis
Tactical MANET (option)	Self-forming mesh, IP, MIL-STD-810 class	Agency-furnished (Silvus/Wave Relay/TSM class)	—	—	Operator-furnished; not catalog-sourced
Gateway / backhaul	Satellite or cellular uplink per domain	*(open)*			One per coordination domain
Antenna	Band-matched, weatherized	*(open)*			

4. Sensing

Function	Requirement	Reference part	Tier	Spec / stencil	Notes
Water pressure	Inlet pressure,	*(open)*			Drives supply-sag

	analog/digital, potable-safe				yield behavior
Flow (optional)	Zone flow verification	*(open)*			
Temperature / RH	Local weather, drying model input	*(open)*			
Wind (weather-station config)	Anemometer for forward node	*(open)*			Expendable forward posting
Pump acoustic (non-contact)	Engine-run detection, zero mod to agency gear	*(open)*			Acoustic DSP
Flame / heat detection (option)	Thermal, per detection payload	*(open)*			Feeds CoT `d-` branch

5. Hydraulics

Function	Requirement	Reference part	Tier	Spec / stencil	Notes
Zone valves	Standalone motorized full-port brass ball valve, 24 V, ISO 5211 pad, position feedback	*(open — industrial actuated-valve class)*			Reference valve; brass body suits the 4X wet section. Solenoid irrigation valve is a permitted budget variant where position feedback is deliberately waived
Sprinkler heads	Impact / rotor, published precip rate	*(open — 25PJ-class impact)*			Reference precip from vendor tables
Manifold / fittings	Poly / brass, standard thread	*(open)*			
Pump (tactical/terrain)	Portable, draft-capable, gph per water plan	*(open — WATERAX class)*			Forestry-standard
Supply line	3/4”+ hose or pipe, friction per planning table	*(open)*			See OSF-2 Appendix A friction figures
Strainer / suction	Draft from static or open water	*(open)*			Millpond / tank / drum

6. Enclosure

Function	Requirement	Reference part	Tier	Spec / stencil	Notes
Residential enclosure	NEMA-rated, oversized for margin, UV-stable	*(open — Hubbell-Wiegmann / AutomationDirect class)*	A	PDF + CAD ✓	Oversize deliberately
Tactical case	Transit-rated, foam-fit, ≤40 lb ready	*(open — Pelican-class)*			Prepositioning / drop
Marine-hardened (Class 2)	316 SS wetted, sealed connectors	*(open)*			Per maritime use cases
Cable glands /	Sealed,	*(open)*			

entries	strain-relieved				
DIN backplate / subpanel	Sized to layout	AutomationDirect subpanel	A	PDF + CAD ✓	Stencil available

How this fills in

1. **Web-planner pass (this document's method):** the panel side is populated from open web configurators and per-part datasheet/CAD downloads (MeanWell, AutomationDirect first). The irrigation side is populated the same way from the sprinkler vendors' public design tools and precipitation-rate tables.
2. **Reference-build pass:** parts are proven in an actual bench/pilot build; anything that flakes is re-specced and noted.
3. **Community pass:** builders report alternatives, regional availability, price tiers, and substitutions under each function; the catalog accretes columns over time.
4. **Sample-hardware (post-attention only):** once the standard has visibility, vendors of Tier-A openly-specced parts are the natural first providers of sample hardware for reference builds — they already publish everything needed to be included, and a reference slot is a placement worth their sample. *No sourcing relationships pre-filing.*

This catalog is a starting place, not a specification. The reference part is the one we could build with confidence and verify in the open — substitute freely against the stated requirement. It fills in over time; that is the design.

The Open Source Firefighter

Coordinated Defense Tactics — A Combined Arms Paper

v1.0 · August 2026 · Companion to the specification family *Document ID:* ***OSF-2*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

This paper describes how coordinated ember defense fights — alone at each tier, and combined with the fire service's existing arms. It is doctrine, not requirement: the specification says what the equipment does; this paper says how the fight uses it. It mirrors the equipment family deliberately: three tiers, one architecture, and the combined operation at the end. It is written for fire officers, planners, and the people who will stand behind the line.

> **A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.**

The premise

Wildland firefighting has always been combined arms. Hand crews cut line, engines hold it, dozers build it faster, aviation cools what the ground cannot reach, and command orchestrates the pieces against a fire that gets a vote. Coordinated ember defense does not replace any arm of that fight. It adds one: **ground that defends itself** — water, timed and placed by machines that never get tired, commanded by people who do. The tactical question this paper answers is what that new arm does at each scale, and what the old arms are freed to do because of it.

One idea runs through every tier: **the receiving surface is the battle.** Structures are lost to embers landing on dry fuel; ground is lost to fire crossing dry fuel. Keep the receiving surface wet during the window when it matters, and the fire arrives at a place that will not have it. Everything below is that one idea at three different sizes.

Tier One — The Structure (the box on the wall)

The unit of action: one node, one structure, one to three zones. The residential wall unit or its commercial sibling, wetting roofs, gutters, wall lines, fences, and the vegetation ring on a rotating cycle driven by live drying conditions.

What it fights: ember cast — the attack that destroys most structures in an interface fire. Firebrands landing on dry receiving surfaces, ahead of and behind the front, in quantities no crew can chase. A wet roof sheds the attack that a dry roof accepts.

How it fights: sequencing. One zone at a time, so an ordinary household supply protects a whole structure; re-wet intervals computed from the fuel's actual drying rate under actual weather, so water is spent exactly as fast as the physics demands and no faster. The node fights alone if it must — no network, no command, no grid.

Tactical effect: a defended structure is subtracted from the list of places a person must stand. That is the whole effect, and it is the most important sentence in this paper. Every structure that holds itself releases the crew that would have held it.

Bounding: a node defends against ember ignition and surface-fuel attack. It does not defeat direct flame impingement or radiant load from a crowning slope at close range — a structure inside that geometry is defended at Tier Three or evacuated, not promised a miracle.

Tier Two — The Community (the mesh and the module)

The unit of action: the coordination domain — a street, a neighborhood, a town — plus the Coordinated Defense Module: one Type 6 engine, a trailer of prepositioned cases, a small element, a drone pilot; ordered and typed like any other resource.

What it fights: the mathematics of simultaneous demand. Every structure needs water in the same hour; the shared

supply cannot feed them all at once; uncoordinated defense collapses the main and protects no one. And it fights the coverage gap — the unprotected parcel, the unarmed house, the vacant lot that turns one ignition into a block.

How it fights: the domain schedules itself — structures take turns on the shared supply automatically, holding total draw within what the mains deliver, yielding instantly when an engine company needs the water (a pressure sag is an order every node obeys without being told). The module extends and repairs the domain: it arrives reading the town's own census off the mesh, seeds nodes into the mapped gaps at open-lid tempo, adds detection overwatch and command presence, and moves on — leapfrogging cluster to cluster, because the nodes hold ground after the module leaves. Prepared communities multiply the module; the load plan writes itself from the domain census in the first minute of radio contact.

Tactical effect: the staffing arithmetic inverts from crews-per-site to sites-per-crew. A town that would have consumed twenty engines in static structure protection is held by its own boxes plus a handful of engines working exceptions. The exceptions are where people belong: the woodpile spot, the pre-code roof, the un-boxed unit.

Bounding: participation density is the controlling variable at suburban spacing — below roughly thirty feet of separation, structures ignite each other, and one protected home in five is a wet spot where three in five is a firebreak. The community tier is only as strong as its adoption map, and the gaps concentrate exactly where money is thinnest. Doctrine must say so, because the fire will.

Tier Three — The Terrain (the wet line)

The unit of action: the pre-surveyed defensive network — nodes emplaced on the ground the fire must cross: the drainages, chokepoints, saddles, and field edges that a century of fire history names in advance. Sited in the quiet season, written into the community's protection plan, cached against the corridors they will hold.

What it fights: the approach itself. Not the fire at the walls — the fire three drainages out, funneling toward town on the same roads it has always used.

How it fights: the reversible fireline. A continuous wetted strip is a constructed line that builds itself faster than hand or dozer line and is held by nodes instead of crews. Its tactical function is identical to fuel-removal line — deny the fire the next foot of receiving fuel — achieved by temporarily raising fuel moisture instead of permanently destroying vegetation. Wetting a fine fuel below ignition threshold is forestry-manual arithmetic; the water needed to hold a narrow line against surface fire is small, and it comes from what the ground offers — a millpond, a stock tank, static storage, a tender loop. The network is **passive until triggered**: emplaced at negligible standing cost, dormant most days of most years, deploying alongside the resident mesh only when conditions and a front arrive. Presence is separated from staffing, which is what makes landscape-scale line economically possible where continuous hose-lay staffing never was.

Green islands are the tier's signature move: a pre-wetted patch ahead of the front is a break point where the fire arrives with less energy — and once committed and safe, it is a helicopter landing zone. The same water builds the line and the airport the incident will need.

Tactical effect: the fight moves off the structures entirely. A front broken or steered at the corridor never becomes a thousand simultaneous ember problems in town. And when it is over, the line evaporates: the vegetation lives, the ground is unscarred, the community was not destroyed to be saved — which dissolves the political resistance that limits destructive precautionary line.

Bounding (mandatory, always stated): the wet line holds surface fire and flanking/backing fire — which is most of a fire's perimeter, most of the time, and where most acres are actually held. It does **not** hold a wind-driven crown run spreading by radiant heat and long-range spotting; no crew stakes its egress on a wet line; the line is best-effort infrastructure that buys probability, not certainty; and employment is human-commanded at every step. The system is a tool a qualified officer wields. It does not fight fire on its own.

The Combined Operation

No tier fights alone in a real event, and the tiers were designed to compose. What follows is the shape of the combined fight — the new arm integrated with the old ones.

Days out — shaping the battlefield. The terrain tier deploys first, because ground does not get wet in an afternoon. Corridors go active off the pre-survey; static sources are verified (*verified* — a walk-around is not an inspection, and the fire will find the February shortcut with terrible precision); the tender loop stands up; green islands saturate and become tomorrow's LZs. Residents arm and self-test; the mesh reports the town's readiness as one picture; modules arrive against the CWPP priority and seed the mapped gaps. Aviation gets its fill points and

landing zones from the same survey. **Water logistics is the campaign** — the boxes are the easy part; the tender driver on the ninth climb of a fourteen-hour shift is the hard part, and the plan must budget roads, fills, and fatigue like the consumables they are.

The smart water grid, sparse by design. The manual water-handling layer remains the backbone; remotely commanded appliances (the hydrant valve actuator, smart gates and wyes, the multi-port manifold) are placed only at selected key junctions where command flexibility pays — chosen by the IC, advised by hydraulics and the system's own ledger history. The signature employment is the **buffered supply point**: portable tanks, a smart manifold in a case, and an observation node watching levels turn a tender shuttle into managed standing storage — a water tower assembled from shuttle loads, with refills and tank swaps triggered on observed actuals. Supply state becomes visible on the command surface, a withdrawing crew can close the hydrant behind itself from the truck, and the supply officer commands the grid instead of driving to its valves. Every smart appliance works manually when dead; the smart version is a powered hand, never a replacement for one.

The approach — fighting at the corridor. The wet line meets the front at the tier-three ground: holds it where the water reached, turns it along the wetted edges, stalls it against the islands. Crews do not stand at this line — they patrol it, tend it, and chase what spots over it, with the drone walking the perimeter when the air allows and the fixed detection carrying the watch when it doesn't. Engines and dozers work where doctrine still needs them — anchor points, contingency line, the places wet line is the wrong tool. Command watches one picture: every node, every tank level, every corridor, live. The forward weather nodes buy the minutes that matter when the outflow comes — and the outflow will come. And command has a second picture available on demand: a thermal pass over the domain reads protection state directly off the ground — cool roofs holding, hot roofs not — so the incident commander sees which structures are actually wet and which have failed or gone unprotected, at range, without pulling a crew off the line to check. It confirms the coordination picture against physical reality: where the map says wet and the roof reads hot, that is where the next problem is, and command knows it before the fire does.

The arrival — the shrunken fight in town. What crosses the corridors arrives at the community tier pre-shrunk: ember cast and leftovers instead of a front. The domain runs its rotation; the mains hold because the schedule holds; municipal-supplied nodes yield the instant an engine drafts. Structure crews — a fraction of what static protection would have consumed — spread out and monitor the whole defense, working the exceptions: the gaps in the adoption map, the pre-code construction, the ember bed the rotation reached one beat late. This is combined arms at its purest: the boxes are the line, the engines are the maneuver element, and each is free to be excellent at its half because the other half is covered.

After — the honest accounting. The ledger holds the whole event: every gallon, every cycle, every yield, every failure logged the hour it happened. The after-action writes itself from a record that cannot be shaded, the wet line disappears into ordinary ground, the cases go home signed against their cards, and the findings — the split tank, the chewed shoulder, the coin-flip coverage map — go back into February, where the next fire is already being decided.

Command integration, stated once: the system presents one command surface at every tier — a residential node and a terrain-line node in the same picture, distinguished by capability, never by separate applications — and publishes into the mapping tools commanders already run. Above it, any battle-management layer the era provides (common operating pictures, drone and sensor fusion) can task and read the network through the same closed vocabulary, because the box is control-system agnostic by construction. And the security is self-sorting by prior design: a node cannot execute a command for hardware it does not carry — the capability bits define each node's executable subset — so the attack surface of any node is exactly its installed equipment and nothing more. Operators prevail over automation at every tier, always, per the charter. The machines hold the water. The people hold the judgment.

The experienced-eyes doctrine. The common operating picture makes seasoned judgment deployable at the cost of a glance, and the service is full of judgment currently trapped behind desks and knees. Three postures, one mechanism: **the desk assist** — the FMO or fire staff officer who came up on the jumps and now runs a small department cannot peel off for every little start, but with the picture already rendering beside the budget spreadsheets, assisting collapses from a half-day drive to a two-minute look and one radio call ("watch that draw — it spots across every time; set your trigger point now"), delivered to every initial attack in the district at zero marginal staffing, in exactly the window — small fire, thin resources, green IC deciding alone — where one experienced set of eyes has always been worth the most. **The defensive seat** — the coordinated defense renders as a screen-runnable domain (nodes, corridors, tank levels, coverage, the wet line: allocation and timing, not chainsaws), and doctrine assigns it as a defensive group under a seasoned officer, including the retired or limited-mobility chief who knows the county better than anyone but can no longer run the woods — freeing the incident commander to concentrate on the main attack while span of control improves for both. **The consultation** — the old hand in front of the live picture, opinion on request; the profession that consumed his knees keeps his judgment. In every posture the mechanism is the same and the boundary is the same: the system does not create the expertise or the responsibility — it deletes the travel between them. Augmentation, never command transfer; on-scene authority retained, per the charter, always.

The one-page version, for the officer in a hurry

- **Tier One:** the box holds the structure. Every structure that holds itself releases a crew.
- **Tier Two:** the mesh holds the town's water math; the module fills the gaps and moves on. Crews-per-site becomes sites-per-crew. Participation density is the controlling variable.
- **Tier Three:** the wet line holds the ground the fire must cross — reversible, pre-surveyed, passive until triggered, human-commanded, honest about crown fire.
- **Combined:** shape days out, fight at the corridor, mop the shrunken fight in town, and read the whole thing off one ledger afterward. Water logistics is the campaign. February is where the fire is actually fought.
- **The point:** not machines replacing firefighters — reach and leverage handed to the people who already do the work, so fewer of the places on fire require a person standing there.

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Appendix A — Supply & Coverage Planning Table

Formerly OSF-2.2, folded into this paper at consolidation; printable as a standalone extract. The register records the supersession.

Conservative planning figures only. Everything is site-specific: wind, slope, elevation gain, and obstructions shrink every number below. The unit's commissioning self-calibration measures your actual supply and governs; this page is for planning, not performance guarantees.

Table 1 — What your supply run honestly delivers

Assumes 50 psi municipal/well pressure at the spigot, level ground. Figures rounded down on purpose.

Supply run	50 ft	100 ft	200 ft
1/2" garden hose	~3 GPM @ ~30 psi	~2–3 GPM @ ~22 psi	~2 GPM @ ~15 psi
5/8" garden hose	~5 GPM @ ~40 psi	~4 GPM @ ~32 psi	~3 GPM @ ~22 psi
3/4" garden hose	~6 GPM @ ~44 psi	~5–6 GPM @ ~38 psi	~4–5 GPM @ ~30 psi
3/4" poly/PEX/pipe	~7 GPM @ ~45 psi	~6 GPM @ ~40 psi	~5 GPM @ ~34 psi
1" pipe	~8+ GPM @ ~47 psi	~8 GPM @ ~44 psi	~7 GPM @ ~40 psi

Rule that teaches itself: over distance, hose diameter is everything. Borrowing supply across the lot line? **Borrow the fat hose.** Subtract ~5 psi per 10 ft of elevation gain.

Table 2 — What each sprinkler class needs and covers

Derated field figures, not catalog still-air numbers. Effective wetted radius assumes light wind.

Head class	Needs (min)	Appetite	Effective radius	Use for
Micro/low-flow spray or mini-impact (plastic)	~15 psi	0.5–2 GPM	8–15 ft	Gutter line, deck, fence line, spot targets
Half-pattern spray head	~20 psi	1–2 GPM	10–12 ft	Wall line, roof edge from below
Small plastic impact	~25 psi	2–3 GPM	15–25 ft	Small roofs, yard corners, thin supply runs
Brass impact, forestry standard (25PJ class)	~40 psi	3–5 GPM	25–35 ft	Roof ridge, broad coverage — the structure-protection workhorse

Table 3 — Spacing: how far apart to put them

The table above gives each head's reach. This one says where the next head goes. Wildland guidance says to place heads so adjacent patterns overlap; irrigation practice states the same rule as a number.

Spacing rule	Distance between heads	When to use
Head-to-head (the standard)	equal to the effective radius	Baseline. Each head throws to the base of the next; patterns overlap by half.
Wind-derated	~90% of the effective radius	Any exposed site, any forecast wind — which is most fire days.
Hard wind	~80% of the effective radius	Ridge tops, canyon mouths, sustained wind.

Do not space at the full diameter of throw. Two heads set a full diameter apart leave a dry hole between them, and the dry hole is where embers sit. Spacing equal to *radius*, not diameter, is the whole rule.

Design pressure is 50 psi. Cache-standard structure-protection heads are built around it. Below roughly 40 psi the radius collapses and every spacing figure above becomes wrong — the layout that overlapped on paper no longer overlaps on the ground. Check Table 1 for what your run actually delivers before trusting a spacing.

Trunk and laterals, not one long hose. Avoid feeding several heads from a single straight run of 3/4 in. hose; the last head starves. Run the largest line you have as a trunk and branch laterals off it. The rule scales down intact: on a single-property install, this means running the fat line to a manifold near the structure and short laterals to each head.

Table 4 — Match line

One zone runs at a time. Plan heads per zone, not per system.

Your supply (from Table 1)	Workable head classes	Heads per zone
~2 GPM @ low psi (long thin run)	Micro/low-flow only	1–2
~3–4 GPM @ ~20–30 psi	Micro, half-pattern, small plastic impact	1–2
~5–6 GPM @ ~35–45 psi	Any above; one brass impact	1 brass or 2–3 small
~7+ GPM @ ~40+ psi	Full forestry pattern	2 brass impacts

Rules box

1. **Receiving sites beat area.** Wet the gutters, roof valleys, fence line, and the junk pile — not the lawn. Embers land on things.
2. **Rotation is the design.** One zone at a time is how a garden spigot protects a whole structure. Do not plan for simultaneous zones.
3. **The forestry anchor:** the brass impact class above is the same sprinkler wildland crews deploy for structure protection. If it reaches your ridge line at your pressure, use it there.
4. **The thin-supply anchor:** if Table 1 puts you in the top rows, small plastic impacts and micro-heads placed *at the targets* out-protect one starving brass head every time.
5. **The undefended parcel next door is coverable:** 200 ft of 3/4" hose and two well-chosen small heads will keep it wet. Not well — enough.
6. **Site specifics govern.** Wind, slope, trees, and your actual measured pressure override every figure on this page. The unit measures truth at commissioning.

Sources: supply figures derive from published hose and pipe friction-loss tables and manufacturer minimum operating specifications, derated for field conditions. Head classes, placement practice, and the overlap rule adopt published wildland structure-protection sprinkler guidance — the national cache kit standards and the structure-protection field literature — by reference to their current editions. Spacing and precipitation-rate relations follow standard irrigation design practice (Irrigation Association / manufacturer design guidance). This page is a planning aid and is not a substitute for the specification.

The Deadman Fire

An Operational Scenario — Coordinated Ember Defense and Terrain Defense at Community Scale

v1.0 · August 2026 · Design-review instrument · constructed operational scenario · notional 2036 fire season, Colorado wildland-urban interface *Document ID:* ***OSF-2.1*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

This is not a case study. It is a constructed scenario used to stress the Open Source Firefighter concept against a realistic event — to see where a mature deployment holds, where it gaps, and what combined operations look like when the system is ordinary infrastructure. The town is unnamed because Colorado is full of it. The gaps are not hidden and not annotated; they are where the reader who built the system should look hardest.

The town

Call it eight thousand people in a valley that opens east onto plains and closes west against the forest. The old residential grid under mature cottonwoods where the town started. A manufactured-home park on the low south ground by the creek, ninety-six units on rented lots. Newer stucco ranch subdivisions on the northeast bench, built defensible-space-aware. A longer commercial strip than the town used to have, a second school with a ballfield complex, and the golf course lots on the west edge, big houses set back in the ponderosa. And the millpond: twelve acres of water behind an old earthen dam at the base of the west slope — a strategic asset most towns would never notice they had.

By 2036 the boxes are furniture. The old grid runs them at roughly eighty percent, the mesh so dense the coverage map shows those blocks as one unbroken green mass. The benches came pre-plumbed. The commercial strip runs maintained commercial units. The park is the honest ground it always is: a county cost-share program, forced along by the absentee owner’s insurer, has put boxes on forty-one of ninety-six units. Forty-one reads as progress in an annual report and reads as a coin flip if you live in the other fifty-five. Renters always know exactly who has what.

The quiet season

The work that decides this fire happens in February, which is the correct month, because in February nobody is afraid.

The district’s terrain-defense coordinator — a part-time position invented for a retired fuels-and-fire-planning hand who failed twice at retirement — pulls every fire perimeter back to 1910 from the state archive. They tell the story terrain always tells: the fires come out of the west and northwest, down three drainages that funnel toward town like the fingers of a hand. Deadman Draw in the center, the big one, the corridor of the 1934 and 1987 fires. Sawmill Gulch to the south, feeding toward the millpond, the creek, and the park beyond it. Bracken Draw to the north, shallow and wide, opening onto the golf course that has been an accidental fuel break since it was seeded in 1961. A hundred years of fires and they all take the same three roads. The fire doesn’t choose; the ground chooses. The only unknown is when.

Sixteen node sites are surveyed and staked along the three corridors — the chokepoints where the draws narrow, the saddle crossings, the field edges where fine fuel meets timber. Hardware is cached in three conex boxes, one per corridor, inventoried on laminated cards. Water is where the budget goes and where it isn’t enough: the millpond covers Sawmill and lower Deadman through pre-graded draft points; upper Deadman and Bracken get static storage — four secondhand 2,500-gallon poly tanks on graded pads, filled by tender in spring, inspected, in theory, annually. Two of the tanks are new-old-stock. Two spent nine years in a field near Calhan before the district bought them, and in February, with snow on the sidewalls and the sun low, they look fine.

Six green islands are surveyed — a hay meadow above the millpond, the school ballfields, an irrigated pivot corner under a signed landowner agreement, three clearings along the corridor lines — each pre-sited for wetting and each surveyed for helicopter approach and departure. Wetted ahead of a fire, a green island does two jobs: it breaks the fire’s run, and once the fire is past, it is the safest ground in the county, which is where the helicopters land. The

plan is not just building fire line. It is building the airport the incident will need, out of grass and water.

All of it goes into the Community Wildfire Protection Plan revision as a terrain-defense annex — corridors named, sites staked, islands surveyed, draft points marked, cache inventories attached — and the federal grant that follows pays for more nodes than the district could otherwise buy in a decade. The plan is drawn when the ground is quiet.

Ignition and buildup

A dry thunderstorm walks the divide on a Tuesday evening in late June and throws eleven strikes. The detection network logs and triages all of them; ten die in rock and wet north-slope timber. The eleventh lands in heavy dead-and-down in upper Deadman Draw, eleven miles out, in country a truck cannot reach. It smolders two days, is confirmed at forty acres with an organizing column, and takes retardant, a hotshot crew, and hope, in that order. By Sunday the Deadman Fire is two thousand acres at zero containment, backing and flanking down its namesake drainage exactly as a century of perimeter maps says it must. Laid over the 1934 perimeter, the match is close enough to trace. Eight to ten days to the saddle if the weather holds.

The terrain line goes in that week — and the argument for starting early is made in one sentence at the incident briefing: the ground doesn't get wet in an afternoon, and in eight days there will be no water trucks to spare. It is logged not as suppression but as construction. The town is building line where the fire has to cross; the line happens to be made of water.

The community meeting fills the second school's gym. The fire behavior analyst talks about the ridge over the Great Basin, failing overnight humidity recovery, and the two-word forecast that moves every firefighter's weight to the other foot: dry thunderstorms — outflow boundaries, wind from a direction the fire hasn't been coming from, at speeds the briefing didn't promise, for twenty minutes that rearrange everything. The operations chief shows the town its own coverage map — the green mass of the grid, the benches, the strip, and the park's forty-one green dots against fifty-five gray — and beside it the three corridors with sixteen staked sites, and explains that crews and water trucks will spend the week working land that isn't burning and isn't near a house, and that this is the line being built where the fire must cross it, days early, out of water instead of bulldozers, on ground that will still be there afterward. Asked from the back what it costs if the fire doesn't come, he answers: wet grass. That is the whole bill. Nobody asks what the other kind of line costs when the fire doesn't come; a man in the room had a precautionary dozer line cut across his pasture in 2029 for a fire that turned north, and the scar is still visible from the road.

Two Coordinated Defense Modules arrive — two, because a community with a terrain-defense annex in its CWPP now carries a resource-ordering priority: a module multiplies in a town that knows itself. Each is one Type 6 engine, a trailer of cases, a small element, a UAS pilot. The town's mesh hands the modules its census the moment their nodes power on — coverage, gaps, cache inventories, draft points — and the load plans write themselves from it. Behind the modules comes the holding force: a seasonal hand-crew detail under qualified leaders, the same labor pool that cuts hand line, assigned to lighter and safer work — carrying cases to the staked sites, running voice-guided commissioning in gloves, laying wetting arcs, and then patrolling the line in pairs with hand tools and radios. A crew boss with nine seasons of hand line, walking the finished wet line at dusk, calls it line construction where the line builds itself and the crew is just the maintenance.

The first failure

The first sun-faded tank is discovered on Thursday, at site D-7 — the saddle site, the one the plan considers load-bearing.

It is one of the Calhan tanks. Nine years of high-plains sun cooked the UV inhibitors out of its south face before the district ever owned it; the winter's freeze-thaw found the weakness; and at some point in the intervening months a seam along the lower third began to weep, then seep, then quietly fail. Twenty-five hundred gallons on the inventory card are nine hundred in the tank, the difference long since drunk by the gravel pad. The annual inspection was a walk-around, and from three feet the tank looked fine, the way it looked fine in February with snow on it.

The failure is logged to the record the same hour it is found — while it stings — and the disposition is arithmetic: D-7 moves onto the tender rotation for daily fill, running like a checking account instead of a savings account. The second Calhan tank, on Bracken, is pressure-checked the same afternoon and found sound, which proves nothing except that coin flips land both ways. It is a solvable problem on Thursday, with the fire eight miles out. It is also, though nobody can see it yet, the first domino.

The water ballet

By the weekend the terrain line is a living thing running on a logistics tempo the public never sees. The boxes are the simple part; the water is the machine. The millpond feeds Sawmill and lower Deadman through the draft points, a

portable pump on the pre-graded pad running lift on a duty cycle the nodes coordinate themselves. Upper Deadman and Bracken run on the static tanks and a continuous tender loop — fill at the millpond or the municipal standpipe, climb the fire roads, top the tanks, descend, repeat. The nodes stretch every gallon — sequencing, one zone at a time, re-wetting on the drying model's arithmetic instead of on fear — but they cannot make water appear. In single-digit afternoon humidity the fine fuels re-dry toward hazard in under two hours, and the corridors settle into their rhythm: wet, dry, wet, dry, tanks falling, tenders climbing, the ledger logging every gallon against every zone.

The green islands go wet on Sunday and hold on maintenance passes. The helicopter manager flies them Monday and approves four of six as landing zones without touching a skid, off the February survey data alone.

Evacuations run staged and orderly — pre-notice, voluntary west side, mandatory for the golf-course lots and upper bench — and the exposed edges empty in daylight, which is the entire point of staging them early. The park goes to pre-evacuation notice. In space 14, a resident who bought her own box before the county program existed runs its monthly self-test one day early and gets the flat voice's READY, which is what she keeps it for.

The second failure

Wednesday night, twenty-one hours before the fire reaches the saddle, a contract tender goes off the Deadman fire road.

Nothing dramatic, and then everything. Eleven hours into a fourteen-hour shift, ninth climb of the day, dark, smoke settled into the drainage with the evening inversion, and a north shoulder that had been fine all week but has been chewed soft by nine days of heavy truck cycles. The tender takes the third switchback a half-foot wide — the same half-foot it survived twice earlier in the shift — and the shoulder that held twice does not hold the third time. The truck does not roll; it slides, settles into the ditch frame-down with 3,800 gallons pinning it exactly where it stopped, and the driver climbs out the high-side door shaken and unhurt.

The cascade is the lesson. The load is lost — pumped into the ditch to lighten the truck. The road is blocked: the only heavy-vehicle route to sites D-5 through D-9, closed until a rotator wrecker can come from the city, four hours out plus three working. The remaining tenders reroute onto the Bracken road and a ranch two-track that adds forty minutes to every upper-corridor cycle. And the tempo the whole upper line runs on begins, quietly, arithmetically, to lose. D-7 goes dry first — it has been living on the daily fill since Thursday because of the split tank. The nodes feel the falling levels and stretch their cycles, but stretching buys hours, not days.

The causal chain deserves stating the way the after-action will state it: the shoulder did not do this. The split tank did this. Six days of extra fill runs to cover a nine-year-old tank's failure put thirty extra truck-cycles on that road, and the shoulder collected the bill. Water logistics fail in ordinary ways. The plan was sound. The tender still went in the ditch — and the failure was cached in February.

The rotator opens the road at 0910. Site D-7 — the saddle, the crossing point the hundred-year maps have always named — goes functionally dry at 0700 and sits with empty arcs for four hours.

The burn

The fire comes off the saddle Thursday evening on the outflow, because of course it does. The forward weather nodes — ridge-sited, expendable by doctrine — read the thunderstorm collapse before any eye does: wind veer, gust spike, humidity crash, on the mesh and the incident display in the same second. The whole west-side defense tightens at once — corridors shortening cycles, the millpond pump stepping up, the green islands running saturation passes.

Where the wet line is wet, the physics does the least dramatic thing imaginable. The surface fire reaches fuel that will not carry it, shortens, lies down, and quits. The front turns along the wetted edge like water finding a channel, flanks south, meets the saturated hay meadow — forty acres, uncrossable — and stalls against it, burning out of fine fuel while crews watch from the black. A wall of nothing: wet grass, drying in the heat of the fire it is stopping.

At the saddle, where the arcs have been dry four hours, the fire crosses — a hundred-yard front through the exact gap the maps always named, dropping down the upper draw with the outflow behind it. For ninety minutes the Deadman Fire has a finger inside the line, and the bounding conditions live or die: doctrine says the wet line holds surface and flanking fire and says a wind-driven run is a different animal, and this run is trying hard to become the different animal. It doesn't — because the mid-corridor sites were sited where the draw pinches between rock, and they are fully wet and fully fed off the restored tender loop. The finger hits the second line at the pinch, splits against it, flanks into rock on one side and the green pivot corner on the other, and dies of fuel starvation over two hours while the drone — flying now, the outflow spent — walks its thermal eye along the perimeter calling spots for the hand crew.

The after-shift analysis is one sentence: the line held everywhere it was wet, and it failed only where the water didn't reach — and the place the water didn't reach traces, link by link, to a tank that split in a field near Calhan nine years before anyone staked the site. The fire found the town's one dry hundred yards with the indifference of arithmetic. It

was never outsmarted. It was out-supplied, everywhere but there.

What reaches the town is not the front. It is the ember cast and the leftovers — exactly the fight the boxes on the houses were built for, arriving pre-shrunk. The old grid runs its rotation and shrugs. The commercial strip never trips a zone, and its ledgers prove health rather than silence, again. The structure crews hold an eight-thousand-person town with four engines in motion, because the boxes are the line and the engines are the exceptions: a woodpile spot, a pre-code shake garage, a cedar fence taking fire at the alley end.

And the park is the honest ground, again. The ember cast comes over the creek with the last of the outflow and falls on the coin flip. Forty-one units run their rotations; fifty-five have what they always had — the engines, the neighbors, and luck. Two engine crews work the park all night, more than the rest of the town requires together, and they are good and fast and not everywhere. The box at space 14 wets its own roof and, by a zone aimed with intent years ago, six feet of the neighbor's. Four rows down, an ember bed takes hold in the skirting of an un-boxed unit whose owner is out of town. The nearest engine is on it in six minutes. Six minutes is fast; it is not instant. The unit loses its skirting, its deck, and a bathroom wall, and is tagged uninhabitable at dawn — the fire's only occupied-structure loss inside town limits, in the place everyone always knew, among the fifty-five.

Final count: one occupied manufactured home damaged, family displaced. Two sheds and a fence line in the park's downwind corner. The shake garage. A hay barn east of town that was never anyone's priority. Zero deaths. Zero injuries beyond a twisted ankle and a tender driver's pride.

The ledger

The record afterward is the town's honest mirror. It holds the nine days entire: every gallon banked and spent; the split tank logged the hour it was found; the tender rotation strained, reroute times and all; D-7's decline written hour by hour in falling levels and stretching cycles; four dry hours at the saddle; the crossing; the pinch that held. It holds the islands' saturation curves, the LZ approvals, the drone's spot calls, the park's coin flip in green and gray, and the six minutes at the un-boxed unit. Nobody reconstructs anything and nobody shades anything; the after-action writes itself out of a record that does not care whose fault anything was, which is the only way after-actions get honest.

The terrain coordinator's appended page says, in part: *The line performed as doctrine — it held surface and flanking fire everywhere water reached it and failed only where water did not, and both failures were logistics, and both were ordinary. A nine-year-old tank split because a walk-around is not an inspection. A tender left a chewed shoulder because six days of extra fills put thirty extra truck-cycles on one road, and the extra fills existed because of the tank. One February shortcut cost the saddle four dry hours, and the fire, which does not miss appointments, arrived inside the window. Recommendations: static water storage shall be pressure-tested at fill and inspected by means that touch the tank; tender routes shall be graded into the plan as consumable infrastructure with traffic budgets; and the district should buy new tanks — the difference between a new tank and a nine-year-old tank is eleven hundred dollars, and we now know precisely what the eleven hundred dollars was for. Everything else held. The ground the fire crossed is wet grass and black. The ground it didn't cross is a town. No apologies for the water bill.*

Morning

The modules reload at the athletic field over two days — cases inventoried against their cards, signed, stacked, the damaged units set aside with faults logged — and roll south toward the next name on the board, because it is that kind of season. The crews recover heads and roll supply line, and the wet line does what reversible line does, which is disappear. By September the only signs the town was ever defended at terrain scale are sixteen graded pads, one new tank at the saddle, and a hay meadow that came back thicker for the watering. The millpond drops four feet and refills by October. The pivot corner is cut for hay in August like every August. Nothing looks like a battlefield. That was the point since February.

In the fall the district pressure-tests every static tank in the county and condemns two more — both nine-year Calhan stock, both looking fine from three feet — and writes traffic budgets for its fire roads. At the state fire conference in November, the slide that stays on screen longest is not the drone footage of the wet line holding, spectacular as it is. It is the logistics slide: the tender loops, the tank levels, the four dry hours, the eleven-hundred-dollar difference. The boxes are the easy part now; anyone can buy the boxes. What cannot be bought is February — and the fire will find a town's February shortcuts with a precision that has to be watched to be believed.

The fire only ever finds what was already broken. Fix it in February.

End of scenario. The gaps in this account are not hidden, and several are not failures of the box at all — which remains the point worth sitting with: at the coin flip in the park, at the tank in the field, at the ninth climb of a fourteen-hour shift, and at everything that was decided, or not decided, in February.

The Open Source Firefighter

Design Philosophy — The Internal Rule Set

v1.0 · August 2026 · The rules for what goes into the box *Document ID:* ***OSF-0*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

These are not requirements for the equipment; those live in the specification. These are the rules the specification was written by — made explicit so that future contributors, and future versions of the authors, apply the same filter.

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1. Invent only at the coordination layer. Everything else is borrowed: valves from industry, radios from their ecosystems, routing from forty years of networking, fire science from the agencies, drying math from NFDRS. If a problem can be solved by citing someone else's solved problem, cite it. Novelty is a budget, and it is spent in exactly one place.

2. Novelty is spent only where the invention lives — everything the hands touch is familiar. The box resembles a fire alarm panel on purpose. Resemblance is inherited trust: inspectors approve what they recognize, trades already know the work, and society already runs the maintenance habits. The enclosure earns trust by familiarity so the invention inside never has to spend any.

3. Time-share the constraint; never pretend it away. A hose bib, a fuel can, a municipal main — the supply that exists is the design input. Sequencing is the same algorithm at every scale: the box time-shares a spigot, the street time-shares a main, the domain time-shares a valley. Honest hydraulics first; the physics is also the fallback delivery mechanism.

4. Enhancement, never dependency. Every capability beyond the core — backhaul, weather feeds, satellite SOS, incident integration — makes the system better when present and changes nothing when absent. The system operates when isolated. This rule is absolute and unweakenable, and it decides most design questions before they are asked.

5. Safety is structural, not promised. A protocol with no string type cannot smuggle payloads. Data with no message category cannot be transported. A policy can be revised; an architecture has to be rebuilt. Where a boundary matters — surveillance, mission creep, enclosure — build it into what the system *can* do, not what it agrees to do.

6. Operators prevail. Automation manages resources; it never overrules the person standing at the equipment. Where policy and operator conflict at the point of use, the operator wins and the system makes consequences visible. The machine informs; it does not refuse.

7. The filter and the floor converge. Design against the hardest user — the expert who will stand on it, overload it, and defeat any lockout. Design for the newest user — the volunteer meeting it for the first time, at night, in gloves. These converge: simple, verbal, tactile, forgiving, heavy-handled, nothing delicate, nothing that requires reading. A choice serving only one has failed both.

8. Commodity wear items, forever. Every part with a service life is replaceable by a stranger with a supply house account and no documentation beyond the label. Single-source dependency is non-compliance by definition. The 2046 unit and the 2026 unit interoperate because both are made of things that never stopped being made.

9. Wrong category on purpose. At every regulatory interface, the equipment deliberately belongs to the open regime, not the moated one: an industrial control panel, not a fire-alarm control unit; unlicensed spectrum, not coordinated channels; a wetting system, not a life-safety device. Look like the panel; certify like the sprinkler timer. The incumbent's moat only works on equipment that enters their category — so don't.

10. Know whose house it is. Dispatch owns the trunks. The COML owns the incident spectrum. The homeowner owns the override. The agency owns its radios and its imagery. The system integrates by respecting existing authority at every tier — leaving the porch light on rather than knocking on doors. Restraint, stated in writing, is its own credential.

11. Degrade to the baseline, which is zero. Every failure mode returns to the unprotected status quo the world already accepts. A dead node is a house as it was; a missed packet is answered by pressure physics; no fix means run conservative and wet. Strictly additive by architecture — which is why the system never needs permission to exist.

12. See the weapon-shape early; weld the safety on while the metal is hot. Every capable system has a second squint. Notice it at design time and answer it in architecture — enumerated messages, mission-bounded charters, operator supremacy, irrevocable openness — so the safeguard is welded, not bolted, and holds when the author is tired, gone, or done.

13. Comprehension is the adoption mechanism. The system shall be fully comprehensible to the people who build, install, inspect, and live with it — the homeowner at their depth, the electrician at the print, the comms tech at the stack. A system with a reachable floor is one people teach each other, improve, and defend as theirs. We were never designing for approval; we were designing for understanding, and approval is what understanding does on its own. Corollary: where an architectural boundary exists, give it a visual expression the relevant inspecting culture already reads (orange wire stays hot; no red jackets on our rail) — legibility is verification at zero cost.

14. Adopt the pattern, drop the vestige. When borrowing a proven frame — a compliance standard, a trade practice, another device's interface — take the principle and shed the parts that solved a different problem. The headphone jack solved the ATM's privacy problem; a device with no private audio does not grow that tail. Cite the frame, inherit the intent, leave the appendages. This is the discipline that keeps borrowing from becoming accretion.

15. If you cannot tell a coherent operational story about it, you have not finished designing it. A design is complete when a realistic scenario can be told end to end — every actor doing something plausible, every failure degrading sensibly, every gap visible — without hand-waving. The places the story resists being told are the places the design has an unwritten assumption. Narrative coherence is an audit: build the story deliberately to find the seams, and populate it with real reviewers in distinct roles where you can, because the decision-architecture is the thing under test, not any single component. The strongest doctrine is usually not invented but remembered — operational memory transcribed and generalized — because remembered doctrine has already survived contact with reality and therefore has no seams.

The pattern under the patterns: constraints are the design engine. The charter, the layer model, and this list exist so that most decisions answer themselves by subtraction — the complicated version is usually a rule violation wearing a feature costume, and it keeps getting simpler because deleting an imported assumption deletes hardware.

The Open Source Firefighter

A coordinated ember defense for the places we live

v1.0 · August 2026 · Public overview

> **A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.** *Document ID:* ***OSF-3*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back. Former: City of Delta Junction VFD Assistant Chief · Donnelly Training Area (DMPBAX) Range Manager · U.S. Air Force Tech Control

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The problem is embers

When a wildfire destroys a neighborhood, it’s usually not the wall of flame that does it. It’s embers — burning fragments carried by the wind, sometimes a mile or more ahead of the fire itself. They land on roofs, in gutters, in the dry mulch beside a wall, and they start small fires that become house fires while the main fire is still far away. In a bad wind event, thousands of embers land across a neighborhood at once, and there are never enough firefighters to stand at every house.

There is a well-understood answer: keep the surfaces wet. A roof that stays damp during the ember window doesn’t ignite. Sprinkler systems that wet roofs and perimeters during fire events have been protecting individual homes and cabins for years. The engineering isn’t exotic — it’s water, pipes, and timing.

What’s missing is coordination. If every house on a street turns its sprinklers on at full blast at the same time, the water system that supplies all of them collapses — pressure drops, and nobody’s sprinklers work. A neighborhood that wants to defend itself has to share its water intelligently: houses taking turns, surfaces staying wet, the supply never exhausted. No product on the market does this. That’s the gap the Open Source Firefighter fills.

The box

The heart of the system is a weatherproof box about the size of a breaker panel, mounted on a wall. Inside is an industrial controller, a small mesh radio, a battery, and three motorized water valves. A garden hose or pipe comes in one side; up to three go out the other, feeding ordinary lawn sprinklers placed on the roof line and around the house.

It works like this:

It takes turns. The box opens one water zone at a time, rotating on a schedule, so an ordinary household water connection — the same pressure that runs your garden hose — can keep a whole house wet. A wet roof stays protective for many minutes between passes; embers need dry surfaces, and they don’t get any. This also means the system works with as little as one sprinkler from the hardware store. Two or three can protect a mobile home, a cabin, or a small house.

It talks to the neighbors. Each box carries a small license-free mesh radio. When several homes on a street have them, the boxes coordinate automatically — scheduling their water use so the shared supply is never overwhelmed and no house goes unprotected. The same taking-turns logic that runs inside one box runs across the whole neighborhood. No internet connection is required. No subscription. No account. The boxes talk directly to each other.

It keeps its own records. Every box keeps a tamper-evident log of what it did and when — when it activated, how much water it used, what the conditions were. After a fire, the neighborhood has a verifiable record of what happened, useful for insurance, for fire investigators, and for making the system better.

It works alone. If a box is the only one on the street, or the radio network is down, it simply protects its own house on its own schedule. Coordination makes it better; nothing makes it stop.

It talks to you. Every unit speaks its status aloud in plain language — “system armed,” “zone two fault,” “low battery” — the way a fire alarm panel does, so nobody needs a manual or an app to know what’s happening. The rugged field version goes further: it talks a first-time user through its own setup, step by step, confirming each connection as it’s made. A volunteer who has never seen one can put it into service in minutes, in the dark, in gloves.

The firefighter is in charge. When professional fire crews arrive, the system can hand them a single unified picture of every participating home — what’s running, what water is available — and accept their direction. And a homeowner always retains control of their own equipment. The system’s rule, written into its founding documents: automation manages resources; it never overrules people.

What it deliberately is not

The Open Source Firefighter is not a smart home system, not a security camera, and not a surveillance network. Its radio carries only short, defined messages about fire defense — activations, water status, coordination. Video, audio, and personal data have no place in its language at all; the system is physically and architecturally incapable of transmitting them. A homeowner’s data stays home. This isn’t a policy promise that could be revised later — it’s built into how the system works, and the program’s charter forbids changing it.

The family

One system, three packages:

Residential. The wall-mounted box described above. Built entirely from standard industrial parts — the same controllers, valves, and batteries used in water treatment plants and fire alarm panels, available from ordinary suppliers. Parts cost is roughly $800–1,500, and a capable homeowner, electrician, or irrigation contractor can build and install it. Every wear item is a commodity: when the battery dies in eight years, the replacement is an $18 brick from any alarm supply house.

Government / Commercial. The same unit in tougher packaging — sealed military-style connectors, marine-grade fittings, higher-end radios where agencies need them, and integration with the mapping and command tools firefighters already use. A participating facility appears on a fire crew’s tablet as a known asset: water, power, communications, and status, at a fixed point.

Tactical. The same system in a rugged transit case with battery and solar power, designed to be stored on a pallet in an emergency warehouse and dropped where it’s needed. Forestry crews already protect threatened structures by setting up portable pumps and sprinklers at remote sites and leaving them running. The tactical unit ties into that existing practice without modifying any of it: it watches the pump by listening to it, monitors the hoses, schedules the water so the fuel lasts the whole event instead of running dry, and reports everything back in real time — so a crew doesn’t drive two hours to discover the pump quit six hours ago. Add a weather station and a thermal detector, and the kit becomes something the fire service lost decades ago and never replaced: a fire lookout — except this one is automated, and it can defend its own ground. Every tactical unit also carries an emergency button that signals for help by four independent paths, including two satellite networks, and an interior fast-charging provision, because a firefighter with a dead phone at night should be able to charge it from the rig. That is an operational capability, not an amenity.

The seatbelt principle

This system will be asked to prove itself before it’s allowed to exist — studies, certifications, market validation, professional sign-off. That demand deserves a direct answer, because history has already graded it.

The Open Source Firefighter is not a product competing for a budget. It is a safety primitive, like the seatbelt or the smoke detector: something whose worth is categorical, whose mechanism is legible from first principles, and whose delay is paid in lives. Both the seatbelt and the smoke detector were made to run the proof gauntlet before adoption, and the studies that eventually arrived mostly measured the cost of the waiting. The honest path for a safety primitive is the one the smoke detector finally took: reason the mechanism, build it, put it in front of the real thing, let it fail occasionally, and improve it in contact with reality — rigorously, in the open, while building.

Two things make that path even safer here than it was for the seatbelt. First, the baseline is zero: a unit that faults or runs dry degrades to exactly what the house had before — nothing. The system is strictly additive by architecture, so occasional failure returns to the unprotected status quo everyone currently accepts without a study. Demanding proof before a wet roof is allowed to exist is defending dry roofs as the evidentiary default. Second, the proof machine ships inside every box: the verifiable event log means every deployment is a public experiment — what ran, when, with what water, to what outcome — checkable by strangers. The answer to “where’s the study?” is: the study instrument is built in. Help us run it.

One more thing has changed since the seatbelt's day, and it is why this project is structured the way it is. Safety primitives used to get a grace period — time to become culture before anyone figured out how to meter them. The smoke detector arrived before the subscription business model that would have eaten it. That grace period is gone: invented today, the smoke detector would arrive as a platform, with a monthly fee, a proprietary battery, a cloud dependency, and an insurance partnership — and the enclosure would masquerade as the safety. The coordination primitive for ember defense is being born right now, and whichever form entrenches first is what "ember defense" means for the next fifty years. This project's openness is not idealism. It is a replacement for the historical luck that seatbelts got for free.

The goal was never to market the seatbelt. It is to establish the floor — after which building in fire country without one becomes unthinkable.

The program

The Open Source Firefighter is not a product for sale. It is a **published open specification** — a complete, freely available description of how to build compliant equipment, which anyone may use: homeowners, contractors, fire departments, manufacturers, and government agencies, without license fees or permission.

This is deliberate, and it comes from watching what usually happens to good ideas in emergency services equipment. An inventor builds a device; the company fails or is acquired; years later the ideas reappear inside proprietary equipment that can't be repaired without a service contract, with batteries that cost ten times what they should because only one company makes them. The knowledge survives. The repairability dies. The people who depend on the equipment pay for that forever.

The Open Source Firefighter is structured so that can't happen to it:

- **The specification is the invention, and it's given away.** The intellectual property strategy exists to keep the coordination system permanently open — so that no one can ever fence it off and sell it back as a subscription.
- **Everything is a standard part.** The specification requires that every component be available from multiple manufacturers through normal supply channels. Equipment that depends on a single company for parts is, in the specification's own words, expressly non-compliant.
- **It is governed like the internet's own standards.** Anyone may propose improvements. Proposals are judged in public against a short charter whose two unbreakable rules are: the system must never require an outside service to do its job, and automation must never overrule the person standing at the equipment. New technology is welcomed — as optional additions, never as new requirements. A box built to this specification in 2026 and one built in 2046 will still work together.

What one street can do

A single box protects a single home — even a single-wide on a hose bib with one dollar-store sprinkler. A street of them protects each other, sharing water no single home could manage alone. A valley of them, with one satellite link on a ridge, becomes visible to incident commanders as an organized, self-defending community with verifiable records. And every home that defends itself is a home a fire crew doesn't have to stand in front of — which is the real point. The system exists to subtract from the list of places we send people into the ember storm.

At ordinary suburban spacing — houses fifteen to twenty feet apart, a wooden fence bridging the gap — fire science has a blunt finding: once one house lights, the next house is the fuel. Your survival is partly a function of parcels you don't control. That's why participation matters more than perfection: three protected homes out of five is a firebreak; one is a wet spot. And coverage doesn't have to stop at the property line — two hundred feet of good garden hose and a couple of well-chosen small sprinklers will keep the vacant lot between neighbors wet. Not well — enough. The gap in the neighborhood's defense is the neighborhood's problem, and now the neighborhood has a tool shaped like the gap.

This county already ran the experiment. The Black Forest fire of 2013 — a few miles from where this specification was written — destroyed nearly five hundred homes and killed two people. The homes that had done mitigation survived at higher rates, and it still wasn't enough, because homes ignited each other faster than individual preparation could answer. The research since has reached a conclusion the fire service now states plainly: the community, not the house, is the unit of survival. What never existed was infrastructure for a community to act as one. That is the entire reason this system exists.

The wetting schedule itself speaks the fire service's language: zones are classed by the federal fuel-moisture system firefighters already use — grass is a one-hour fuel; a fire-rated roof's real fuel is the debris in its gutters — and the system re-wets each zone on the actual drying rate of its fuel under live weather, from a local sensor, a neighbor's, or the public fire-weather feeds. Smart water, spent exactly as fast as the physics demands and no faster.

What one crew can do

Initial attack has always required putting people where the danger is, because eyes and water both required bodies at the point of contact. This system separates them. Picture a neighborhood with a few dozen boxes, a small drone overhead watching for spot fires, and a crew coordinating from a truck at a safe anchor point — a package small enough to be one engine, one trailer of cases, and a handful of people, deployable like any other fire resource: the boxes are the hands, the drone is the eyes, and the people direct water across the whole neighborhood without standing in the ember cast. A handful of firefighters becomes the initial attack for an entire community — from cover.

The same idea scales past the neighborhood. Where a fire is coming days out — and a century of fire history tells us, for most ground, exactly where it will run — nodes can be placed ahead of it on the land itself: the drainages, the field edges, the chokepoints a fire has to cross. A pre-wetted strip of ground is a fire line that builds itself and needs no crew to hold it, and unlike a bulldozed line it does the job by making the fuel briefly too wet to burn instead of by destroying it. When the fire has passed, the water dries, the vegetation lives, and the town is still a town — not a place that was torn up to be saved. It works against grass and surface fire in the flanking and backing phases where most ground is actually held; it is not a wall against a wind-driven crown run, and it makes no such claim. It is held and commanded by people, always — seasonal crews and their leaders, doing lighter and safer work than cutting hand line, because the water holds the line they would otherwise dig.

That is the heart of it: this is not machines replacing firefighters. It is a technical advantage handed to the firefighting community that already exists — reach and leverage for the people who do the work, built by one of them. When it’s over, the crew goes home and leaves nothing behind but wet ground drying in the sun.

No system prevents every tragedy, and this one makes no such claim. Wildland firefighting will remain dangerous work done by people of uncommon courage. What this system changes is what we have to ask of them — how many of the places on fire require a person standing there. The American West is getting drier. There will be more fire, and there will never be enough crews. The honest answer to that arithmetic is ground that fights for itself, so the people we send face less of it. That is what this system is for, and it is why it is free.

The specification, hardware annex, and governance charter are published under permissive open license. Build one. Improve it. Propose changes. It belongs to everyone it protects.

The Open Source Firefighter specification family: main specification, Hardware Annex A, governance charter, design philosophy. Published under permissive open license.

The Open Source Firefighter

A Community Adoption Briefing for Towns, Fire Districts, and Departments

v1.0 · August 2026 *Document ID:* ***OSF-3.1*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

To the councils, boards, and chiefs reading this

You have seen documents like this one before. A company arrives with a product, a subscription, and a model ordinance, and asks your community to adopt something you will then depend on them for. This document follows that format because the format works — but it is asking for something different, because it is not selling anything.

The Open Source Firefighter is a published open standard for coordinated neighborhood ember defense. There is no company behind it. There is no subscription, no license fee, no vendor to depend on, and no data leaving your community. The specification is free, permanent, and belongs to everyone it protects. What it needs from your community is not a purchase order. It is recognition, a pilot, and a place in your planning.

The problem your community already knows

When wildfire reaches a community, most homes are not lost to the wall of flame. They are lost to embers — burning debris blown ahead of the fire, sometimes by miles — landing on roofs, in gutters, in dry landscaping, starting small fires that become house fires while the front is still distant. In a wind event, thousands of embers fall across a neighborhood at once. No department, anywhere, has enough crews to stand at every house.

The defense is simple and proven: surfaces that stay wet during the ember window do not ignite. Home sprinkler protection has existed for years. What has never existed is coordination — because if every house on a street opens its sprinklers at once, the water system that supplies all of them collapses, and then no one is protected. Defending a neighborhood means homes taking turns on a shared supply, automatically. That is what this system does, and it is the only thing like it.

If your community is in Colorado, you don’t need the theory — you have the Black Forest fire. Nearly five hundred homes, two lives, and a lesson the research has since made explicit: mitigated homes survived at higher rates and it still wasn’t enough, because at neighborhood spacing, homes ignite each other. The community, not the house, is the unit of survival — and until now, no infrastructure existed for a community to defend itself as one. Homeowner planning is supported by a one-page supply and coverage table (conservative figures, ordinary garden-hose arithmetic), and every unit verifies its own actual supply at commissioning.

What it is

A weatherproof box the size of a breaker panel, mounted on a wall. Water in one side, up to three zones of ordinary lawn sprinklers out the other. Built entirely from standard industrial parts — the same controllers, valves, and batteries in your water plant and your fire alarm panels — for roughly $800–1,500 in components, by any competent electrician, irrigation contractor, or capable homeowner.

- **It takes turns.** One zone at a time, rotating, so an ordinary household water connection protects a whole structure. The minimum installation is one garden hose and one hardware-store sprinkler. Two or three protect a mobile home — the housing your most vulnerable residents live in, and the housing proprietary systems ignore.
- **It coordinates the street.** Each box carries a small license-free radio. Boxes on a street schedule around the shared water supply automatically — no internet, no cloud, no account. They talk directly to each other, and they work alone if they have to.
- **It answers to your firefighters.** Crews arriving on scene see every participating home as one picture, on the mapping tools many departments already use, with command authority — including a one-button order for all municipal-supplied units to yield the mains when an engine company needs the water. Homeowners always keep physical control of their own equipment.

- **It keeps records.** Every unit maintains a tamper-evident log. After an event, your community has a verifiable account of what ran, when, and with what water — for your review, your residents' insurers, and your planning.
- **It speaks.** Every unit announces its status in plain language, like a fire panel. The rugged field version talks a first-time user through its own setup — a mutual-aid volunteer can put one in service in minutes, in the dark, in gloves.
- **It is not a surveillance system.** This deserves plain words, because your community has been asked to adopt camera networks before. This system's radio protocol has a short, fixed vocabulary of fire-defense messages. Video, audio, and personal data have no message type — the network is physically incapable of carrying them. That is architecture, not policy, and the standard's public governance charter forbids changing it. Nothing about your residents leaves their property.

A rugged, palletized version of the same system serves your department directly: stored for years in a warehouse, dropped alongside the pump-and-hose structure protection forestry crews already deploy, it watches the pump site in real time, stretches a fuel load across the whole ember window instead of running dry, and reports back — so nobody drives two hours to find the pump quit six hours ago. Every field unit carries an emergency button that transmits distress by four independent paths, and a charging point, because a firefighter's dead phone at 2 AM is a safety problem, not a convenience problem.

What it changes for your department

A few dozen participating homes, a small drone overhead, and a crew coordinating from a truck at a safe anchor point can hold a neighborhood: the boxes are the hands, the drone is the eyes, and your people stay out of the ember cast. That same drone, flown with a thermal camera, shows you *whether it is working* without anyone leaving the truck: a wetted roof reads cold and a dry one reads hot, so a pass over the neighborhood is a live map of which homes are protected and which have a problem — the coordination picture confirmed against the actual heat coming off the actual roofs. The deployable form of this has a shape your department already understands — a **coordinated defense module**: one Type 6, a trailer of cased units, three or four people, and a drone operator, seeding persistent protection at a threatened community in about an hour and then moving to the next, because the boxes hold the ground after the crew leaves. Nothing about it is new equipment doctrine — it is a new resource type built from resources you already have. And where homes in the community already run their own units, the arriving module's radios join their network automatically: **a prepared community is a cheaper community to defend, measurably.** No system prevents every tragedy, and this one claims no such thing. What it changes is the arithmetic — how many burning places require a person standing there. The West is drying, fire seasons are lengthening, and there will never be enough crews. Ground that defends itself is the honest answer, and every home that holds is a home your people don't stand in front of.

Why open matters to your budget

Commercial coordinated-defense systems are entering the market at $20,000–$80,000 per home with subscriptions, sold through insurance partnerships, priced for your wealthiest residents and no one else. The engineering inside them is standard industrial equipment. An open standard means: your residents build or contract protection at parts-plus-labor cost; your local electricians and irrigation contractors do the work and keep the money in town; every replacement part comes from a normal supply house forever; and no company's bankruptcy, acquisition, or pricing decision can ever strand your community's investment. The standard's charter makes single-source dependency non-compliant by definition. A unit built in 2026 and one built in 2046 will work together.

What we are asking your community to consider

1. **Recognize the standard in your permitting.** Installations are ordinary plumbing and electrical work under existing codes — backflow prevention, NEC compliance, standard permits. A council or district resolution recognizing the specification as an accepted basis for ember-defense installations removes ambiguity for your building department and your residents.
2. **Pilot it.** One street, one water district conversation, one season. The specification includes everything your public works department needs to evaluate coordinated draw against your supply — and the coordination exists precisely to protect your system, not stress it.
3. **Put the field version in your continuity planning.** Prepositioned units are a line item, not a program: no subscription, no activation service, commodity batteries on a shelf-inspection cycle, deployable by whoever is available. And verifying a whole neighborhood of installed units before a season is one overflight, not one property visit at a time — a thermal pass during a test cycle turns the hot roofs into the work order, so maintenance scales to the community instead of the calendar.

4. **Connect it to your existing programs.** Firewise USA communities, community wildfire protection plans, and mitigation grant applications all have natural places for neighborhood-scale coordinated defense. The specification’s verifiable event logs are the documentation those programs and your residents’ insurers increasingly want.
5. **Send your firefighters’ knowledge back.** The standard is governed publicly — anyone may propose improvements, and the people who fight fire are exactly who it needs to hear from. Your department’s experience is worth more to this standard than any purchase order.

On the question of proof

Someone at your table will ask — correctly, it’s their job — where the studies and certifications are. The direct answer: this is a safety primitive, not a product, and the seatbelt and the smoke detector already ran this experiment. Both were held to proof-before-existence, and the studies that eventually arrived mostly measured the cost of the waiting. Two things make the cautious path even safer here: the system’s failure mode is the status quo (a unit that faults degrades to exactly the unprotected house you have today — the baseline is zero, so a pilot risks nothing your community isn’t already living with), and the proof instrument ships inside every unit — the verifiable event log makes each deployment a public experiment your department can read. The rigorous move is not to wait for someone else’s study. It is to run a pilot street and generate your own.

What this costs your community

The specification, all documentation, and the governance process: nothing, permanently, by design. A resident’s installation: roughly the cost of a good appliance. A pilot street: the cost of the units plus your staff’s attention. The alternative — waiting for the proprietary version to entrench — costs your community the difference between $1,500 and $40,000 per protected home, forever, and protects only the homes that can pay it.

Who is behind this

The Open Source Firefighter was developed in Colorado by a former firefighter and fire officer with a background in military communications, range operations, and industrial fabrication, and is maintained as a public standard under an open governance charter. It is dedicated to three Colorado firefighters killed on the Knowles Fire during the development of this work. It is free because the people it is built to protect — and the crews it is built to relieve — are not a market. They are the point.

Specification family: main specification, Hardware Annex A, governance charter, and public overview — published under permissive open license.

The Open Source Firefighter

Public Roadmap

v1.0 · August 2026 · Living document — revised at each release *Document ID:* ***OSF-3.2*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

This document describes where the specification family goes next: what is settled, what is being built, and what the community is expected to carry. It is a direction of travel rather than a schedule — the field and the fire season set the pace.

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1. Specification Core

The coordination system's defining documents.

- **Settled.** Main specification (architecture, coordination protocol, incident command interface, cryptographic event log)
- **Settled.** Layered architecture, event interface and message enumeration, command broadcast, single command surface, backhaul and gateway nodes, expeditionary deployment, governance charter
- **Settled.** Specification assembled and complete
- **Ahead.** Variation taxonomy (systematic capability-axis enumeration; publication-day companion)
- **Ahead.** Front-matter hardening pass (deterrents and category boundaries stated summarization-proof, per red-team findings)

2. Hardware Annex A

The buildable system, three enclosure classes, one architecture.

- **Settled.** Three-class family (Residential / Government-Commercial / Tactical), common architecture, sequencing behavior, self-calibration
- **Settled.** Power, GNSS, radio/network, gateway roles, spectrum posture, export-control boundary
- **Settled.** Coverage engineering: wetting classes (NFDRS-anchored), cycle equation, weather source hierarchy, application rates
- **Settled.** Fasteners, mounting, grounding, wiring palettes, enclosure water management
- **Settled.** Tactical doctrine: weight/carry, joint inventory, self-test, prepositioning, satellite emergency module
- **In progress.** Continuous revision from review findings (continuous)
- **Ahead.** Reference layout figures (see Workstream 5)
- **Ahead.** Freeze/climate appendix expansion; marine environment notes consolidation

3. Protocol & Firmware

The wire format and the code that speaks it.

- **Settled.** Wire format and vocabulary: envelope, byte budget, closed two-dozen-verb vocabulary, quantization rules, capability bits
- **Ahead.** Wire format: field layouts per verb, worked byte-level examples
- **Ahead.** Reference firmware: sequencer + self-calibration (controller)
- **Ahead.** Reference firmware: router/publication policy + mesh coordination
- **Ahead.** Voice annunciation manifest (spec artifact; translations as community contributions) and guided-setup state machine

- **Ahead.** Ledger/snapshot implementation and witness tooling
- **Ahead.** Incident command / TAK (Cursor-on-Target) publication capability

4. Documentation & Guides

Everything a builder, installer, inspector, or council needs.

- **Settled.** Public overview (plain-language, full program)
- **Settled.** Community adoption briefing (councils, districts, departments)
- **Settled.** Supply & Coverage Planning Table (OSF-2 Appendix A) (one-page conservative hydraulics)
- **Settled.** Inspector’s Review Draft (code-format shall-paragraphs;)
- **Settled.** Design philosophy (the fifteen rules)
- **Ahead.** Workmanship & Best Practices guide (ferrules, dressing, labeling, wire palettes — “not required; expected in quality work”)
- **Ahead.** Guided-assembly sequence (numbered build order per class)
- **Ahead.** Code-Format Edition (full NFPA-patterned rendering; post-publication)
- **Ahead.** Commissioning & annual test checklist (printable, lives in the door pocket)

5. Reference Build Program

Patterns, not products. Process-agnostic, multi-factory by construction.

- **In progress.** First bench infrastructure exists (electronics review/education bench)
- **Ahead.** Four-vendor differential layout pass → convergence table
- **Ahead.** Reference layout drawings per class (manufacturer CAD, dimensioned; double as spec figures)
- **Ahead.** Pattern library: valve brackets, tactical insert set, panel plates, compartment divider — geometry first, printable derivatives first-class
- **Ahead.** Open panel-planning utility (browser-based fit/BOM calculator, no account, no server dependency)
- **Ahead.** Bench prototype, Residential class (first full build against the documents)
- **Ahead.** Review station protocol (community builds graded against checklist; findings to change log)

6. Governance & Program

How the standard stays open, forever.

- **Settled.** Governance charter (six clauses + operators-prevail; Clauses 2 and 7 unweakenable)
- **Settled.** Message-category enumeration as the mission boundary (structural, not policy)
- **Settled.** Equipment category and certification pathway declared (industrial control panel class; UL 508A route; not a life-safety listing regime)
- **In progress.** Intellectual property protection and permanent-open-release instruments (in preparation; the standard publishes free and stays free)
- **Ahead.** Public repository and project site (git-backed; commit history as the change log)
- **Ahead.** Proposal/disposition process stood up in public
- **Ahead.** Steward succession and continuity documentation

7. Validation & Testing

Prove it, in public, with the instrument built into every box.

- **Settled.** Adversarial review round 1 (hostile-reader analysis; findings dispositioned into the documents)
- **In progress.** Inspector review (installation requirements under review by a career building inspector)
- **Ahead.** Adversarial review at document freeze (real documents, multiple models, hostile personas)
- **Ahead.** Bench hydraulic validation (self-calibration against real supplies; planning-table verification)
- **Ahead.** Fold test (tactical case abuse trial by the intended population)
- **Ahead.** Pilot street (first coordinated multi-home deployment; ledger data published)

- **Ahead.** Empirical re-wet tables (wetted-surface persistence by material, from fleet ledger data)

8. Community & Adoption

The standard belongs to the people it protects.

- **Settled.** Activation instruments written (overview + adoption briefing)
- **Settled.** Dedication (Rifle Helitack crew, Knowles Fire, June 27, 2026)
- **Ahead.** Publication day release (M1)
- **Ahead.** Activation outreach: Firewise communities, extension services, VFD networks, maker/trade channels
- **Ahead.** Contribution intake (war stories welcome; the record learns to hear practitioners)
- **Ahead.** Configuration library seeded (first two or three known-good configurations — see §9)
- **Ahead.** Build registry open (what people built, what they substituted, what broke)
- **Ahead.** Safety-defect reporting channel published
- **Ahead.** Model council resolution / permitting recognition template
- **Ahead.** First adopting jurisdiction; first insurance-recognition conversation
- **Ahead.** Translations of voice manifest and public documents

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9. The Development Commons — what has to exist for the community to work

The specification can be finished by one person. The thing that makes it *usable* cannot be, and it is worth naming precisely what the community is for, because “build a community” is not a plan.

The goal: known-good configurations

The target state is that installing a node on a common structure is **trivial** — not because the system got simpler, but because someone already worked out that configuration, published it, and a hundred people refined it.

The precedent worth studying is the open engine-management community. An open ECU is a general-purpose device and, on its own, an expert-only proposition: you would need to derive fueling and timing from theory for your specific engine. What made it ordinary was not the hardware. It was that the community produced and maintained **per-platform base configurations** — a tested starting map for a Miata, for a Civic, for whatever you had. A newcomer stopped starting from theory and started from something that already ran, then tuned from there. Strapping one to a common car became a weekend job that thousands of people had already documented.

That is exactly what this project needs, and the axis is structure and site rather than engine. The configurations the community should converge on look like:

- Single-wide manufactured home, municipal supply, hose-bib class
- Ranch house on well and cistern, with pump control
- Hillside home, gravity-fed tank, no pump
- Manufactured-home park — multiple nodes on a shared supply, coordinated
- Outbuilding, barn, or detached shop
- Expeditionary case, tactical class
- Utility right-of-way corridor segment

with regional variants layered on top, because radio allocation, freeze exposure, and water-supply practice all differ by jurisdiction.

What a known-good configuration contains

A configuration is a citable artifact, not folklore. Each one carries:

- **The archetype and its assumptions** — roof geometry, supply class, climate, jurisdiction
- **Coverage layout** — head placement and count, zone division
- **Supply math** — worked against the Planning Table (OSF-2, Appendix A)
- **Parts list at a price tier**, with the substitutions someone has actually verified
- **Panel layout and wiring variant**

- **Firmware configuration** — zone count, timings, regional radio parameters
- **Its evidence** — who built it, what was tested, what failed, and any thermal-verification pass
- **Known gotchas** — the things that bit the people who built it

Two properties are non-negotiable. **Configurations are versioned and attributed**, so a builder knows what they are trusting and can go ask the person who wrote it. And **configurations carry their test status honestly**, including "untested" — because the real hazard here is not a missing configuration, it is a plausible-looking one that under-waters and gives a homeowner false confidence. A configuration that has never been run must say so in the same breath as its parts list. Thermal-signature verification (Annex A) is the field check that closes this loop: a configuration is proven when the roof reads wet.

The first two or three configurations are seeded by the author. **Every one after that is the community's**, which is the point — the same substitution-permitted doctrine the parts catalog already runs on, applied to whole installations.

The infrastructure that produces them

Functions, not products — platforms change, these do not:

- **A canonical record that no single company owns.** The documents are plain-text and version-controlled already, which is most of the work. Mirror it: Charter Clause 2 says the *system* carries no mandatory external dependency, and the same discipline applies to the project. If a forge disappears, the standard must not.
- **A change-proposal process.** The most important item on this list, because interoperability is the product and the charter routes vocabulary additions through governance. It needs a template, a discussion window, and a decisions-of-record log. Without one, "governance" is one person answering threads.
- **An archived asynchronous forum.** The record. Real-time chat is for velocity and is ephemeral by design; a decision that lives only in chat did not happen.
- **A build registry.** What people built, what they substituted, what broke. This is the feedback loop that fills in both the parts catalog and the configuration library — and it is the item most easily forgotten, because it has no software analogue.
- **A conformance suite and a way to claim conformance.** This is the anti-fork instrument: it defines what *is* an OSF node, and it makes contributing easier than forking.
- **A private safety-defect channel** with coordinated disclosure. Most projects add one after an incident. Safety-adjacent hardware does not get that grace.
- **Regional working groups**, because compliance, climate, and supply practice are local and no central maintainer can hold them all.

The name as the conformance instrument

The design is free. The **name** is the thing that carries a promise, and it is the only practical lever for keeping that promise honest.

This is the structure mature open-hardware projects converge on, and it separates two things that are usually confused. The specification, the drawings, and the reference implementation are open under their licences: anyone may build, modify, sell, and fork without asking. The **mark** — the project name and its identifying logo — is held, and its use is granted on stated terms. A builder who meets the conformance suite may say the thing is an OSF node. A builder who does not may still build, sell, and ship whatever they like; they simply may not call it that.

Why this matters more here than in most projects: a non-conforming node in a coordination domain does not merely underperform, it **draws water out of turn**. A device that ignores the schedule and calls itself compliant is a hazard to every neighbour on the supply. Conformance is a safety property, and a conformance claim needs something behind it.

Two provisions distinguish a workable mark from a hollow one:

- **The grant is objective, not discretionary.** Meet the published conformance suite and the grant follows; fail it and the grant lapses. It is not a relationship to be managed, a fee to be paid, or a favour to be granted. Discretionary marks become gatekeeping, and gatekeeping is the failure this project is built to avoid.
- **The grant is not revocable at will.** Revocable-at-any-time marks are the same failure as revocable patent pledges: they look like a commitment and function as leverage. A conformance grant that can be withdrawn without cause is worth nothing to the builder relying on it.

A secondary mark for compatibility — *works with*, *interoperates with* — costs nothing to offer and lets adjacent

projects state a relationship without claiming conformance they have not tested for. Established projects in this space do exactly that, and it defuses most of the friction a mark would otherwise create.

What this is not: a licence fee, an approval queue, an endorsement, or a warranty. It is a claim about testable behaviour, and nothing more.

Roles, decision rights, and succession

Written down, publicly, in ordinary language: who decides, how a contributor becomes a maintainer, how disputes resolve, and what happens when a maintainer — including the author — is unavailable. Founder-led projects most reliably fail at the transition from founder to community, and the failure is social rather than technical.

The succession question and the estate instrument are the same problem, and belong in the same conversation with counsel: the instrument that guarantees the work stays open should also say who carries it.

A code of conduct belongs here too, and is best understood as a safety brief rather than an administrative formality — the same document type as the briefing given before anyone steps onto a live range.

Licensing — three decisions of record

Hardware, documentation, and software are different license families and need separate, deliberate choices, recorded with their reasoning. Ambiguity here is the most common reason a willing contributor walks away.

What this is not

None of this gets pre-built. An empty forum is worse than no forum, and infrastructure without people is a maintenance burden that signals abandonment. **This section is a specification for what to stand up when there is traffic to justify it** — in roughly the order listed, since the record and the change process must exist before the first outside contribution arrives, and everything else can wait for demand.

Where this goes

- The twelve-month development window (M2) exists precisely because the remaining work — firmware, builds, testing, adoption — is designed to be done in the open, by many hands, against a published standard. Everything on this list is buildable by strangers from the documents; that is the point of the documents.*

The Open Source Firefighter

Inspector’s Review Draft — Installation Requirements, All Enclosure Classes

v1.0 · August 2026 · Installation requirements in code-adoption language · Derived from the Hardware Annex and Main Specification *Document ID:* ***OSF-3.3*** *· one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

Reviewer’s note: This draft renders the installation-facing and inspection-facing requirements of the OSF hardware specification in code-format numbered paragraphs for plan-check style review. Mark it up the way you would mark up a submittal. Comments on anything are welcome — enclosure, mounting, access, wiring, plumbing, labeling, documentation, and especially anything that would generate a correction notice in the field. Network protocol and coordination behavior are out of scope for this review and are summarized only where an inspector would encounter their physical evidence.

Chapter 1 — Scope and Application

1.1 This document covers the installation, mounting, power, plumbing, marking, and documentation requirements for Open Source Firefighter (OSF) units in three enclosure classes: Class 1 (Residential), Class 2 (Government/Commercial), and Class 3 (Tactical/Transportable).

1.2 All three classes shall contain identical control, radio, and coordination internals. Class distinctions shall be limited to packaging, connectors, power sourcing, and environmental rating.

1.3 Installations shall comply with locally adopted building, plumbing, and electrical codes. Nothing in this document shall be construed to waive local permitting requirements.

1.4 The unit is a supplemental exterior wetting system for ember defense. It is not a fire alarm system (NFPA 72 / UL 864 class), not a fire suppression system as defined by the sprinkler codes (NFPA 13, 13D, 13R), and not a life-safety notification system, and shall not be represented as any of these. It is supplemental to, and replaces no requirement of, the wildland-urban interface provisions a jurisdiction has adopted — commonly the International Wildland-Urban Interface Code, NFPA 1140, or NFPA 1144.

Chapter 2 — Definitions

2.1 Zone. A valved water discharge circuit serving a defined coverage area.

2.2 Zone Manifold. The basic intake manifold header carrying the standalone zone valves, inlet gallery, check valve, drain, and pressure transducer, each valve individually mounted and individually replaceable.

2.3 Coordination Domain. The set of units sharing radio coordination. Physical evidence at inspection is limited to antennas and their penetrations.

2.4 Sequencing. Default operation in which one zone is open at a time on a rotating schedule.

2.5 Assembled Penetration. An enclosure penetration formed by a sealed component’s own mounting stub rather than by a discrete bulkhead fitting.

Chapter 3 — General Requirements (All Classes)

3.1 Enclosure

3.1.1 Wall-mounted enclosures shall be rated NEMA 4X or better.

3.1.2 Enclosures shall be hinged and lockable.

3.1.3 Where a sealed component forms its own enclosure penetration, the penetration shall be evaluated against the

enclosure rating as assembled.

3.1.4 All penetrations shall be gasketed or otherwise sealed consistent with the enclosure rating.

3.1.5 Batteries shall be mounted on a dedicated shelf, positioned per 3.1.6 through 3.1.9.

3.1.6 Interior hydraulic/electrical cohabitation shall follow established sprinkler valve-enclosure and irrigation-controller practice: control electronics shall be mounted above the wet works, and no electrical component shall be mounted beneath a mechanical plumbing joint.

3.1.7 All conductors descending toward the hydraulic zone shall include drip loops.

3.1.8 A drainage path shall be provided from the enclosure low point to the exterior.

3.1.9 Batteries shall not be placed below the hydraulic zone without a drained tray.

3.1.10 Sectioned two-compartment enclosures, or dual-enclosure formats separating electrical and hydraulic equipment, are equally compliant and are recommended for first-time builds.

3.2 Electrical

3.2.1 The internal bus shall be 24 VDC.

3.2.2 Power supply and battery charging shall be provided by standard DIN-rail power modules. Charging circuitry integrated onto control hardware shall not be permitted.

3.2.3 A low-voltage disconnect shall be provided to prevent battery deep discharge.

3.2.4 Batteries shall be commodity form factor, field-replaceable without special tools, and replaceable without disturbing control wiring.

3.2.5 Electrical installation shall comply with the locally adopted edition of the National Electrical Code or local equivalent.

3.2.6 Standby capability shall be a minimum of 72 hours for controller, radio, and positioning functions.

3.3 Plumbing

3.3.1 All wetted connections shall be NPT. Garden-hose-thread adapters are permitted as accessories on Class 1 units.

3.3.2 A spring check valve and a manual isolation valve shall be provided on the inlet gallery.

3.3.3 Backflow prevention shall comply with local water district requirements.

3.3.4 A manually valved drain-down port shall be provided at the manifold low point.

3.3.5 In freeze climates, the installation shall support a winterization procedure consisting of: all zone valves opened, supply isolated, manifold drained, system depressurized. The procedure shall be firmware-assisted and physically completed at the drain valve.

3.3.6 The zone manifold shall be a basic header feeding standalone valves, each valve individually removable without disturbing the others or the header.

3.4 Controls and Operator Interface

3.4.1 A local operator interface shall be provided on the unit, by which an operator physically present can, at minimum: initiate protection manually; disable or lock out automatic operation; command each zone independently; and signal distress to the coordination domain by a deliberate action of approximately three to five seconds' duration. These functions are controller inputs and are not emergency-stop devices; no safety category is claimed for them. No software configuration shall disable an operator control (Charter Clause 7). The hardware means of removing power is the disconnect; loss of power drives the auto-return valves closed.

3.4.1.1 The interface shall be operable by an operator wearing structural firefighting gloves, in darkness, without instruction, without a personal device, and without any network or account. It shall present its state without requiring the operator to interpret a display. *The reference implementation is discrete panel-mounted pushbuttons with indicator lamps; any implementation meeting the requirements of 3.4.1 and 3.4.1.1 is compliant.*

3.4.2 The unit shall indicate, at minimum, the following states in a manner readable at a glance without a display or a connected device: power, radio link, readiness, fault, active operation, and distress. *Panel indicator lamps are the reference implementation.*

3.4.3 The unit shall annunciate status audibly in plain language, including specific fault conditions and a periodic

maintenance reminder.

3.4.4 Configuration shall be by local direct connection only. No cloud service, remote account, or subscription shall be required for any function of the unit.

3.4.5 Operator hardware controls shall not be subject to automated lockout.

3.4.6 Operator hardware controls shall be sized and spaced for actuation by a gloved hand. The reference case is a structural firefighting glove; where a lighter wildland or extrication glove is the design basis, the installation documentation shall state the glove class assumed, and control sizing shall follow from it.

3.5 Antennas and Positioning

3.5.1 Radio antennas shall be external via weatherproof RF bulkheads or assembled penetrations.

3.5.2 Each unit shall carry a GNSS receiver. The antenna shall be placed with open sky view: external active antenna via bulkhead, external sealed puck on data cable, or integrated through-panel unit per 3.1.3.

3.6 Marking and Labeling

3.6.1 Each unit shall bear a permanent label identifying: specification and annex version built to; supply voltage; battery type and replacement rating; and the drain-down location.

3.6.2 The notification button shall be labeled with words to the effect that it signals participating systems and is not a substitute for contacting emergency services.

3.6.3 Zone outlets shall be individually and permanently identified.

3.7 Accessory Water Appliances

3.7.1 Coordination-capable water appliances (remotely actuated valves, gates, wyes, manifolds, and supply-point assemblies) are accessories, not required equipment. A compliant installation of any class requires none of them.

3.7.2 All wetted appliance connections shall use national hose thread (NH/NST) or NPT as appropriate to the connection point. Proprietary couplings shall not be used.

3.7.3 Appliances attaching to hydrants, standpipes, or utility fittings shall be clamp-on or thread-on, shall not permanently modify the fitting, and their removal shall restore the fitting to its original condition and operation.

3.7.4 Every actuated appliance shall preserve manual operation. Failure or de-energization of the actuator shall leave the appliance manually operable by standard tools or standard practice.

3.7.5 Battery-powered appliances shall use a major open tool-platform battery interface per 6.5. Appliances requiring proprietary single-source battery systems shall not be permitted.

3.7.6 Appliance installation on utility infrastructure (hydrants, district valves) shall be subject to the utility’s or district’s approval and standards; nothing in this document waives that authority.

Chapter 4 — Class 1: Residential

4.1 The enclosure shall be NEMA 4X, polycarbonate or stainless. The reference size for a residential node is nominal 16 in. × 20 in. × 8 in.; larger enclosures are compliant provided they house the specified internal panel layout and preserve the wet/dry separation of §3.

4.2 Penetrations shall consist of: one 3/4 in. NPT inlet; three 3/4 in. NPT outlets; one N-type RF bulkhead (top); one SMA RF bulkhead (top); and one cord-grip strain relief.

4.3 Power shall be 120 VAC cord-and-plug through the strain relief.

4.4 Batteries shall be two 12 V, 7–9 Ah sealed lead-acid units in series, or a drop-in 24 V LiFePO4 equivalent.

4.5 Zone valves shall be standalone motorized full-port brass ball valves with actuator pads and position feedback, fed by the manifold header per 3.3.6, one per independently commanded zone. *The reference residential node uses three 3/4 in. valves at 24 VDC; zone count and valve size follow from the supply and coverage plan of the installation.* Solenoid valves are a permitted budget variant where position feedback is deliberately waived.

4.6 External connections shall use commodity means: cord grips, DIN terminal blocks, NPT/GHT. Keyed or proprietary connectors shall not be used.

4.7 The unit shall self-calibrate to available supply by observing inlet pressure under single-zone flow. No hydraulic configuration by the installer or owner shall be required.

4.8 Default operation shall be sequencing per 2.4. Simultaneous zone operation shall not be assumed in design or represented in installation documents.

Chapter 5 — Class 2: Government / Commercial

5.1 Class 2 units shall meet all Chapter 3 and Chapter 4 requirements except as modified by this chapter.

5.2 External connections shall use sealed circular connectors of an open multi-vendor standard; M12 is the reference class. Deutsch DT and MIL-DTL-38999 are permitted where procurement requires.

5.3 For marine and coastal exposure, wetted fittings shall be 316 stainless steel and polymer enclosures are preferred.

5.4 GNSS shall be an integrated through-panel marine-grade unit per 3.1.3 as standard.

5.5 An internal DIN-rail unmanaged Ethernet switch, minimum 5 ports, 24 VDC, shall be standard equipment. Managed switches are permitted where agency network policy requires; the reference installation shall not require network configuration to commission.

5.6 Power shall be hardwired or cord-and-plug per site requirements, on the same DIN/commodity battery architecture as Class 1.

5.7 Agency-furnished radio and transport equipment shall integrate via standard Ethernet interfaces and shall be installed per the furnishing agency’s requirements. Units shall not transmit on fire voice networks; agency radio equipment operates under the furnishing agency’s authority and applicable interagency listings.

Chapter 6 — Class 3: Tactical / Transportable

6.1 The enclosure shall be a ruggedized transit case. Interior components are not required to be individually sealed; the case provides the environmental boundary in transport and the open lid is the service interface in operation.

6.2 The case, as shipped ready to commission, shall not exceed 40 lb (18 kg). Kit components carried separately (power station, solar panel, gateway terminal) shall each individually not exceed the same limit.

6.3 Every case shall provide handles sized and positioned for both one-person and two-person carry.

6.4 Primary power shall be battery/solar: a portable power station of an open commodity class or internal LiFePO4 with charge controller.

6.5 An auxiliary power dock shall be provided accepting a major open tool-platform battery pack, selected at order to match the operating department’s existing battery fleet. Equipment requiring proprietary single-source battery systems shall not be permitted at this interface.

6.6 One or more interior USB Power Delivery outlets, minimum 45 W each, shall be provided. Outlets shall deliver rated power whenever energy is physically present; automated policy shall inform but shall not refuse. A charging cable of the prevailing consumer interface shall ship in the case and shall be inventoried as a consumable.

6.7 Hydraulic sensing shall be by an inline sensor spool with 1.5 in. NH forestry couplings. Engine sensing shall be by non-contact acoustic means. No baseline sensor shall touch, mount to, or modify pump equipment.

6.8 Units shall commission by voice-guided setup: each instruction shall gate on a sensor-verified condition, and the unit shall announce readiness on completion.

6.9 Prepositioned units shall require no external activation, registration, provisioning service, or connectivity to enter service. Warehouse maintenance shall be limited to battery inspection/replacement and consumable replacement on standard storage intervals.

6.10 A satellite emergency module (dual-constellation) is permitted as the expected configuration; the unit’s fire defense function and local distress signaling shall not depend on satellite service.

Chapter 7 — Documentation

7.1 Each installation shall be documented with: a written narrative of system intent and coverage; a zone layout identifying coverage areas; the water supply type and measured or rated delivery; and the electrical supply arrangement.

7.2 Completion documentation shall record the self-calibration result (supply behavior and cycle timing as set) and a functional test of each zone valve, each hardware control, and the audible annunciation.

7.3 The unit’s internal event log shall be regarded as the operational record. The installer shall verify at completion

that the log is recording.

7.4 A copy of the installation documentation shall remain with the unit (interior document holder) and with the owner.

End of document. All requirements above derive from the Hardware Annex and Main Specification; discrepancies found in review supersede nothing automatically but will be dispositioned against the source documents. This draft is structured for direct use as public input to relevant standards-development processes.

The Open Source Firefighter

The Training Range — A Vision for the Program’s School

v1.0 · August 2026

*Document ID: **OSF-3.5** · one document of the Open Source Firefighter specification family.*

> **A treatise on a new interface firefighting technology that enables the incident commander with a strategic resource at the line of contact.**

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

They were the first to arrive. This system exists so that the ground they arrive on is already fighting back.

What this document is

The rest of this family specifies a system: the box, the mesh, the water, the doctrine. This document answers the question that follows every good box onto the pallet: **how do people at every level learn to use it?** The homeowner’s answer is easy — the Overview and the Adoption Briefing carry it. The commercial installer’s answer is the default — the trades already read specifications for a living. This document is written for the harder and more valuable audience: **the fire and forestry professional**, the district office, the volunteer department, the burn boss, the range manager — the people who will run the coordinated defense at scale, and who need somewhere to train on it.

It is a vision document. The program it describes does not exist yet. It is published now, with the rest of the family, because the ground it describes already exists, sits idle at the exact moments it is most relevant, and belongs — in the ways that matter — to more people than currently realize they are its stakeholders. If you work in a forestry office or run a volunteer department anywhere near a military installation, this document’s purpose is that you finish it knowing something you did not know when you started: **you have an avenue onto that ground, and reasons to use it.**

The campus already exists

The Department of Defense operates the largest inventory of instrumented, maintained, fire-relevant training land in the country. Every terrain type. Every fuel condition. Every size, from a pistol range to maneuver areas the size of counties. Roads, water points, impact-area discipline, and — on modern instrumented ranges — installed thermal camera networks and observation towers, maintained on someone else’s budget, watching ground where fires start routinely because starting fires is what tracer, artillery, and demolition training does.

And here is the fact this document turns on: **during red flag conditions and high fire danger, those ranges shut down.** Live fire stops. Some years, even blanks stop. The installations’ training capacity goes idle at precisely the moment fire training has maximum urgency and realism — and the crews reduced to dry-fire theater on a silent range are sitting in the middle of the best wildland-fire classroom in the country, on the exact days it can teach.

A range that cannot shoot can host node emplacement drills, wet-line lays, buffered-supply-point exercises, UAS patrol circuits, mock-fire coordination problems, and the full surveillance-and-release curriculum described below — none of which require a single live round or an open burn window. The fire restriction that suspends the range’s primary mission creates its availability.

The rent is the fuels backlog

Every range in the system is behind on land management. Grass and fuels work is the orphan task of range operations — the busy-work everyone pretends is not their responsibility, deferred every year because the people and money are never there. Meanwhile the interagency burn crews that could treat it arrive on a schedule set months in advance: a few days per facility per spring, weather permitting. Some years the grass is wet and the visit produces nothing. The good years, the crew burns what is reachable in the time available — the flat infield grass, the course roads — and moves to the next post. The hard ground stays hard forever: the draws, the tracer fans under the gun-target lines, the terrain that actually sets fires all summer.

The arithmetic is the problem, not the people. **The scarce resource is not the burn crew; it is the burnable window** — the scattered handful of days each year when a given problem area is safely treatable — and a traveling crew scheduled months ahead almost never overlaps it. A defense that must be present when the window opens

cannot be built from a scarce expert resource that travels. It has to be standing, local, and ready — which is this specification's entire thesis, pointed at prescribed fire.

So the school pays its rent in the host's own currency: **every training evolution leaves the range with fuels work done.** Wet-line exercises hold real burns on real backlog. Node emplacement drills pre-position real terrain defense. The range's thousand-acre problem is the curriculum, and the range officer stops tolerating the program and starts requesting it.

The gap the school closes

Coordination between the fire world and the range world exists today — but as deconfliction, not integration. The burn crew takes the range for its scheduled days; range control clears the space and steps back; everyone follows the process. The information flows one way. Nobody asks the range office where the fires usually start, which draws always get away, or what instrumentation is buried under which grass — and so target pits and lifters have burned under professionally executed prescribed fire, not because anyone failed, but because the professional-courtesy structure defined the locals as a scheduling entity rather than a knowledge source. Every range manager can draw the ignition map of his own ground from memory. Almost no one from the fire side has ever asked.

Deconfliction keeps two operations from colliding. Integration lets them inform each other. The missing piece is not a coordinator — the coordination slot is filled. The missing piece is *standing*: someone the fire side recognizes as a peer, who can put local ground truth inside the burn plan instead of outside the tape. Credentials are the carrier wave for knowledge; the same warning lands differently from "a local" than from a dual-qualified professional whose input the burn boss must weight.

That is the school's product. Not a facility. Not a technology. **A profession**: operators fluent in both dialects, minted deliberately instead of occurring by accident.

The two formations that never overlapped

The wildland fire tradition produces superb ground-readers: professionals who read fuel, weather, slope, and fire behavior as a first language, formed by seasons on the line. That formation is analog and terrestrial by lineage, and it is honorable. What it never includes — because the equipment does not exist in that world — is the instrumented battlespace: mesh networks, thermal camera grids, observation towers, coordination protocols, and the discipline of running a surveillance picture.

The range world produces the complement: operators who run multi-camera thermal networks as a daily instrument, who detect a developing situation in a wood line before it becomes a problem, and who live inside the loop of *detect, correlate, decide, task*. On instrumented ranges, that discipline extends to its full form — an observer sensor, a decision authority, and an effector as separate nodes on a network, with positive identification established across independent sensors and consequential action released only by a human voice, and held the instant anything is ambiguous. That reflex cannot be read out of a book. It is built by running it, and the only places it can be run are exactly the ranges this document describes.

The curriculum, in one line: hand the ground-reader the grid. Take the fire professional who already reads the land and teach the surveillance picture, the correlation call, the release discipline, and the coordinated water employment this specification defines — talk the student onto the thermal signature, require agreement across independent sensors, require the words said out loud, and teach that holding everything when unsatisfied is not caution but the job. The military spent a century writing this pedagogy. The school ports it to fire — dry, then rehearsed, then live — and the dry tier runs on any instrumented range on any red flag day, when nothing else can train at all.

What standing local capability unlocks

A range operating this specification's terrain layer — pre-surveyed node sites, reversible wet line, buffered supply points fed by ordinary porta-tank orders through local departments and forestry, the installed thermal net as overwatch, a resident cadre of dual-fluent operators — can prosecute its own fuels program opportunistically: hit the hard draw on the exact right Tuesday in November, one problem area at a time, all year, with the traveling professional crews reserved for the genuinely difficult prescriptions where their depth is irreplaceable. The lighting was never the hard part of prescribed fire; the *holding* is — and pre-positioned water collapses holding risk on exactly the ground nobody will otherwise touch: the deep masticated mats, the smoldering-prone cleared units where grass grows through years-old hydroaxe work, the category of ground that needs fire most and receives it least.

The sequence is deliberately modest: **assist first.** The program shows up as the holding and safety element that makes the interagency crew's scheduled days count. Capability migrates the way it always migrates — toward whoever does the work best with the least friction — and no one needs to be displaced for the backlog to start shrinking.

You are a stakeholder in that ground

This is the part written directly to the reader in the forestry office and the volunteer fire station. The access avenues exist now:

- **Any volunteer fire department in the country has a path onto a military installation** — mutual aid relationships, installation support agreements, and joint training arrangements are established machinery, exercised routinely.
- **Federal and cooperating agencies have partnership avenues** to installation ranges as a matter of standing policy; the ground is government ground, and cooperative use is what the machinery is for.
- **The specialized facilities sit empty.** Urban training complexes — dozens of varied structures, furnished, built for search and maneuver training — stand unused the overwhelming majority of the year, while departments an hour away improvise structure-search training in whatever they can borrow.
- **The ranges' fire problem is real and unowned**, which means the agency or department that shows up with capability is not competing for a resource — it is answering a need the host cannot currently meet.

The ask of this document is small: the next time you drive past the installation boundary, do not think of it as someone else's fence. Think of it as the best-instrumented fire classroom in your county, idle on red flag days, behind on the exact work your profession exists to do, reachable through agreements your organization is already entitled to pursue. The coordinated defense this family specifies will be run, at scale, by people — and the people will be trained somewhere. The somewhere already exists. It is waiting for its stakeholders to recognize themselves.

Different colors of the same money

One more thing the reader in the district office already knows, said out loud: the volunteer department, the range, the Army, city fire, EMS, police, forestry, the hotshot crew, BLM, state fire, federal fire, county fire, the HOA, the FAA, the water department, and the power company are **different colors of the same public money.** The citizen writes one check, and it comes back wearing twenty uniforms that each defend a slice as though it were separate funds. It never was. The fragmentation everyone treats as natural law is accounting made visible as turf — and the public that pays for all the sleeves does not care whose patch is on one when the ridge is burning.

This specification's coordination layer is built for exactly that reality, in exactly one role: **the box is the secretary.** Not a commander, not a new agency, not anyone's boss — the secretary that did the interagency coordinating in advance, mechanically, so that every professional simply does their job. The hydrant, the tender, the terrain node, the range's thermal net, and the incident commander's tablet cooperate through a protocol that already reconciled what the org charts never will, without a memorandum summit, without a turf negotiation, without anyone yielding authority to anyone. Coordination as infrastructure instead of meetings. If the public wants the red boxes — and understands that it is all the same game, played by the same team, on the same dollar — then the remaining work is what this document describes: train the team on the shared ground the team already owns.

What this document is not

It requests no facility, no construction, no program office, and no budget line. It proposes tenancy, not ownership — conducted in the tenant ethic this specification's author practiced for eleven years as a range manager: honor the host's load-bearing rules absolutely, leave the ground better than found, and pay rent in solved problems. The program described here begins the way everything in this family begins: with existing parts, existing avenues, and the arithmetic of assembling what is already paid for.

The graduate this school produces does not exist yet. The ground that will produce them does. That asymmetry is an invitation.

The Open Source Firefighter

Software Requirements & Development Roadmap

v1.0 · August 2026 · Public working document · OSF-4

*Document ID: **OSF-4** · one document of the Open Source Firefighter specification family.*

Andrew Morton · Palmer Lake, Colorado

This work is dedicated to Emily Barker, Nick Hutcherson, and Sydney Watson of the Rifle Helitack crew, killed in a burnover during initial attack on the Knowles Fire, western Colorado, June 27, 2026 — and to the two crewmates who survived it, and to all wildland firefighters who stand between fire and the rest of us.

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0. How to read this document

This is the software lane of the Open Source Firefighter. The specification documents are written to hold still; this one is written to move. Firmware architecture, language choice, libraries, and milestone ordering will all change as real implementations teach real lessons. **The specification is the contract; this is the plan.** Where they disagree, the specification and the register govern.

It is self-contained. A competent embedded developer — or an AI coding agent working from this text — should be able to read this alone, set up a development environment for the cost of two dev boards, and produce mergeable work without first reading the whole family. Where deeper authority is needed, this document names the source (the specification's vocabulary appendix for the wire vocabulary, the main specification for coordination behavior and the charter) rather than restating it.

Scope boundary, stated once: the software does not detect fire, locate fire, or aim anything at fire. It manages a shared water resource, coordinates turns among neighbors, reports honestly, and stays out of the operator's way. Detection events, where sensors produce them, are *reported* — never acted on offensively. This is a charter requirement, not a v1 simplification.

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1. The problem, stated for software people

A node is a box on a wall with a controller, a low-power mesh radio, a few sensors, and a small number of motorized valves that put water on a roof and the ground around it. Many such boxes exist in a neighborhood. They share a water supply that cannot serve all of them at maximum draw simultaneously.

The software in each box must, on its own hardware and without any server:

1. **Know its own state** — supply pressure, zone positions, power condition, faults — and act on it.
2. **Announce itself** on the mesh so neighbors know it exists and what it can do.
3. **Negotiate a rotating schedule** so aggregate draw stays inside the supply while no structure goes uncovered too long during the ember-cast window.
4. **Obey a scoped, signed, expiring directive** from an incident commander when one arrives, and operate sanely when one never does.
5. **Degrade honestly** — radio gone, water gone, neighbor vanished, battery sagging — to a defined lower capability rather than an undefined failure.
6. **Take the operator's word over its own** wherever a physically present human touches it.
7. **Record what it did** so it can be verified by someone who does not trust the box.

The hardware is not the hard part and neither is the radio; both are commodity and both are solved. The hard part is (3), (5), and (7): distributed scheduling under partial information with nodes that come and go, degradation that stays predictable when assumptions break, and a record that means something without a trusted third party.

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2. The layer model — a rough OSI stack

The build follows a rough OSI layering, deliberately. This is not academic tidiness; it is the mechanism that makes the system's central architectural promise enforceable. The coordination protocol must remain transport-independent — no revision may bind it to a specific physical, link, or network technology (main specification, coordination layer). Layering is how that promise is kept structurally rather than by good intentions: if coordination

logic cannot name a radio, it cannot depend on one.

OSI	OSF layer	What lives here	Module
1–2 Physical / Data link	**Physical & link**	Radio hardware, framing, the wire itself. LoRa today; anything tomorrow.	`hal/`, `link/`
3 Network	**Network**	Mesh addressing and relay. Meshtastic protocol at residential class; agency MANET at tactical class.	`link/`
4 Transport	**Envelope**	The compact binary envelope, fragmentation discipline, best-effort delivery semantics.	`proto/`
5–6 Session / Presentation	**Vocabulary & trust**	The closed verb enumeration, quantized field encoding, signing and verification, publication policy.	`proto/`, `policy/`
7 Application	**Coordination & operation**	Schedule negotiation, node state machine, degradation behavior, operator interface, ledger.	`coord/`, `state/`, `ui/`, `ledger/`, `app/`

The practical consequence: swapping LoRa for a different radio should touch layers 1–3 and nothing above. If a radio change requires editing `coord/`, the layering has been violated and the violation is the bug — not the radio.

Two departures from textbook OSI, both intentional:

- **The trust layer sits at 5–6, above transport, not bolted to the application.** Signing and publication policy are properties of the *boundary*, so they belong where every message must pass regardless of which application-layer feature produced it. This is why `policy/` is a chokepoint (§5).
- **There is no session state in the usual sense.** Delivery is best-effort by design; nodes exchange soft-state announcements with expiry rather than maintaining connections. A missed message is a normal event, not an error, and supply-pressure physics is the fallback coordination mechanism when messages do not arrive.

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3. Fixed invariants — what software may not violate

From the charter and the specification. These are not implementation preferences; they are properties an implementation must preserve to be called compliant. Breaking one is a fork, not a contribution.

#	Invariant	Software consequence
I-1	**The system operates when fully isolated.** No mandatory external network, cloud, account, subscription, or license server for any core function. *(Charter Clause 2, absolute.)*	No boot-time network dependency. No feature behind a remote call. Provisioning is local. A node that has never seen the internet is fully functional.
I-2	**The inbound command vocabulary is closed and enumerated.**	A verb outside the enum has no representation and cannot be constructed, transported, or acted on. Enforced at the type level, not by validation-after-parse.
I-3	**Translation is outbound-only, at the boundary.**	The mesh never carries a rich external format. CoT/CAP/Contact-ID rendering happens at a gateway, above coordination, never between nodes.
I-4	**The operator physically present prevails.** *(Charter Clause 7.)*	No software configuration may disable an operator control. No automated lockout of an operator's own resources. Conflicts at the point of use resolve to the operator — in code, not in

		documentation.
I-5	**No autonomous suppression or targeting.**	The software never decides what to attack. It decides when it is this node's turn to wet its own assigned coverage.
I-6	**Transport independence.**	Coordination imports an interface, never a radio. See §2.
I-7	**Failure degrades to baseline, never to worse-than-nothing.**	Every failure path has a defined, tested destination state. Loss of power drives auto-return valves closed by physics; software must not fight it or assume it can prevent it.
I-8	**The record is verifiable without trusting the node.**	Ledger entries signed and chained. A node that lies must be detectable by someone holding only the published chain.

4. Compliance boundaries to design against

Read this before you build, not after. Nothing here is difficult to accommodate if you know about it early; several items are expensive or impossible to retrofit. This section exists because an implementer can otherwise spend months on a design that is unshippable for a reason no software person would think to check.

Orientation, not legal advice. Requirements vary by jurisdiction and change between editions. Per the family's adoption-by-reference discipline, verify against the issuing authority's current edition for your location — this document deliberately does not restate values that will age.

4.1 The three traps that bite hardest

1. Do not accidentally become a fire alarm system. This is the big one. A device that detects fire conditions and reports them to a monitoring station is, in most jurisdictions, a *fire alarm system* — which pulls in NFPA 72, product listing (UL 864 class), monitoring-station standards, and an entirely different regulatory posture with certification costs to match. OSF sits deliberately outside that boundary: it is a water-application resource-coordination system that reports its own operational state. **The distinction to protect in software:** the system reports *what it is doing* (zone open, supply degraded, node in distress), not *that a building is on fire*. Detection events, where present, are situational telemetry for coordination and command awareness — not alarm signals to a monitoring authority. If a feature request pushes toward "notify the alarm company that there is a fire," that request changes the product's regulatory class. Escalate it; do not implement it quietly.

2. Cross-connection and backflow. The node connects to a potable supply and can create a path from irrigation-side plumbing back toward the main. Backflow prevention and cross-connection control are enforced by water purveyors and plumbing codes essentially everywhere, and a water district can refuse an installation over it. This is primarily a hardware and installation matter (Annex A), but it lands in software one specific way: **do not design control logic that assumes it can be the safety barrier.** Backflow protection is a physical device. Software must never be positioned as the thing preventing contamination, and no failure mode may depend on the controller behaving correctly to protect the potable supply.

3. Radio equipment authorization. The mesh radio is an intentional radiator. In the United States, equipment marketed with an intentional radiator requires FCC equipment authorization; elsewhere the equivalent regime applies. **The practical path is a pre-certified radio module, unmodified in its RF characteristics** — changing transmit power, antenna, or duty cycle beyond the module's certified conditions can void the certification of the finished device. In software: power and duty-cycle limits are not tunables to optimize freely. Treat the certified operating envelope as a constraint the code respects, and keep region-dependent parameters in one clearly marked place (§4.2).

4.2 Radio and spectrum

- **Band and power follow the jurisdiction, not the specification.** The reference implementation uses the North American 902–928 MHz ISM allocation; other regions use their own (863–870 MHz in Europe, for example) at their own permitted power. Some regions impose **duty-cycle limits that materially affect protocol design** — a heartbeat cadence legal in one region may not be in another. Build region parameters as configuration with enforced ceilings, not constants scattered through the code.

- **Stay off public-safety spectrum.** Interoperating with agency radio happens at the network interface (Ethernet/IP at a gateway), not by transmitting on public-safety frequencies. Licensed-band operation is a different regulatory world and is not part of this design.
- **Stay off amateur spectrum, and understand why.** The trust layer signs every outbound message. Encryption and obscured messaging are prohibited on amateur allocations — in the United States under FCC Part 97.113(a)(4) — and amateur operators must identify by call sign at intervals. A signing node and the amateur service are therefore mutually exclusive by regulation. Do not build an amateur-band mode. If a downstream implementation wants one, it must disable signing, and a node with signing disabled is not an OSF node.
- **Tactical-class radios are their own regime.** Agency MANET radios may carry export-control and procurement restrictions. The node integrates with them at the data interface and does not embed them.

4.3 Electrical and installation

- **Industrial control panel and enclosure practice** (UL 508A class listing, NEC/NFPA 70 wiring, enclosure type ratings) governs the box. This is Annex A territory; the software consequence is small but real: **do not create a design that requires the panel to be opened during normal operation**, and do not assume access to internals a listing would require to stay closed.
- **Low-voltage control is where the software lives.** Keeping control and communication in the low-voltage domain keeps most electrical compliance in the hardware lane, where it belongs.

4.4 Interfacing with government and public-safety systems

These apply when a deployment publishes into an official system — not to a standalone neighborhood domain, which is the common case.

- **Public alerting is authorization-controlled.** CAP is the format for public alerts, but **the ability to originate a public alert into a national alerting system is granted, not assumed** — origination authority is held by qualified authorities under a formal process. A node or gateway may *produce* CAP-formatted output; it may not self-authorize into an alerting network. Design the boundary so an authorized operator's system consumes OSF output, rather than OSF injecting into an alerting network directly.
- **Incident-command interoperability follows NIMS/ICS conventions.** Where the system presents information to incident command, it should fit existing command structure and terminology rather than inventing a parallel one. The tactical picture publishes as CoT.
- **Federal systems bring federal requirements.** If a deployment ever hands data to a federal system, expect requirements around cryptographic module validation (FIPS-class), information-security controls, records handling, and accessibility for any government-facing interface. Do not build toward this speculatively — but **choose cryptographic primitives that have validated implementations available**, so the door stays open. Selecting an exotic cipher because it is elegant closes it permanently.
- **Records and evidence.** The ledger is designed to be evidentiary (I-8). Where output may become part of an official record — a utility's mitigation record, an after-action report — integrity and retention matter more than convenience.

4.5 Privacy and data

- **The node has no camera and no recording microphone by design**, and that is a compliance asset, not a limitation. Imagery and audio recording pull in privacy law, retention obligations, and consent questions a fire-defense appliance has no reason to accept. Acoustic sensing, where used, produces *derived state* (machinery present, engine running), never a recording. **Do not add an imagery or audio-capture path.** If one is ever proposed it is a specification change with a compliance review, not a feature.
- **Position data is coarse by design** — adequate for coordination and command, not precise tracking of people.
- **Detection events report source, bearing class, and confidence class** — deliberately minimal, deliberately not personal.

4.6 Licence inheritance from upstream dependencies

A dependency's licence is a constraint you inherit, not a choice you make. The reference mesh implementation is Meshtastic, whose firmware is distributed under **GPLv3**. Firmware that links into, embeds, or derives from that codebase is GPLv3 by operation of that licence, and no downstream decision changes it.

The boundary is worth stating precisely, because it is narrower than it first appears:

- **Captured:** firmware built on or linked against the upstream mesh implementation.
- **Not captured:** the OSF specification documents themselves, the coordination protocol, the closed vocabulary, and the envelope format. These are specifications. A specification describing what a message contains is not a derivative work of any program that happens to carry that message. A clean-room implementation written from the specification alone inherits nothing.

Two practical consequences. **Check the licence of every dependency before adopting it**, and prefer dependencies whose terms are compatible with the project's own licence selection. And **keep the protocol implementation separable from the transport implementation** — which the layer model (§2) already requires for transport independence, and which turns out to serve a licensing purpose as well. A coordination layer that cannot name a radio also cannot inherit that radio's licence.

Upstream trademark terms are a separate matter from upstream licence terms and bind independently: a project may be free to use the code and still not free to use the name.

4.7 The rule of thumb

If a proposed feature would make the system (a) tell an outside authority that there is a fire, (b) originate a public alert on its own authority, (c) record imagery or audio, (d) act on the supply in a way that only software prevents from contaminating it, or (e) transmit outside the radio module's certified envelope — **stop and escalate.** Each changes what the device legally is. None are necessary to do the job.

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5. Decomposition — the code map

The layering below mirrors §2, which is what makes the invariants enforceable by structure rather than discipline. Names are suggestions; boundaries are not.

Module	Responsibility
`app/`	Main loop, task scheduling, wiring the layers together
`ui/`	Operator inputs (debounce, deliberate-press timing) and indicator state
`policy/`	Publication policy — the trust boundary. Nothing leaves except through here
`ledger/`	Signed, chained snapshots
`coord/`	Schedule negotiation, neighbour table, domain view
`proto/`	Envelope, codec, verb dispatch, vocabulary definitions generated from Appendix C
`link/`	Transport interface — one implementation per radio, one for simulation
`state/`	Node state machine, timers, fault model
`hal/`	Hardware abstraction: valves, sensors, power, controls, indicators — with a real implementation and a simulated one

Module names are a suggestion. The boundaries between them are not.

Dependencies point downward only. `coord/` may call `proto/` and `state/`; `proto/` may not know `coord/` exists. `hal/` knows nothing above it. This is what makes I-6 structural.

`policy/` is a chokepoint, not a library. Every outbound byte passes through it. It is the only module permitted to emit. If a developer can transmit without traversing `policy/`, that is a defect of the highest severity — it means the trust layer is advisory, and an advisory trust layer is not one.

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6. The development environment — start with two dev boards

The full physical node involves plumbing, valves, an enclosure, and a wall. None of that is needed to write, run, or test the software. Valves and flow sensors are trivially simulated; the interesting behavior is entirely in scheduling, degradation, and protocol.

6.1 The simulation contract

Everything physical sits behind a hardware abstraction layer with two implementations. This is the single most important architectural decision in the project, because it is what lets a contributor with no hardware do real work.

The interface is deliberately narrow. It exposes, and exposes only: **zone position** as read from independent feedback rather than from commanded state; **zone command**; raw readings for **supply pressure**, **reservoir level** where fitted, and **flow**; **power condition** (mains, battery, low, critical); the state of each **operator control**; **indicator output**; and a **monotonic clock**. Nothing above `hal/` may reach the hardware by any other route.

Two implementations satisfy it. One drives real hardware. The other simulates it — and the simulator is where most development happens, so it deserves as much care as the driver.

The simulated implementation should be a small physical model, not a stub: a valve that takes realistic seconds to travel and *can fail mid-travel*; a supply whose pressure sags when simulated neighbours draw; a battery that discharges; a flow sensor reading zero on a kinked line.

The simulator should be able to lie and to break. A simulator that models only the happy path produces software that works only on the happy path. `hal_sim/` supports scripted faults — valve seizes, feedback disagrees with command, pressure collapses, battery falls off a cliff, sensor reads implausible values — because embedded failure is the credibility mechanism of this project (§14) and belongs in the harness, not only in the prose.

6.2 Two build targets from day one

- **Host build (native).** Runs on a laptop. Many simulated nodes in one process or several, over a simulated link with configurable loss and latency. Ninety percent of development happens here: fast iteration, real debuggers, property tests, fifty-node domains.
- **Device build (ESP32).** Same source, `hal_real/` and the LoRa link. Where timing, memory, power, and radio reality get checked.

CI runs the host build on every commit; the device build on a cadence.

6.3 Minimum development hardware

Purpose	Item	Note
Node compute	Any ESP32 dev board	Two is meaningfully better than one — a mesh cannot be tested with a single radio.
Radio	LoRa module for the local ISM allocation	Combined ESP32+LoRa boards are ideal. Band and power follow the jurisdiction (§4.2).
Everything else	*(nothing)*	Valves, sensors, pumps, enclosure, plumbing: simulated.

A contributor who owns nothing can still work productively on the host build alone.

6.4 Language and toolchain

Not yet fixed, deliberately — this should be decided and defended by the first serious implementer, not inherited from a document. Constraints the choice must satisfy:

- Runs on ESP32-class hardware within its memory budget.
- Compiles for device and host without source forks.
- Has a workable property-based testing story (the §3 invariants want property tests, not examples).
- Approachable enough that a competent hobbyist can contribute — the adoption model depends on it.
- **Has validated cryptographic implementations available** (§4.4).

C or C++ with ESP-IDF is the conservative answer. Rust is credible and buys real safety. MicroPython lowers the contribution barrier at a cost in determinism. **Pick one, write the reasoning down, make it a decision of record.**

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7. The node state machine

State	Meaning	Leaves when
`BOOT`	Power-on, self-test, hardware inventory	Self-test completes → `IDLE`, or fault → `FAULT`
`IDLE`	Armed and quiet. Heartbeating. Valves closed.	Activation (command, local, or schedule) → `ACTIVE`; disarm → `DISARMED`
`DISARMED`	Operator has locked out automatic operation. Still heartbeats and reports.	Operator re-arms → `IDLE`
`ACTIVE`	Protection running. Executing this node's slot.	Deactivation, expiry, or supply loss → `IDLE/DEGRADED`
`YIELDED`	Deliberately holding off — another node's turn, or a directive said cease.	Slot arrives, or directive expires → `ACTIVE/IDLE`
`DEGRADED`	Reduced capability. Supply gone, zone failed, or battery low.	Condition clears → prior state; worsens → `REPEATER`
`REPEATER`	No water capability. Still carries mesh traffic.	Water capability restored → `IDLE`
`ISOLATED`	No mesh contact within timeout. Autonomous conservative operation.	Neighbor heard → rejoin coordination
`DISTRESS`	Operator signaled distress. Latched full protection, status published.	Operator clears
`FAULT`	Cannot perform core function; reporting the reason.	Fault cleared or serviced

`ISOLATED` is not an error state. It is a normal operating mode with conservative defaults, and it must be *good*. Most nodes will spend most of their lives with intermittent mesh contact. A node that panics when alone is a broken node.

`REPEATER` is the graceful-degradation showcase. A node whose water has failed is still worth having on the wall because it carries its neighbors' traffic.

`DISTRESS` latches. It does not time out; a human clears it.

Every transition is a ledger event (§13).

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8. The wire protocol and vocabulary

the specification's vocabulary appendix (Appendix C) is the authority for field layouts, quantization, and byte-level examples. The vocabulary itself is reproduced here because it is the data dictionary, the dispatch switch, and most of the software design — a developer should not need a second document open to know what the system can say.

8.1 Envelope constraints

The wire format is a compact binary envelope sized to the most constrained transport in the family. LoRa mesh payloads live in the ~200-byte range. The signature is the large field (Ed25519 class, 64 bytes) and is non-negotiable — the trust layer rides in the envelope. **Realistic body budget after header and signature: 60–100 bytes.** That number is the discipline; every verb must fit it.

Readings on the wire are **single-byte quantized classes**, not precise values: coordination needs "pressure sagging," not 47.3 psi. Full-resolution values stay in the local log.

8.2 The V1 verb list

Group	Verbs	Notes
Presence	`HEARTBEAT`	status flags, battery class, supply class, capability bits, coarse position
State	`ARM`, `DISARM`, `ACTIVATE`, `DEACTIVATE`, `ZONE_OPEN`, `ZONE_CLOSE`, `FAULT`, `MAINT_REQ`	`FAULT` carries a fault-code enum

Supply	`PRESSURE`, `FLOW`, `LEVEL`	quantized single-byte classes
Coordination	`SCHED_STATE`, `SCHED_PROPOSE`, `YIELD`, `RESUME`	schedule negotiation per main specification
Command	`CEASE_OPS`, `RESUME_OPS`, `PRIORITY_PROTECT`, `BROADCAST_ALERT`	scope filter, authority, expiry; short, audited vocabulary
Safety	`DISTRESS`	node, position, timestamp — deliberately minimal
Events	`DETECT`, `ENGINE_STATE`, `ENERGY`	detection events (source / bearing class / confidence class); acoustic-derived machinery state; power transitions
Ledger	`SNAPSHOT_COMMIT`	hash chain link; the payload is the commitment

Roughly two dozen verbs. **The vocabulary is closed and short.** Additions go through charter governance; smallness is a protected property of the protocol, not a v1 limitation.

Capability bits ride in the heartbeat (supply class, stored water, pump control, detection payload, gateway role). This is how the domain knows what each node is without a registry, and how directive scope filters resolve locally. Get this right early; much depends on it.

8.3 Implementation requirements

- **Generate, do not transcribe.** The Appendix C enum table should compile to the header mechanically. Hand-copying a wire vocabulary is how two implementations silently diverge.
- **The verb enum is the dispatch switch.** A `default:` case doing anything but count-and-drop is a bug.
- **Assert the byte budget in tests.** A verb exceeding its maximum encoded size must fail at build or test time, not on a congested mesh at 0300.
- **Round-trip fuzz everything.** Encode/decode identity for all verbs, plus decode-arbitrary-bytes-without-crashing. The radio will deliver corrupt frames; the parser must be boring about it.
- **Golden vectors.** Appendix C specifies byte-level examples for `HEARTBEAT`, `ZONE_OPEN`, `CEASE_OPS`, `DISTRESS`. These are the conformance fixtures a second implementation tests against first.

8.4 Coverage math the node computes

Coverage and supply planning is not incidental to the software — a node self-calibrates its delivered supply at commissioning, sizes its zones against it, and reports coverage capability in its capability bits. Implementers should not have to go find these relations. They are reproduced here for the same reason the verb list is: they are the working data dictionary of this part of the system.

The Supply & Coverage Planning Table (OSF-2, Appendix A) is the authority for planning figures, head classes, and derated field values. This section carries only the relations a node computes.

Quantity	Relation	Notes
Effective radius at pressure	`r ≈ r_design × √(P / P_design)`	Design pressure for structure-protection heads is 50 psi. Below ~40 psi the radius collapses and derived spacing no longer overlaps.
Flow at pressure	`q ≈ q_design × √(P / P_design)`	Same square-root pressure relation.
Head-to-head spacing	`S = r`	The baseline. Spacing equals *radius*, never diameter.
Wind-derated spacing	`S = 0.9 × r` (hard wind `0.8 × r`)	Most fire days are wind days; derate by default.
Precipitation rate	`PR = 96.25 × GPM ÷ S²`	Inches per hour, square spacing, GPM per head. Published head data assumes zero wind.
Time to target depth	`t = (depth ÷ PR) × 60`	Minutes. Planning targets are light dampening through pre-front soak.

Sector wetted area	`A = (arc ÷ 360) × π × r²`	For arc-limited heads; the basis for coverage-fraction calculations.
Stored-water duration	`t = tank_gallons ÷ zone_GPM`	The number that makes rotation necessary rather than optional.

Two implementation notes with teeth. **A node must not report coverage it cannot deliver** — computed coverage is derived from *measured* supply at commissioning, never from the planning figures, and where the two disagree the measurement governs. And **the arc matters to the numbers**: an arc-limited head applies its full flow to a fraction of the circle, so precipitation rate rises as arc narrows. A coverage calculation that ignores arc will under-report application rate on exactly the part-circle heads the placement doctrine prefers.

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9. Coordination — the scheduling problem

This is the invention, and the most interesting code in the project.

The problem: N nodes share a supply sustaining fewer than N at full draw. Each needs its coverage wetted often enough that no structure has an extended dry gap during the ember-cast window. Nodes join and leave. Messages get lost. There is no coordinator and there must never be one.

Inputs, arriving via heartbeat and coordination verbs: each node's supply class, current supply state, coverage pattern, activation state, battery condition.

Output: each node's own decision about whether it runs right now. **Not** a global schedule computed and distributed — a local decision that, taken by every node from shared information, produces a sane global pattern.

Properties the implementation must have (these are the acceptance tests):

1. **Supply safety.** Aggregate committed draw stays within sustainable capacity. Violating this is the one unforgivable failure — it empties the tank and defeats everyone.
2. **Coverage floor.** No participating structure exceeds a maximum dry interval while the domain is active.
3. **Convergence under churn.** Nodes appearing, vanishing, reappearing must not produce oscillation, thrash, or a stuck domain.
4. **Graceful partial participation.** A domain where only some structures have nodes still works for the participants — the honest and common case.
5. **No coordinator, no election.** Leader election is a failure mode waiting for a fire. Prefer algorithms needing no leader at all.
6. **Physics as backstop.** A node that misses coordination and draws anyway is answered by pressure sag and must yield to it. Falling back to hydraulic reality is designed behavior, not accident.

In one embodiment, the schedule is computed as follows (this is one method of satisfying the properties above, not the only one): each node computes a deterministic priority from its identity, supply class, and time-since-last-wetted, seeded so all nodes computing from the same neighbor table reach the same ordering. Nodes take slots in that order, sized to the supply budget. Time-since-wetted rising monotonically prevents starvation. Stored-water nodes bias toward running during municipal pressure sags, since they are not competing for the same resource. Deterministic-from-shared-state means convergence without negotiation traffic in the common case; `SCHED_PROPOSE/YIELD` handle disagreements.

Build this in simulation first, fifty nodes on a lossy link. Far cheaper to find an oscillation with a hundred simulated nodes than three real ones on a bench.

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10. Trust layer and publication policy

`policy/` implements the boundary between what the node knows and what the world sees.

Outbound: nothing transmits except through this module. It is where the enumerated-message discipline becomes real, where signing happens, and where — at a gateway only — rendering into CoT, CAP, or Contact-ID occurs. A non-gateway node never speaks an external format.

Inbound: commands are verified before they are anything else. A directive carries issuing authority, scope filter, and expiry. The implementation must:

- **Verify the signature** before the message influences any state.
- **Resolve scope locally** against this node's capability bits — no registry lookup, no asking permission.
- **Honor expiry and resume automatically on it.** A directive not refreshed dies. *A stale silencing order must not*

outlive the attention of the commander who issued it. This is a safety property; implement the timer before the directive handling.

- **Never allow an inbound message to disable an operator control** (I-4). A directive constrains automated behavior; it does not reach the panel.

Key handling is an open design question (§17), not a solved one. What is settled: verification must be possible offline by a node that has never contacted any authority server (I-1), and primitives should have validated implementations available (§4.4).

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11. The degradation ladder

Each rung is a defined behavior with a test. This table *is* the graceful-degradation architecture in software terms.

Failure	Node behavior	Domain behavior
Mesh lost	→ `ISOLATED`; conservative autonomous cycle	Remaining nodes reschedule around the absence
Neighbor vanishes	Drop from neighbor table after timeout	Reschedule; no stall waiting for the missing
Supply pressure sags	Yield draw; report `PRESSURE`	Others sense the same and back off; physics coordinates
Water supply fails	→ `DEGRADED`, then `REPEATER`	Node keeps carrying traffic; coverage gap reported
One zone fails (clog, kink, seized valve)	Report `FAULT` with code; continue on remaining zones	Coverage map updates; neighbors may compensate
Valve feedback disagrees with command	Trust feedback, report fault	Never report protection that is not physically happening
Battery low	Shed load: reduce cycle, prioritize reporting over wetting	Domain sees reduced capability in heartbeat
Power lost	Auto-return valves close by physics; nothing to do	Node disappears; domain reschedules
Directive missed	Continue local operation, yield via pressure	Best-effort delivery is the design, not a defect
Controller crash	Watchdog reset → `BOOT` → self-test	Brief absence, then rejoin

Design rule: every one of these degrades to *the unprotected baseline the world already accepts*, and preferably a bit better. No failure may produce a state worse than having no system at all. A proposed feature that can fail into worse-than-nothing does not belong in the node.

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12. Operator interface

Per the capability requirements (specification and Inspector's Draft §3.4): a physically present operator can initiate protection, lock out automation, command each zone independently, and signal distress — in structural gloves, in darkness, without instruction, without a personal device, and without any network or account.

Software requirements regardless of the physical control form:

- **Debounce everything**, then apply intent timing. The distress signal requires a deliberate press of roughly three to five seconds; brush contact and bumps must not trigger it.
- **Controls are inputs the controller reads.** They are not emergency-stop devices and no safety category is claimed. The hardware kill is the disconnect. Never write software as though a button bypasses it.
- **Indicators report physical reality, not intent.** A zone lamp shows *valve feedback*, not *commanded state*. If those disagree, the lamp tells the truth and a fault is raised. This single rule prevents the most dangerous class of lie the system could tell.
- **State readable at a glance without a display:** power, radio link, readiness, fault, active, distress.
- **No software configuration may disable an operator control** (I-4). There is no settings flag for this, and there must not be a code path that could grow one.

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13. The ledger

Signed, chained snapshots — `SNAPSHOT_COMMIT`, where the payload is the commitment.

- Every state transition, directive received, zone command, and fault is an event.
- Events batch into periodic snapshots; each snapshot commits a hash chaining to the previous.
- The chain makes an after-action report evidentiary rather than narrative — a record whose integrity does not depend on the credibility of whoever holds the box.
- **Complement, not substitute, for physical verification.** The ledger attests what was *commanded*; a thermal overflight attests what physically *happened* (Annex A). Make the two easy to correlate: timestamps and zone identities that line up with an external observation.

Implementation notes: flash is small and write cycles finite. Snapshot cadence is a tunable. Chain verification must be possible from the published chain alone, by a third party with no access to the node.

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14. Test strategy

Property tests for the invariants. The §3 table is a test list. *No path leaves valves commanded open through a power-loss sequence. `DISARM` is reachable from every state. An unenumerated verb cannot be constructed. Nothing transmits without traversing `policy/`.* These are properties, not examples.

Conformance fixtures. The Appendix C golden vectors plus a growing corpus of encoded messages. The goal is the interoperability property: *two independent implementers working from the vocabulary table produce interoperable nodes without coordination.* That claim needs a test suite to be true.

Scenario tests — the project's unusual asset. The Deadman Fire scenario (OSF-2.1) is a complete, physically honest, hour-by-hour operational narrative with realistic embedded failures. **It should be encoded as an automated integration test:** a simulated domain matching its geometry, driven through its timeline, with its failures injected at the right moments — the seized valve, the flat battery, the burnt-through hose, the missed directive — and assertions that the domain behaves as the scenario describes. When the software passes Deadman, it has been tested against a scenario a fire chief already read and called realistic.

Fault injection is mandatory. Per §6.1 the simulator must break. A suite that exercises only success has verified nothing about a system whose credibility rests on failing gracefully.

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15. Roadmap

Ordered by dependency, sized to be individually completable, each with an acceptance criterion that can be demonstrated rather than argued.

#	Milestone	Acceptance criterion
M0	**Skeleton and simulation harness.** Repo, host build, CI, `hal/` interface with `hal_sim/`, scriptable faults.	`make test` runs on a laptop with zero hardware. A simulated node boots and logs a main loop. Faults injectable from a script.
M1	**State machine.** §7 implemented with timers and fault model. No radio.	Property tests pass for the §3 invariants. Every state reachable and exitable in simulation.
M2	**Wire format.** `proto/` generated from Appendix C; envelope, codec, dispatch.	Golden vectors for `HEARTBEAT`, `ZONE_OPEN`, `CEASE_OPS`, `DISTRESS` match the specification's vocabulary appendix byte-for-byte. Round-trip fuzz clean. Byte budget asserted in CI.
M3	**First real hardware.** ESP32 + LoRa, `hal_real/` with simulated plumbing.	Two physical boards exchange heartbeats and decode each other's capability bits.
M4	**Coordination.** §9 algorithm in simulation at scale, then on real radios.	50 simulated nodes on a lossy link: supply budget never exceeded, coverage floor held, converges under churn. Then three real nodes rotating correctly.
M5	**Trust layer.** Signing, verification,	Unsigned and expired directives

	`policy/` chokepoint, command broadcast with scope and expiry.	rejected. Scope resolves locally. Automated test proves nothing transmits outside `policy/`. Expiry auto-resumes.
M6	**Operator interface and degradation.** §11 ladder and §12 interface complete.	Every rung of the degradation table demonstrated with injected faults. Indicators track feedback, not command.
M7	**Ledger and event interface.** Chained snapshots; outbound rendering at a gateway.	Tampering with any historical event detectable from the published chain. An external consumer reads the domain picture.
M8	**Deadman as integration test.** The scenario encoded and passing.	The simulated domain runs the Deadman timeline with its failures and behaves as written.
M9	**Conformance and second implementation.** Published test suite; an independent implementation interoperating.	Two implementations, written from Appendix C without coordinating, exchange traffic and coordinate a shared schedule.

M0 through M2 are a solo weekend-scale effort for someone competent and require no hardware. That is deliberate: the barrier to a first useful contribution should be a laptop and an afternoon.

M4 is the research risk. Everything before it is engineering; the scheduling algorithm is where genuine design work lives. Budget accordingly and expect to throw the first one away.

M9 is the real finish line for the software lane, because it proves the claim the whole specification rests on: that the documents are sufficient for independent implementation.

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16. Getting started

1. **Read this document, then the specification's vocabulary appendix (Appendix C).** Together they are enough for M0–M2.
2. **Skim §4 before you design anything.** The compliance boundaries are cheap to respect early and expensive to retrofit.
3. **Skim the main specification's coordination sections** before M4; the Deadman scenario before M8, or earlier if you want to understand what you are building.
4. **Do not start with the radio.** Start with `hal/` and the state machine on the host build. Radio-first is the most common way embedded projects stall.
5. **Do not start by optimizing.** The node has enormous time budgets by software standards — a valve takes seconds to travel, a schedule slot is minutes. Clarity beats cleverness everywhere in this codebase.
6. **Write the decision down when you make one.** Language, scheduling algorithm, key handling, snapshot cadence — these are decisions of record. The next implementer's ability to work depends on knowing why, not just what.

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17. Open questions

Unresolved. A contributor who resolves one well has made a real contribution.

- **Language and toolchain** (§6.4). Needs a decision and a written rationale.
- **Key management and identity.** How does a node get an identity a neighbor will trust — offline, with no authority server, in a community where units are bought at different times by different people?
- **Command authority.** How does a node know an incident commander's signature is an incident commander's, without a registry it cannot reach? *In one embodiment,* the authority public key for an incident is distributed with the incident communications plan by the same means as radio frequency assignments — the analogue of an ICS-205 — and is accepted by nodes for the duration of the incident, bounded by the directive expiry already required by the trust layer. *In a further embodiment,* a domain designates one or more steward keys at commissioning, and an incident authority key is accepted when countersigned by a steward, so that a node can verify command authority entirely offline against material it already holds.
- **Scheduling algorithm** (§9). The deterministic-priority method described in §9 is disclosed as one embodiment

and is expected to be refined or replaced as implementations gather field data; the properties it must satisfy, listed there, are the requirement.

- **Time.** Coordination and expiry need a shared-enough clock. GNSS where present — but what about a node without it, in a valley, after a week?
- **Snapshot cadence and flash budget.** Ledger granularity against device lifetime.
- **Configuration and commissioning.** Local-only by charter — but what is the actual ritual for a homeowner, in gloves, at dusk?

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18. Governance

The protected core — coordination protocol, closed vocabulary, trust layer — may not be bound to any single transport, vendor, or mandatory external dependency by any revision. Vocabulary additions go through charter governance; smallness is a protected property of the protocol.

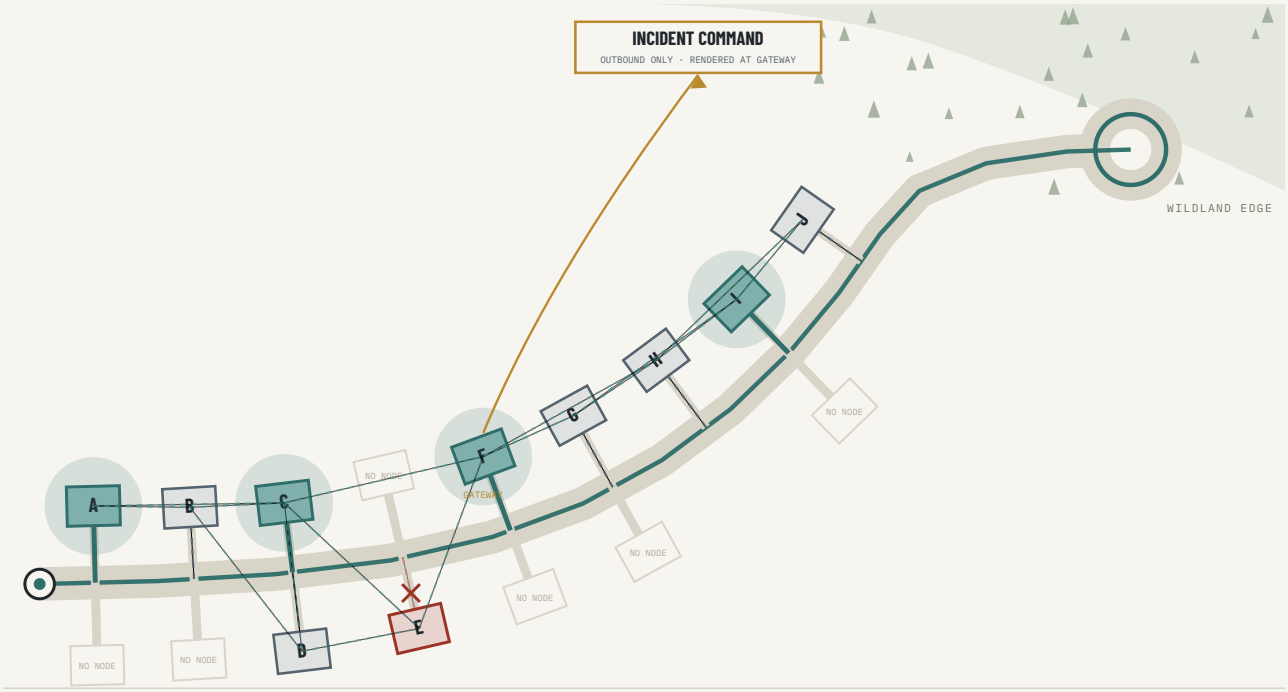
Contributions that implement this document are welcome without ceremony. Contributions that change what the software *must do* are specification changes and belong in the register, not in a commit.

The community infrastructure this project needs — the change-proposal process, the build registry, the conformance claim, the safety-defect channel, and the library of known-good configurations that is the real goal — is described in the Public Roadmap (OSF-3.2, §9, The Development Commons). Software contributors should read it before the first substantial pull request, because where a decision gets recorded matters as much as what the decision was.

This document is expected to change faster than any other in the family. That is its purpose. The specification holds still so that this one can move.

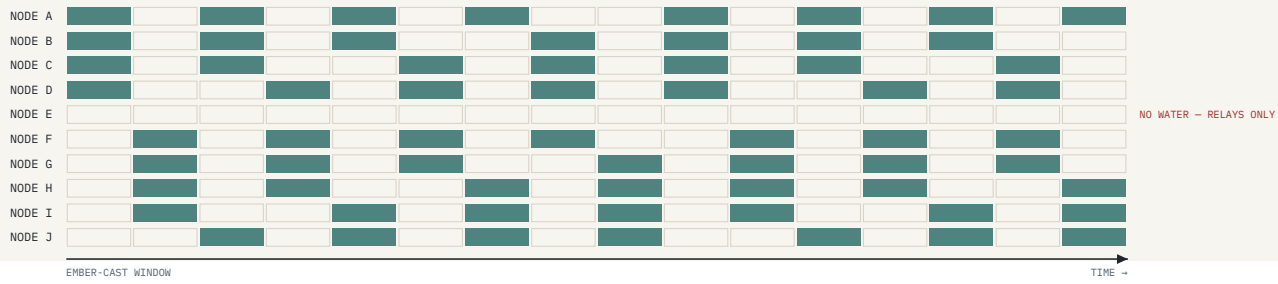
FIG. 1 — THE COORDINATION DOMAIN

A NEIGHBORHOOD SHARING ONE SUPPLY · PLAN VIEW · PARTIAL ADOPTION



THE ROTATING SCHEDULE

EACH NODE DECIDES LOCALLY FROM SHARED STATE · NO COORDINATOR · NO ELECTION · FOUR MAY DRAW AT ONCE



SUPPLY SAFETY
AGGREGATE DRAW NEVER EXCEEDS SUSTAINABLE CAPACITY

COVERAGE FLOOR
NO STRUCTURE EXCEEDS ITS MAXIMUM DRY INTERVAL

PARTIAL ADOPTION
UNEQUIPPED STRUCTURES DO NOT STOP THE DOMAIN WORKING

FIG. 2 — THE LAYER MODEL AND THE TRUST BOUNDARY

COORDINATION NEVER NAMES A RADIO · NOTHING LEAVES EXCEPT THROUGH PUBLICATION POLICY

DEPENDENCIES POINT DOWNWARD ONLY

OSI 7 — coord/ state/ ui/ ledger/ app/

APPLICATION

Coordination · state machine · degradation · operator UI · ledger

OSI 5-6 — proto/ policy/

VOCABULARY AND TRUST

Closed verb enumeration · signing · publication policy

SUBSTITUTION LINE — SWAP ANYTHING BELOW

LORA · MANET · WIRE · WHATEVER COMES NEXT. NOTHING ABOVE CHANGES.

OSI 4 — proto/

ENVELOPE

Compact binary envelope · best-effort delivery

OSI 3 — link/

NETWORK

Mesh addressing and relay

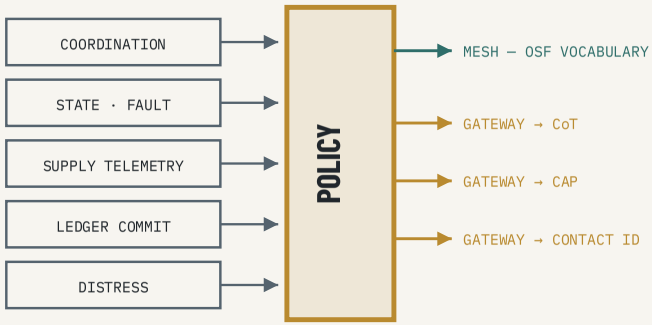
OSI 1-2 — hal/ link/

PHYSICAL AND LINK

Radio hardware · framing · the wire itself

THE CHOKEPOINT

EVERY OUTBOUND BYTE PASSES ONE MODULE



INBOUND

VERIFIED BEFORE IT IS ANYTHING ELSE

1. SIGNATURE VERIFIED
2. SCOPE RESOLVED LOCALLY
3. EXPIRY HONOURED — AUTO-RESUME
4. CLOSED VERB, OR DROPPED

NO INBOUND MESSAGE MAY DISABLE AN OPERATOR CONTROL.

TRANSPORT INDEPENDENCE

COORDINATION IMPORTS AN INTERFACE, NEVER A RADIO.
A LINK CHANGE CANNOT REACH LAYER 7.

PUBLICATION POLICY

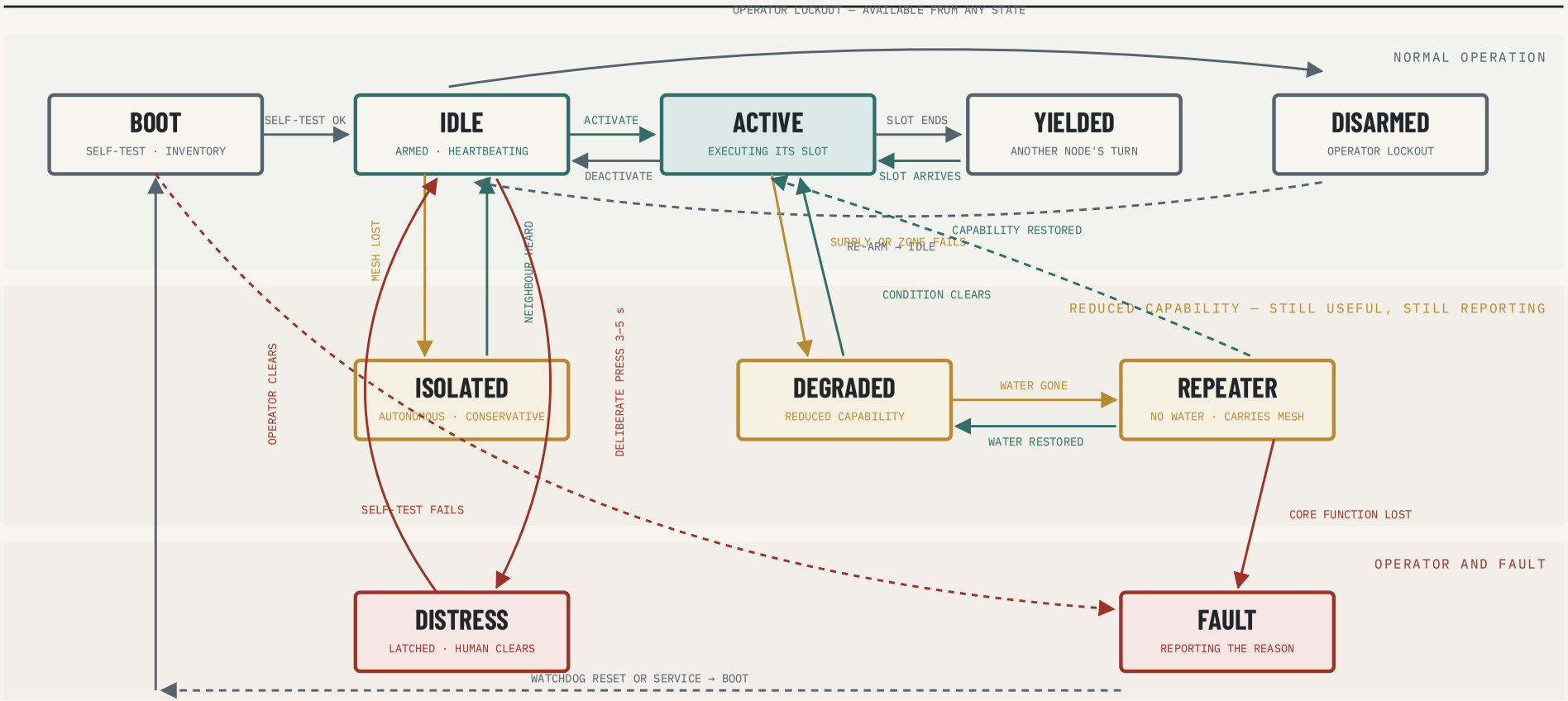
THE TRUST LAYER SITS ABOVE TRANSPORT BECAUSE IT IS
A PROPERTY OF THE BOUNDARY, NOT OF A FEATURE.

NO SESSION STATE

BEST-EFFORT BY DESIGN. A MISSED MESSAGE IS NORMAL;
SUPPLY PRESSURE IS THE FALLBACK.

FIG. 3 — THE NODE STATE MACHINE

EVERY FAILURE HAS A DEFINED DESTINATION · EVERY DESTINATION HAS A WAY BACK



POWER LOSS IS NOT A STATE

SPRING-RETURN VALVES CLOSE BY PHYSICS. SOFTWARE DOES NOT PARTICIPATE AND MUST NOT ASSUME IT CAN.

ISOLATED IS NOT AN ERROR

MOST NODES SPEND MOST OF THEIR LIVES WITH INTERMITTENT CONTACT. A NODE THAT PANICS ALONE IS BROKEN.

REPEATER EARNS ITS PLACE

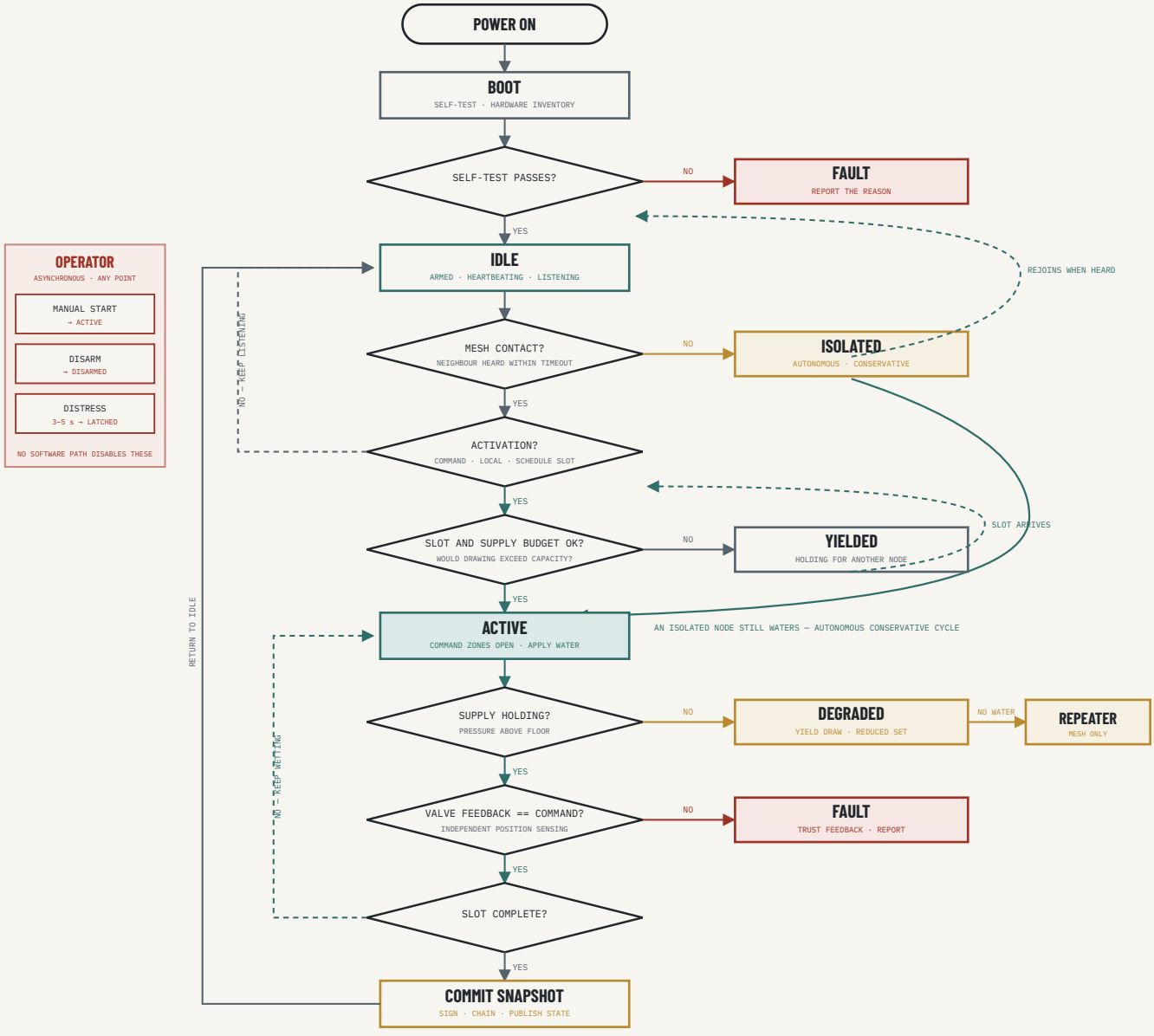
A NODE WITH NO WATER STILL CARRIES ITS NEIGHBOURS' TRAFFIC. IT IS WORTH LEAVING ON THE WALL.

NOTHING IS A DEAD END

EVERY DEGRADED STATE HAS A RETURN PATH. DEGRADATION IS A ROUTE, NOT A TERMINUS.

FIG. 4 — NODE OPERATION, CONTROL FLOW

THE SAME STATE MACHINE READ AS SEQUENCE · COMPANION TO THE STATE DIAGRAM

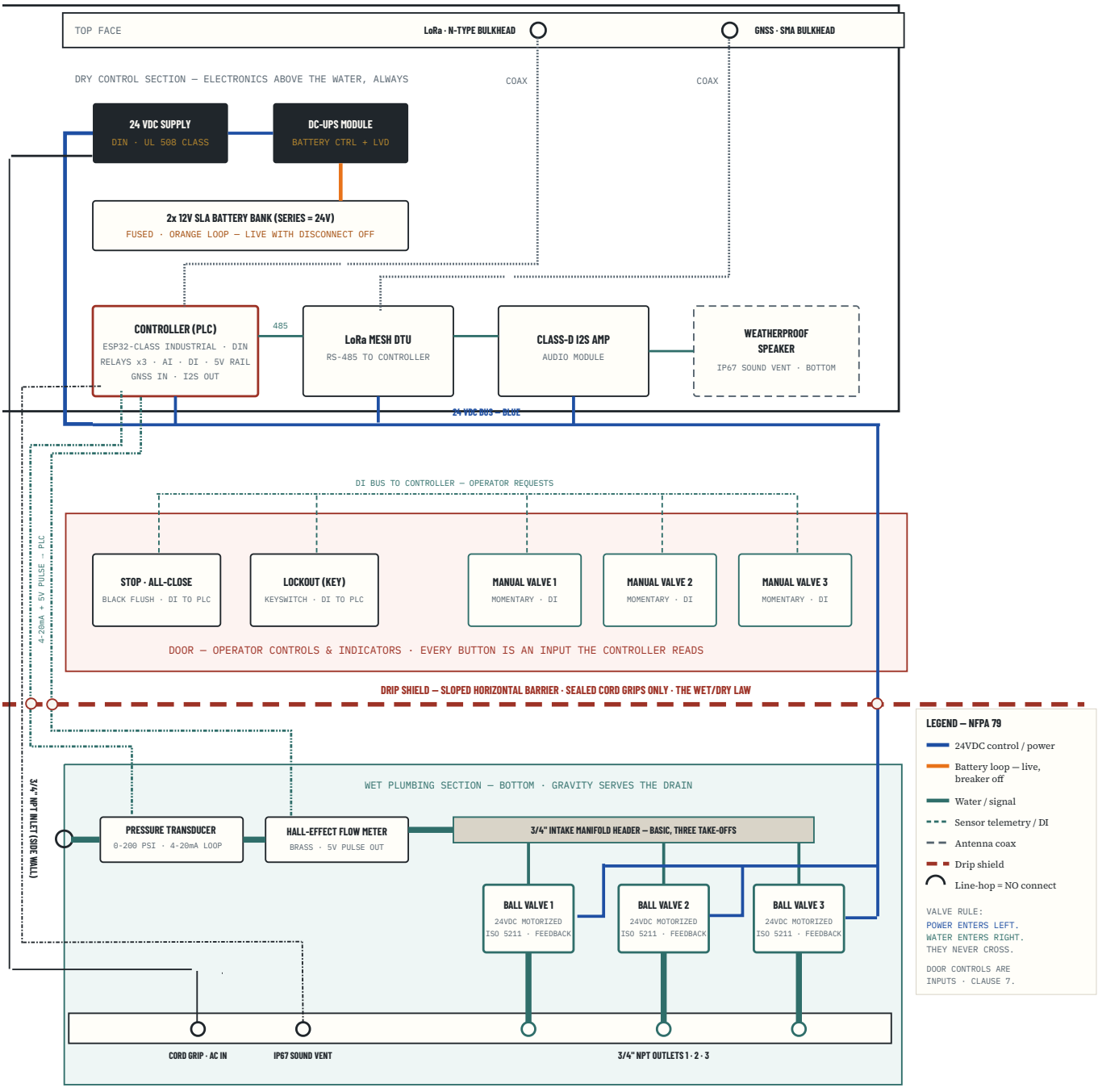


POWER LOSS DOES NOT APPEAR IN THIS FLOW

AUTO-RETURN VALVES CLOSE BY PHYSICS. THERE IS NO SOFTWARE STEP, AND SOFTWARE MUST NOT ASSUME IT CAN PREVENT OR DELAY IT.

EVERY DECISION THAT FAILS HAS A DEFINED DESTINATION, AND EVERY DESTINATION RETURNS. THE FLOW HAS NO TERMINAL FAILURE.

THIS SHEET IS THE ENCLOSURE. WHAT IS HIGH ON THE PAGE IS HIGH IN THE BOX. THE RED DASHED LINE IS REAL. POWER RUNS LEFT OF EACH VALVE, WATER RUNS RIGHT — THEY NEVER CROSS.



INSTRUCTIONAL — BUILD TO THE HARDWARE ANNEX · NAMED PARTS LIVE IN THE ANNEX PARTS CATALOGUE · THE REGISTER GOVERNS

FREE LIKE A FIRE CODE

THE STRIP IS THE CONTRACT: PANEL WIRING LANDS ON TOP IN NFPA COLORS, FIELD WIRING LANDS BELOW IN FACTORY COLORS.

PANEL SIDE (TOP) — BLUE CONTROL · WHT/BLU COMMON

TB1	TB2	TB3	TB4	TB5	TB6	TB7	TB8	TB9	TB10	TB11	TB12	TB13	TB14	TB15	TB16	TB17	TB18	TB19	TB20	G1	G2
M+	M−	LCOM	OPN	CLS	M+	M−	LCOM	OPN	CLS	M+	M−	LCOM	OPN	CLS	XD+	XSIG	5V+	PLS	FCOM	0V	PE
VALVE 1 TRACK					VALVE 2 TRACK					VALVE 3 TRACK					XDCR			FLOW METER		GROUNDS	

FIELD SIDE (BOTTOM) — FACTORY PIGTAIL COLORS THROUGH IP68 GRIPS

LANDING SCHEDULE — VALVE 1 TRACK (REPEAT PATTERN FOR TRACKS 2 AND 3)

TB1 M+ · field: valve RED (drive +) · panel: BLUE from PLC RELAY 1 OUT — the only drive path

TB2 M− · field: valve BLACK (drive −) · panel: WHT/BLU to 0V common (G1)

TB3 LCOM · field: valve GRN/WHT (limit common) · panel: BLUE from 24V+ bus — wets the switches

TB4 OPN · field: valve YELLOW (open limit) · panel: BLUE to PLC IN_0, teed to ACT lamp group (see box)

TB5 CLS · field: valve BLUE (closed limit) · panel: BLUE to PLC IN_1, teed to RDY lamp chain (see box)

TRACK 2 → IN_2/IN_3 · TRACK 3 → IN_4/IN_5 · SAME PATTERN, NEXT CHANNELS

SENSOR BLOCKS

TB16 XD+ · transducer RED · panel: BLUE from 24V+ bus (4-20mA loop power)

TB17 XSIG · transducer BLACK/shield · panel: BLUE to PLC AI_0 — the loop return IS the signal

TB18 5V+ · panel: from PLC 5V logic rail — noise-isolated feed for the meter

TB19 PLS · flow meter pulse out · panel: BLUE to PLC IN_6 — the totalizer counts here

TB20 FCOM · flow meter common · panel: WHT/BLU to 0V (G1)

LAMP LOGIC — WIRED TRUTH, THREE VALVES

RDY (white): all three CLOSED switches in SERIES → lamp lights only when EVERY valve confirms closed.

ACT (red): all three OPEN switches in PARALLEL → lamp lights when ANY valve is physically open.

RELAY LOGIC IN COPPER — THE LAMPS DESCRIBE THE SYSTEM, NOT ONE VALVE, AND NEVER THE SOFTWARE.

GROUND BLOCKS — TWO GROUNDS, NEVER MIXED

G1 0V DC MASTER (WHT/BLU block) — supply negative, DC-UPS negative, valve and sensor returns tie here

G2 CHASSIS EARTH (GREEN rail-bond block) — AC mains ground, enclosure shell, door braid. Bonds to the DIN rail itself.

FIG. 7 — REFERENCE INSTALLATION, CONCEPT VIEW

ILLUSTRATIVE RENDERING OF AN INSTALLED RESIDENTIAL-CLASS NODE UNDER EMBER CAST



CONCEPT ART — NOT A CONSTRUCTION DRAWING

This view illustrates a residential-class node as installed: the sealed wall-mounted enclosure with its external supply connection, the distribution line routed to the roof, and ridge-mounted heads wetting the roof plane and the structure's immediate surroundings during an ember cast. Dimensions, component placement, finish, head count, and head spacing are illustrative. The governing technical disclosure is FIG. 5, the node master schematic, and the Hardware Annex.